

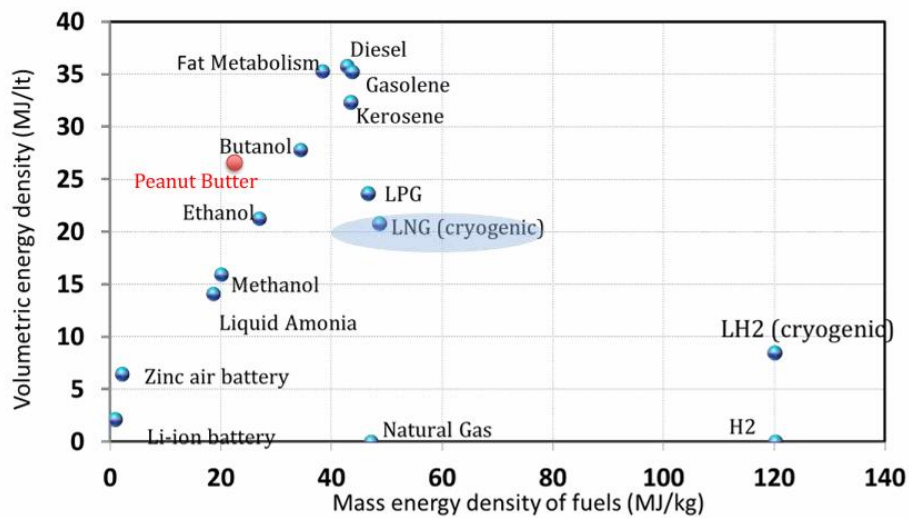
Aero engine technology

Lecture 1

A propulsion system is made up of not only the engine but many different components:

- The energy source, which provides energy to the engine
- Energy to work convertor (e.g. the combustion chamber)
- Thrust producer: takes the work generated and converts it into thrust

When it comes to energy sources, for aviation the most critical characteristic is the energy density of the source w.r.t. volume and mass.



We see that normal batteries have low energy density in both cases, while conventional fuels have much higher energy density, especially per unit volume. On the other hand hydrogen has very low energy density per unit volume, however this can be increased by condensing it (liquefying it).

Many factors need to be considered in the choice of an energy source:

- Volumetric energy density
- Specific energy density
- Emissions
- Policy
- Compatibility
- Safety
- Infrastructure
- Availability
- Cost

Depending on the mission requirements there are different types of engine with varying power outputs:

Type	Power Range	Speed Range (Mach No)	Power to weight (KW/kg)	Approx. Efficiency (%)
Military Engines	< 20 MW	Upto Mach 2	~20	30
Large Turbofans	~ 50 MW	~ Mach 0.9	~10	40
Micro gas turbines	~ 50 kW	< Mach 0.5	~6	10
Turbo Props	< 10 MW	< Mach 0.7	~6	40
Electric Motors	< 250 kW	< Mach 0.5	~5	95
Wankel Engine	< 200 kW	< Mach 0.5	~2.5	25
Radial engines	< 3000 kW	< Mach 0.5	~1.8	30

For aero engines, this power needs to be then converted into thrust. This can be done in two ways:

- Nozzle: the exhaust gasses pass through a convergent channel and their speed is greatly increased.
- Propeller: the power produced is used to spin a propeller which generates thrust. Typically characterized by very high propulsive efficiency but only works for low Mach numbers/

Thermodynamics summary

To fully understand how an aero engine works it is important to recap some basic thermodynamic concepts.

There are 4 fundamental laws in thermodynamics, which state:

Zeroth Law Thermal equilibrium & the concept of temperature

First Law Conservation of Energy and its interconvertability

Second Law Entropy of any isolated system always increases

Third Law Entropy of a system is zero at absolute zero

The first law states the conservation of energy in a closed system, in other words energy is converted from form to another, however the total value is conserved. This can be expressed as:

$$\Delta E = \Delta KE + \Delta PE + \Delta U$$

Where E is the total energy, U is the internal energy (which is a function of temperature), KE is the kinetic energy in the system and PE is the potential energy. The internal energy is a state variable, which means that it's value depends on the first and last thermodynamic states of the system and not on the transformation. It can be expressed as:

$$du = c_v \cdot dT$$

Another important thermodynamic quantity is enthalpy, which can be defined as:

$$h = u + p \cdot v$$

Just like the internal energy it can be expressed as:

$$dh = c_p \cdot dT$$

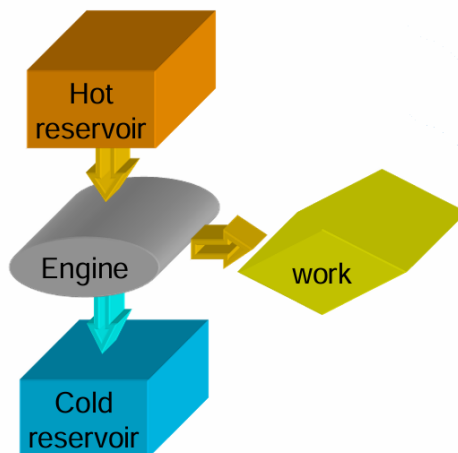
The second law introduces an additional quantity called entropy, which can be thought of as a measure of the disorder of a system. Just like enthalpy and internal energy, entropy is also a state variable so its value only depends on the initial and final thermodynamic states of a transformation. Specifically the second law states that the entropy of the universe always increases in a spontaneous (natural) process and remains unchanged in an equilibrium process (an ideal process in which all thermodynamic states are equilibrium states).

In other words the second law defines the concept of irreversible processes, or processes during which the entropy of the system increases. These processes cannot be reversed and are caused by dissipation and losses.

Entropy can be evaluated as:

$$dS = (\partial Q/T)_{\text{rev}}$$

Going back to the engine, let's now look at the thermodynamics of a heat engine. A heat engine is a machine that operates between two heat reservoirs and completes a thermodynamic cycle. A generator is a heat engine which takes heat from the hot source, produces work and releases heat into the cold source.

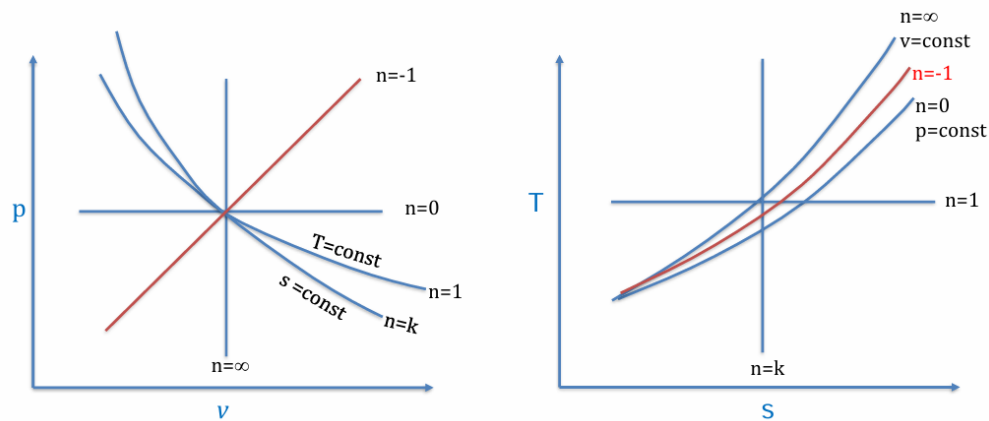


Thermodynamic cycles are typically made up of different thermodynamic processes. These are usually polytropic processes, which can be defined by the following relation:

$$pV^n = C$$

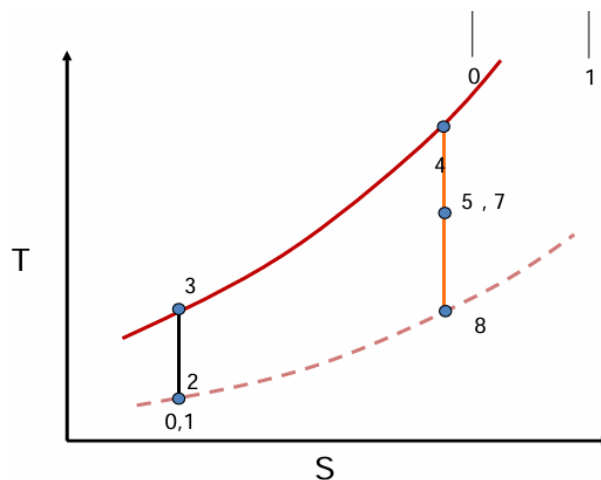
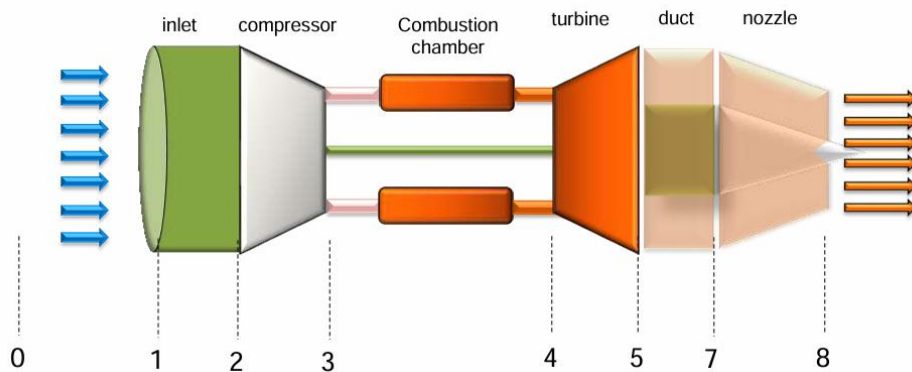
Where n depends on the process:

- $n = 0$: isobaric process
- $n = 1$: isothermal process
- $n = k$: isentropic process (where k is the specific heat capacity coefficient ratio)
- $n = \infty$: isocoric



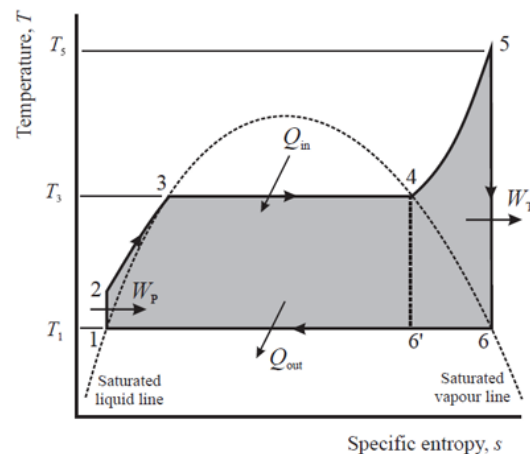
An aero engine is nothing more than a heat engine in which a fluid (air) completes a thermodynamic cycle. This cycle is the Joule-Brayton cycle which is ideally made up of:

- Isentropic compression: this occurs in the inlet and the compressor. In reality the process is adiabatic since there is an associated entropy increase due to losses. Work is added to the cycle with no heat involved.
- Isobaric combustion: this occurs in the combustor. In reality it is not perfectly isobaric since there are pressure losses. Heat is added to the cycle with no work involved.
- Isentropic expansion: this occurs in the turbines and the nozzle. In reality the process is adiabatic since there is an associated entropy increase due to losses. Work is extracted from the cycle with no heat involved. Part of the work extracted is used to run the compressors, while the remaining part is used to generate thrust.
- Isobaric refrigeration: this is the last process required to close out the cycle, however it is not executed by the aero engine. It represents what occurs in the atmosphere: the hot exhaust gasses release heat into the atmosphere at constant atmospheric pressure.



Another example of a thermodynamic cycle is the Rankine cycle, which is made up of:

- Isentropic compression: this occurs in the liquid phase
- Isobaric heating: heat is added into the system. The fluid transitions and becomes vapor at a constant temperature and pressure. Once all of the vapor has transitioned more heat can be added. This last phase is called superheating.
- Isentropic expansion: work is extracted by letting the vapor expand in a turbine
- Condensation: heat is extracted from the system. All of the vapor transitions back to its liquid form.



Some power plants used these two cycles combined to obtain very high thermodynamic efficiencies. The exhaust gasses coming out of a Joule-Bryton cycle are quite hot, so they can be used to heat up the liquid in the Rankine cycle.

Lecture 2

Brayton cycle

The assumptions of the ideal Brayton cycle are the following:

- Ideal gas with constant C_p , C_v and composition
- Mass flow constant
- Isentropic expansion and compression
- No pressure losses in ducts or during heat supply
- No change in kinetic energy per each component (no energy losses)

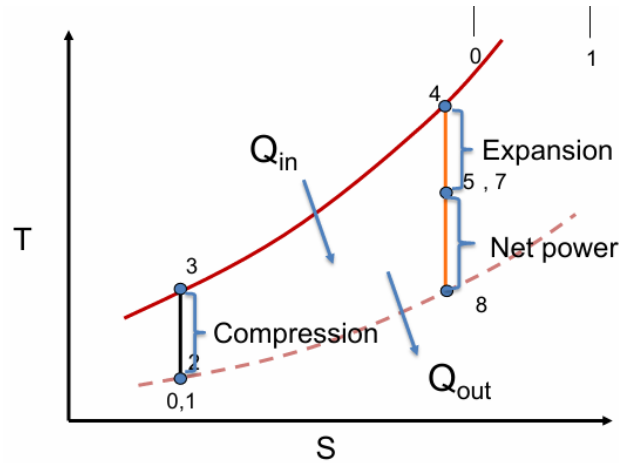
If we consider the whole cycle, from the first law:

$$\Delta U_{2-3-4-8} = (Q_{in} - Q_{out}) + (W_{in} - W_{out}) = 0$$

The change in internal energy is zero since it's a state variable and the initial and final thermodynamic state coincide.

By again using the first law, the work (compression and expansion) and the heat (heat supply and rejection) coming in/out the system can be evaluated:

$$\begin{aligned} \text{Compression} \quad W_{in} &= W_{2-3} = \dot{m} \cdot c_p \cdot (T_3 - T_2) \\ \text{Heat supply} \quad \dot{Q}_{in} &= \dot{m} \cdot c_p \cdot (T_4 - T_3) \\ \text{Expansion 1} \quad W_{4-5} &= \dot{m} \cdot c_p \cdot (T_4 - T_5) \\ \text{Expansion 2} \quad W_{net} &= W_{5-8} = \dot{m} \cdot c_p \cdot (T_5 - T_8) \end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} W_{out}$$



With these relations a power balance equation can be setup:

$$W_{2-3} + \dot{Q}_{3-4} = W_{4-5} + W_{5-8} + \dot{Q}_{8-2}$$

Since the power extracted by the turbine is used to spin the compressor:

$$W_{4-5} = W_{2-3}$$

$$\dot{Q}_{3-4} = W_{5-8} + \dot{Q}_{8-2}$$

The heat we need to provide to the gas in the combustion chamber is equal to the power generated in the nozzle and the heat that is released into the environment by the exhaust gasses.

By using these expressions, the thermodynamic efficiency of the Brayton cycle can be evaluated. This is defined as the ratio between the net power output and the total energy supply.

$$\begin{aligned} \eta_{thdy} &= \frac{W_{net}}{\dot{Q}_{in}} = \left(\frac{W_{4-8} - W_{2-3}}{\dot{Q}_{3-4}} \right) \\ &= \left(\frac{c_p \cdot \dot{m} \cdot (T_4 - T_8) - c_p \cdot \dot{m} \cdot (T_3 - T_2)}{c_p \cdot \dot{m} \cdot (T_4 - T_3)} \right) \\ \eta_{thdy} &= \left(\frac{(T_4 - T_8) - (T_3 - T_2)}{(T_4 - T_3)} \right) \end{aligned}$$

Since compression and expansion are assumed to be isentropic, we can use the isentropic relations for temperature and pressure:

$$\left(\frac{T_4}{T_8} \right) = \Pi^{\frac{k-1}{k}} \quad \left(\frac{T_3}{T_2} \right) = \Pi^{\frac{k-1}{k}}$$

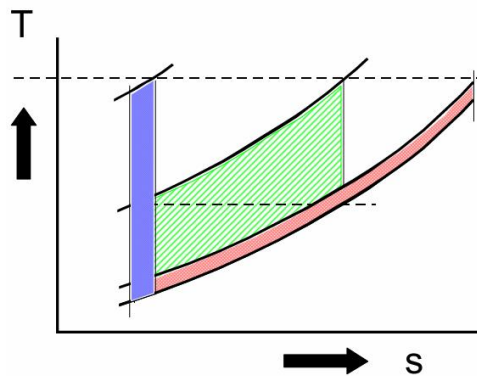
$$\eta_{thdy} = \frac{\left(T_4 \left(1 - \frac{T_8}{T_4} \right) - T_3 \left(1 - \frac{T_2}{T_3} \right) \right)}{(T_4 - T_3)} = \frac{\left(T_4 \left(1 - \frac{1}{\Pi^{\frac{\gamma-1}{\gamma}}} \right) - T_3 \left(1 - \frac{1}{\Pi^{\frac{\gamma-1}{\gamma}}} \right) \right)}{(T_4 - T_3)}$$

$$\eta_{thdy} = \frac{\left((T_4 - T_3) \cdot \left(1 - \frac{1}{\Pi^{\frac{\gamma-1}{\gamma}}} \right) \right)}{(T_4 - T_3)}$$

$$\eta_{thdy} = \left[1 - \frac{1}{\Pi^{\frac{\gamma-1}{\gamma}}} \right]$$

We can see that the thermodynamic efficiency of the Joule-Brayton cycle is only dependent on the pressure ratio, the pressure difference between the max pressure (reached after the isentropic compression) and the min pressure (reached after the isentropic expansion).

If we only cared about the maximum thermodynamic efficiency then we would need to maximize the pressure ratio and the ideal cycle would be the purple one.



However, additional factors need to be considered. The net power is defined as the total power generated in the expansion process, minus the power required to fuel the compressor:

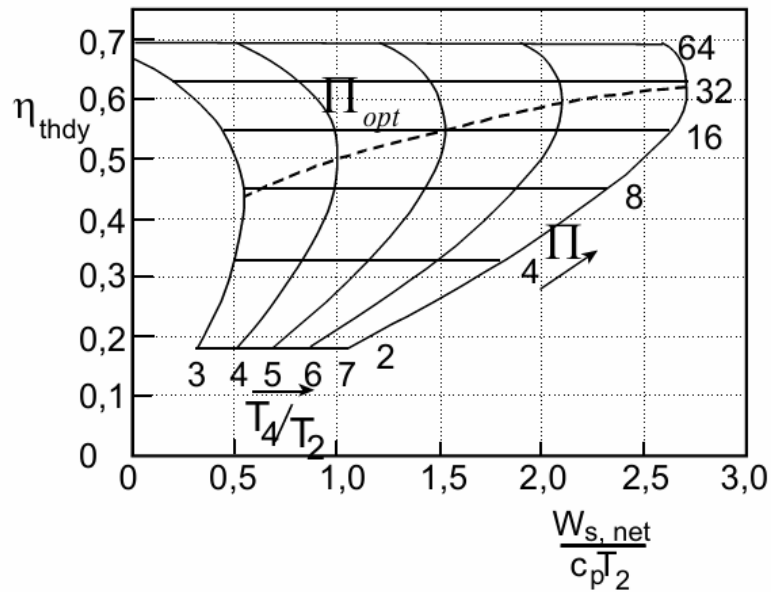
$$W_{net} = W_{4-8} - W_{2-3}$$

$$= \dot{m} \cdot c_p \cdot (T_4 - T_8) - \dot{m} \cdot c_p \cdot (T_3 - T_2)$$

With a similar procedure as the one used to obtain the thermodynamic efficiency:

$$\frac{W_{s,net}}{c_p T_2} = \frac{T_4}{T_2} \left[1 - \frac{1}{\Pi^{\frac{\gamma-1}{\gamma}}} \right] - \left[\Pi^{\frac{\gamma-1}{\gamma}} - 1 \right]$$

Where $W_{s,net}$ is the specific net power, or the power generated per unit of mass flow, which is then normalized w.r.t. the ambient enthalpy. We can see that this parameter, which is as important as the thermodynamic efficiency is not only dependent on the pressure ration but also the temperature ratio. For an ideal cycle we have that the pressure ratio and the temperature ratio have to be chosen at the optimum:



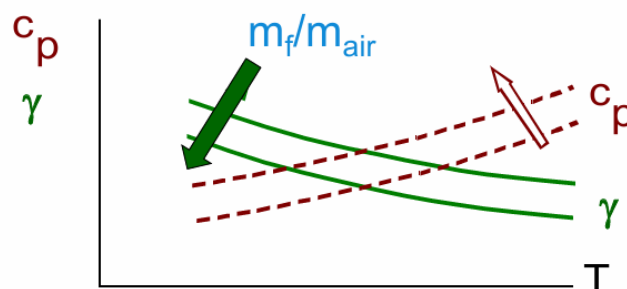
As the temperature ratio increases the net power also increases, however for a given temperature ratio there is a specific pressure ratio which maximized the net work. The thermodynamic efficiency depends only on the pressure ratio.

Real Brayton cycle

All of the previous results are based on the many assumptions listed at the beginning, which are:

- Ideal gas with constant C_p , C_v and composition
- Mass flow constant
- Isentropic expansion and compression
- No pressure losses in ducts or during heat supply
- No change in kinetic energy per each component (no energy losses)

Let's look at the validity of all these assumptions. The assumption of constant C_p and C_v is not valid for a real cycle since they both depend on temperature and also composition. The addition of fuel in the combustion chamber changes the composition of the fluid, typically reducing the heat capacity ratio and increasing the heat capacity at constant pressure.



$c_{p,a} = 1000 \text{ J/kg.K}$	$\gamma_a = 1.4$
$c_{p,g} = 1150 \text{ J/kg.K}$	$\gamma_g = 1.33$

This can lead to over/underestimations of up to 5% for the efficiency of an engine. The addition of the fuel flow also breaks the assumption of constant mass flow rate. However, typically this addition is so small that it can be neglected.

Another important assumption for the ideal Brayton cycle is that there are no kinetic energy changes in each component, however this is not true for the real cycle. To take into account these kinetic energy variations, since we do not know the velocity of the fluid in all the components total quantities are used.

$$T_t = T_s + \frac{v^2}{2} \quad p_t = p_s \left(\frac{T_t}{T_s} \right)^{\frac{\gamma}{\gamma-1}}$$

For a real Brayton cycle we need to introduce efficiencies which take into account losses in both the compressor and the turbine. These efficiencies are defined as isentropic efficiencies since the ideal power is defined along an isentropic expansion/compression.

$$\eta_c = \frac{W_{c,ideal}}{W_{c,real}} = \frac{\dot{m} \cdot (c_{p3'} \cdot T_{t3'} - c_{p2} \cdot T_{t2})}{\dot{m} \cdot (c_{p3} \cdot T_{t3} - c_{p2} \cdot T_{t2})}$$

$$\eta_t = \frac{W_{t,real}}{W_{t,ideal}} = \frac{\dot{m} \cdot (c_{p4} \cdot T_{t4} - c_{p5} \cdot T_{t5})}{\dot{m} \cdot (c_{p4} \cdot T_{t4} - c_{p5'} \cdot T_{t5'})}$$

For both compression and expansion, in the real case an increase in entropy is associated. For the real compressor more work is required to achieve the same pressure ratio, while for the turbine less work is extractable for the same pressure ratio.

When solving a real Brayton cycle these efficiencies can be used to evaluate the final temperature/pressure reached. For the compressor:

$$T_{t3} = \frac{1}{\eta_c} \cdot T_{t2} \cdot \left(\frac{T_{t3'}}{T_{t2}} - 1 \right) + T_{t2} = T_{t2} \cdot \left(\frac{\pi_c^{\frac{\gamma}{\gamma-1}} - 1}{\eta_c} + 1 \right)$$

For the turbine:

$$P_{t5} = P_{t4} \cdot \left(1 - \frac{1}{\eta_t} \cdot \left(1 - \frac{T_{t5}}{T_{t4}} \right) \right)^{\frac{\gamma}{\gamma-1}}$$

The total temperature at the turbine outlet is measured through a power balance with the compressor:

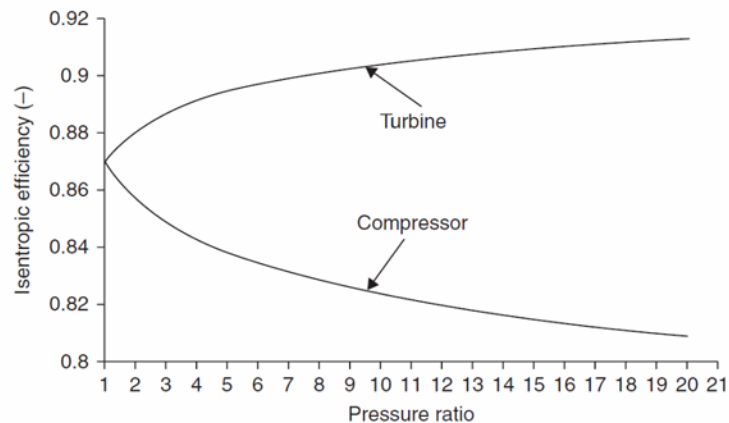
$$W_{2-3} = W_{4-5}$$

$$\dot{m} \cdot (c_{p3} \cdot T_{t3} - c_{p2} \cdot T_{t2}) = \dot{m} \cdot (c_{p4} \cdot T_{t4} - c_{p5} \cdot T_{t5})$$

$$\dot{m} \cdot c_{pa} \cdot (T_{t3} - T_{t2}) = \dot{m} \cdot c_{pg} \cdot (T_{t4} - T_{t5})$$

$$T_{t5} = T_{t4} - \frac{\dot{m} \cdot c_{pa} \cdot (T_{t3} - T_{t2})}{\dot{m} \cdot c_{pg}}$$

In general as the pressure ratio increases the efficiency of the turbine increases as well, while the efficiency of the compressor decreases, so the benefits of increasing the pressure ratio are reduced.



Mechanical losses occur between the turbine and the compressor, so a mechanical efficiency can be defined (although it is typically very high ~99.5%).

$$\eta_m = \frac{W_t - \text{mechanical losses}}{W_t}$$

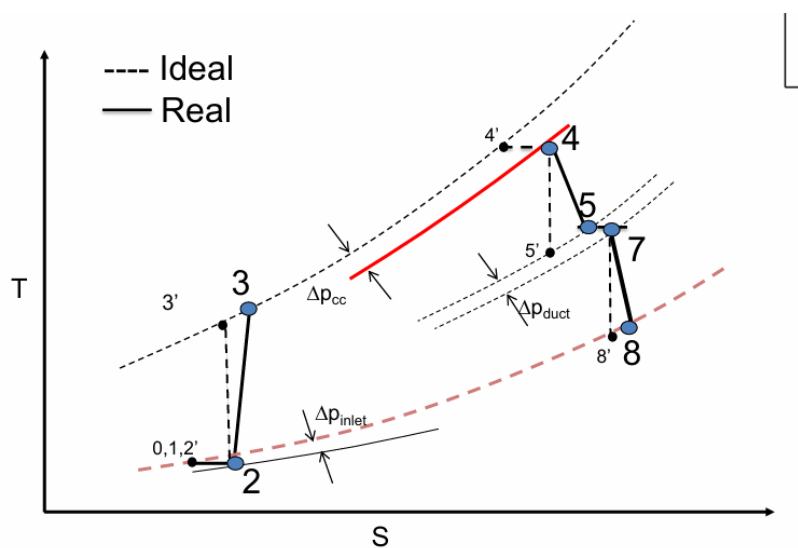
Pressure losses occur in every component and have to be accounted for by introducing pressure ratios.

Inlet: $\Delta p_{inlet} = p_{t1} - p_{t2}$

Combustion chamber: $\Delta p_{cc} = p_{t3} - p_{t4}$

Duct: $\Delta p_{duct} = p_{t5} - p_{t7}$

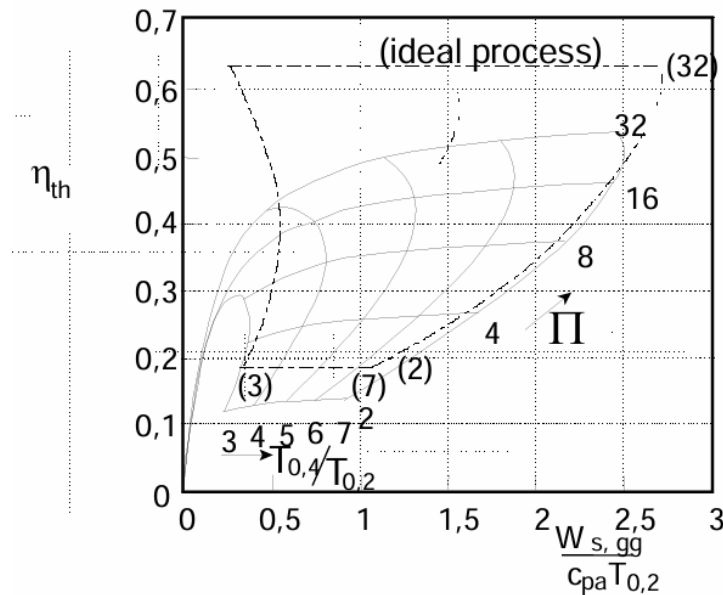
Overall the real Brayton cycle is:



Pressure losses in the inlet bring state 2 to a lower pressure. Losses in the compressor tilt the compression curve so that entropy increases and the power required increases. In the combustion chamber, due to pressure losses, we move to a curve with lower pressure. In the turbine losses dictate

an increase in entropy and a decrease in the total work that is extracted. Then pressure losses in the duct reduce the pressure from 5 to 7. Finally the expansion in the nozzle is also tilted to a higher entropy.

By introducing losses the relations obtained for an ideal Brayton cycle break down. We see that for high pressure/temperature ratio they still roughly hold, but for lower temperature ratios this is not the case:

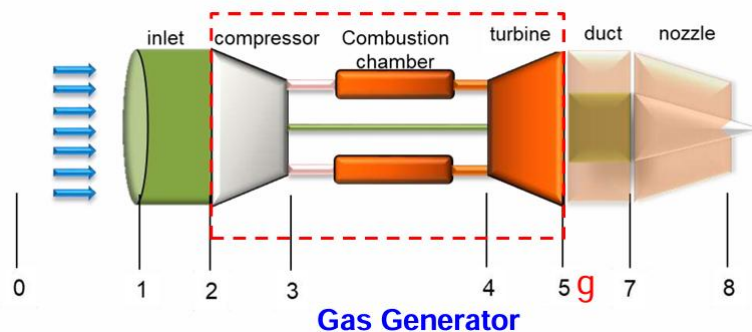


For lower temperature ratios, very little heat is being added into the system, while at the same time high pressure ratios are trying to be obtained. This is not possible since the cycle cannot maintain itself (work extracted is not enough to fuel the compression) and therefore both the net power and the efficiency go to zero.

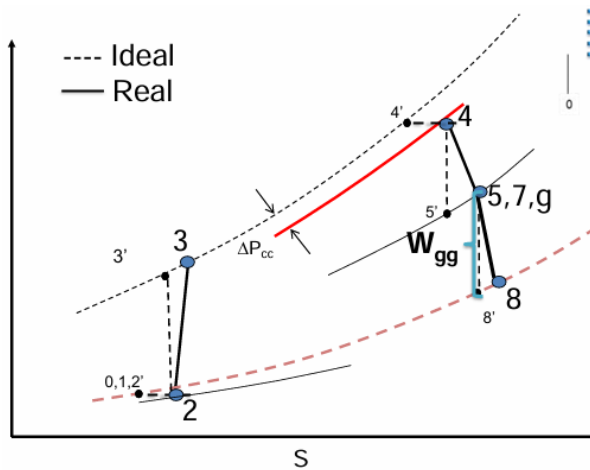
Lecture 3

Gas Generator

The "gas generator" part of the engine is the part which generates the gas power, or the power that is then transformed into kinetic energy. This transformation process into kinetic energy occurs after station g.

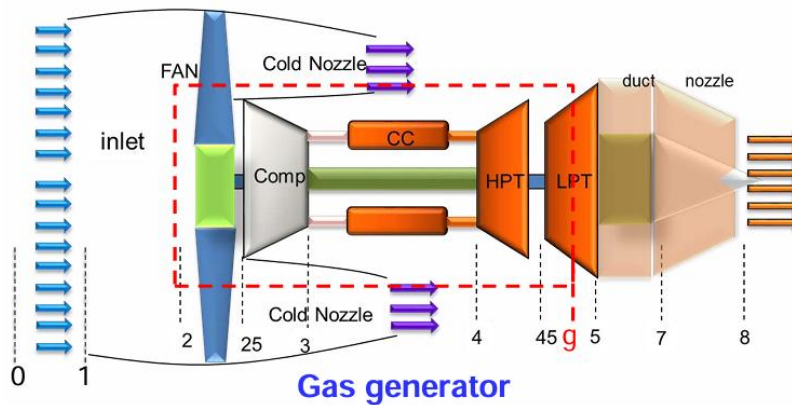


We can evaluate the power generated by this section as the gas power, or the power generated by the equivalent isentropic expansion from station g to 8'.

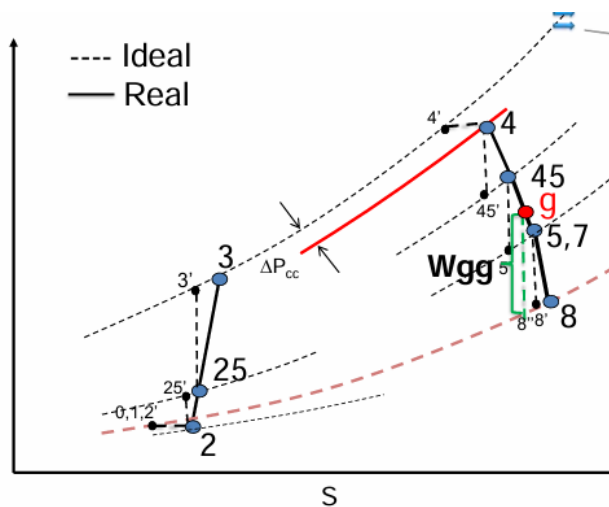


$$W_{gg} = \dot{m} \cdot c_p \cdot (T_g - T_{8'})$$

A turbofan differs from a turbojet due to the presence of the fan in front of the compressor, which compresses both the bypass and the core flow. Thus the gas generator part is also different.



In this case station g, where the gas generator ends, is positioned in the middle of the low pressure turbine. This is because the HPT drives the compressor of the core flow, while the LPT drives the fan. This energy is divided in gas generation (compresses the core flow) and thrust generation (compresses the bypass flow). Therefore the station g is not exactly at the end of the LPT but slightly before. This also complicates the equation for the gas power as we need to determine the exact position of position g in the T-S diagram.



$$W_{gg} = \dot{m} \cdot c_p \cdot (T_g - T_{8'})$$

Some useful definitions for aero engines (assuming choked nozzle conditions):

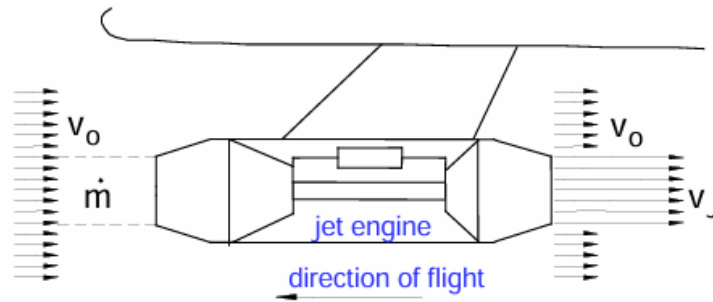
Gross thrust $F_G = \dot{m} v_j$

Net thrust $F_N = \dot{m} * (v_j - v_o)$

Propulsive jet power $\dot{W}_{prop, jet} = \frac{1}{2} \dot{m} (v_j^2 - v_o^2)$ Kinetic energy that is can be transformed into thrust

Thrust power $\dot{W}_{thrust} = \dot{m} (v_j - v_o) v_o$ Given by the product of net thrust and flight velocity

Power loss $P_{loss} = \frac{1}{2} \dot{m} (v_j - v_o)^2$ Residue kinetic energy at the nozzle exit that is not transformed into thrust



From the these equations we can see that we can increase the net thrust by increasing either the mass flow rate or the exhaust velocity. However, by increasing the exhaust velocity the energy losses also increase with the square of the exhaust velocity, so it is much more convenient to increase the mass flow rate. This is the basic idea of the turbofan engine.

If the nozzle is in the unchoked condition the pressure at the nozzle exit is different from the environment pressure and this difference generates some thrust, which is proportional to the nozzle exit area and the pressure difference. This pressure difference can also be accounted for by defining an effective exhaust velocity $v_{j,eff}$.

Gross Thrust $F_G = \dot{m} \cdot v_j + A_{exit} \cdot (p_j - p_o) = \dot{m} \cdot v_{j,eff}$

Net Thrust $F_N = \dot{m} \cdot (v_j - v_o) + A_{exit} \cdot (p_j - p_o) = \dot{m} \cdot (v_{j,eff} - v_o)$

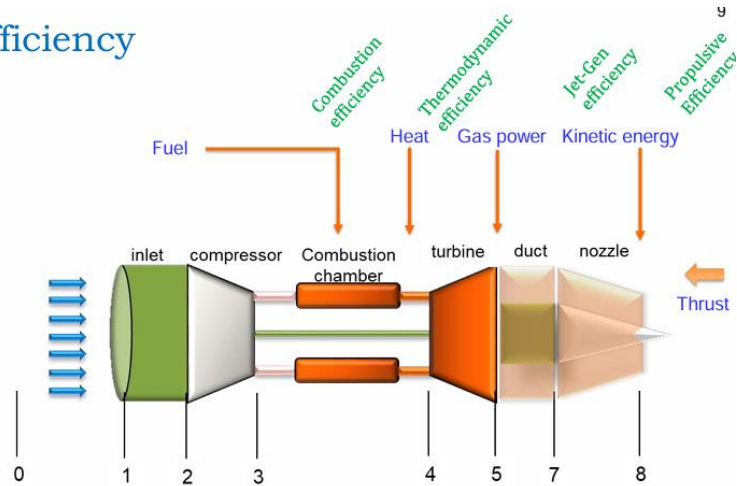
Thrust power $\dot{W}_{thrust} = F_N \cdot v_o$

Power loss $P_{loss} = \frac{1}{2} \dot{m} (v_{j,eff} - v_o)^2$

Aero engine efficiencies

Every station of the turbojet engine introduces losses that reduce the overall efficiency.

Efficiency



- **Combustion efficiency:** the process that occurs is the transformation of chemical energy into heat. This takes into account the losses that occur in the combustion chamber due to incomplete combustion. The efficiency is given by extra amount of heat produce divided by the chemical energy contained in the fuel.

The process that occurs is the transformation of chemical energy into heat.

$$\eta_{comb} = \frac{(\dot{m}_{gas} \cdot c_{p,g} \cdot T_4 - \dot{m}_{air} \cdot c_{p,air} \cdot T_3)}{\dot{m}_f \cdot LHV_f} \approx \frac{\dot{m}_a \cdot c_{p,a} \cdot \Delta T_{comb}}{\dot{m}_f \cdot LHV_f}$$

Fuel Type	Lower Heating Value (LHV) [MJ/kg]
Kerosene	43
Hydrogen	120

- **Thermodynamic efficiency:** : the process that occurs is the transformation of heat into gas power. This determines the part of heat that is transformed into gas power and it's given as the ratio between gas power and total heat produced in the combustion process.

$$\eta_{thdy} = \frac{W_{gg}}{\dot{m}_{air} \cdot c_{p,a} \cdot \Delta T_{comb}}$$

Where the gas power can be evaluated as follows:

$$W_{gg} = \dot{m}_{gas} \cdot c_{p,g} \cdot (T_g - T_{8'})$$

$$W_{2-3} = W_{4-5} \rightarrow T_{t5}$$

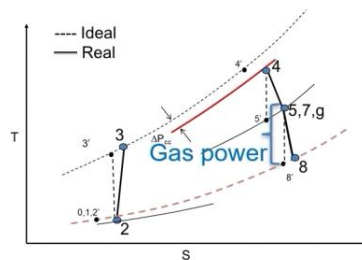
$$T_g = T_{t5}$$

$$P_g = P_{t5}$$

$$P_{8'} = P_{ambient}$$

Isentropic process

$$\frac{T_g}{T_{8'}} = \left(\frac{P_g}{P_{8'}} \right)^{1-1/\gamma} = \left(\frac{P_g}{P_{ambient}} \right)^{1-1/\gamma}$$



- Jet generation efficiency: the process that occurs is the transformation of gas power into kinetic energy. This takes into account the losses that occur in the duct and nozzle during the expansion.

$$\eta_{jet, gener.} = \frac{\frac{1}{2} \dot{m} (v_j^2 - v_o^2)}{W_{gg}}$$

- Propulsive efficiency: the process that occurs is the transformation of kinetic energy into thrust power. This takes into account the kinetic energy that cannot be transformed into thrust.

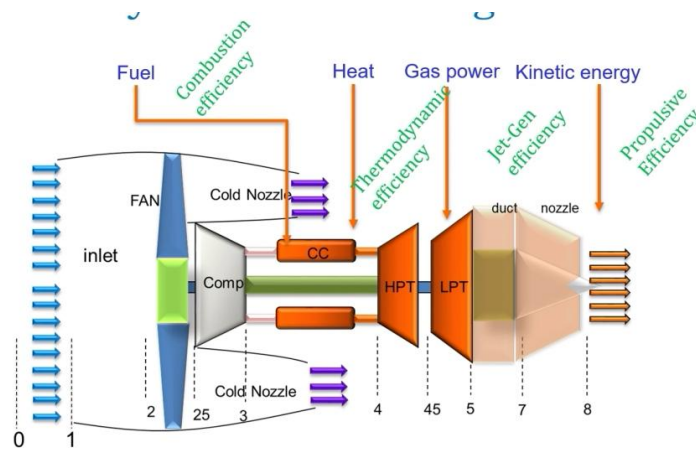
$$\eta_{prop} = \frac{W_{thrust}}{W_{kinetic}} = \frac{\dot{m} (v_j - v_o) v_o}{\frac{1}{2} \dot{m} (v_j^2 - v_o^2)}$$

$$\eta_{prop} = \frac{2}{1 + \frac{v_j}{v_o}}$$

- Total efficiency: takes into account all of the losses from the combustion chamber to the generation of thrust. It quantifies how efficiently the gas turbine transforms the chemical energy of the fuel into thrust power.

$$\eta_{total} = \frac{\dot{m} (v_j - v_o) v_o}{\dot{m}_f LHV_f}$$

For a turbofan the same efficiencies can be defined and evaluated with some caveats:



- Combustion efficiency: defined and evaluated in the same way as we did for the turbojet.
- Thermodynamic efficiency: defined and evaluated in the same way as we did for the turbojet. However, the gas power is different in this case, since station g changes as well as the final temperature T_8'' . Again the temperature at the station g is evaluated by setting up the power balance between the expansion from 4-g and the power needed to drive the core compressor (including the part of the fan that compresses the core flow).

To define the pressure at station g we need to evaluate it from the turbine efficiency.

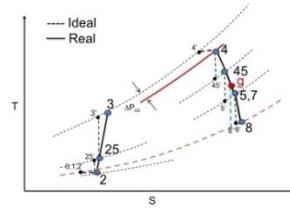
Again the temperature at the exhaust after an isentropic expansion used to evaluate the gas power is obtained through the isentropic relations since $p_8'' = p_8$.

$$W_{gg} = \dot{m}_{gas} \cdot c_{p,g} \cdot (T_g - T_{8''})$$

$$W_{2-3} = W_{4-5} \Rightarrow T_g$$

$$\left. \begin{array}{l} P_g = f(\eta_t, T_g) \\ P_{8''} = P_{ambient} \end{array} \right\} \text{Isentropic process}$$

$$\frac{T_g}{T_{8''}} = \left(\frac{P_g}{P_{8''}} \right)^{1-1/\gamma} = \left(\frac{P_g}{P_{ambient}} \right)^{1-1/\gamma}$$



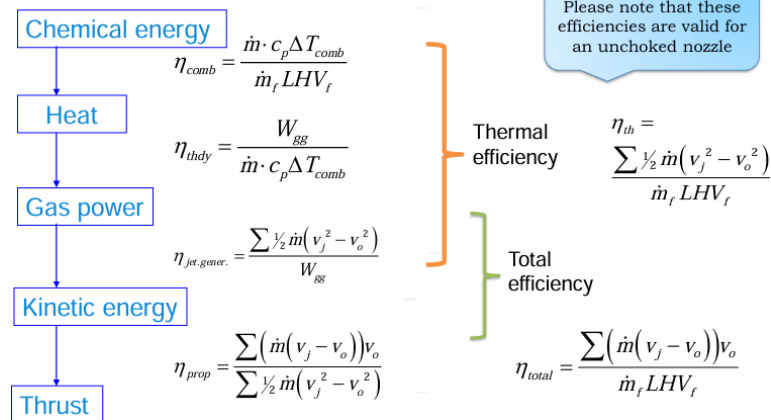
- Jet generation efficiency: defined in the same way as we did for the turbojet. In this case the kinetic energy not only takes into account the core flow but also the bypass flow.

$$\eta_{jet, gener.} = \frac{\sum \frac{1}{2} \dot{m} (v_j^2 - v_o^2)}{W_{gg}}$$

- Propulsive efficiency: defined in the same way as we did for the turbojet. In this case the both flows need to be taken into account.

$$\eta_{prop} = \frac{W_{thrust}}{W_{kinetic}} = \frac{\sum (\dot{m} (v_j - v_o)) v_o}{\sum \frac{1}{2} \dot{m} (v_j^2 - v_o^2)}$$

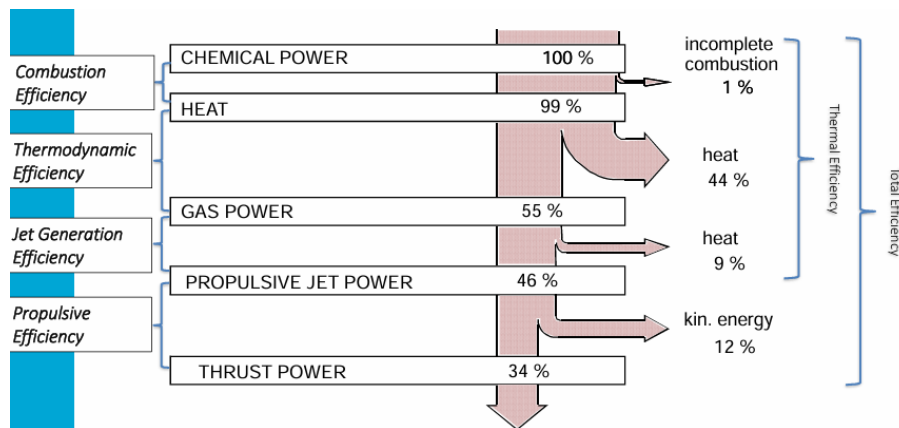
Overall for a turbofan:



Where the thermal efficiency can be obtained from the product of the combustion, thermodynamic and jet generation efficiencies. The global efficiency is still given by considering all of the losses encountered from the combustion chamber to the production of thrust.

For an unchoked nozzle the exhaust velocity needs to be substituted by the effective exhaust velocity.

For a generic jet engine we can visualize all of the losses through a Sankey diagram:



Although the actual values might change from engine to engine, the largest amount of energy will always be lost through heat and will be taken into account by the thermodynamic efficiency.

Let's now introduce some useful parameters to compare engines:

▪ Net Thrust $F_N = \sum [\dot{m}(v_j - v_o) + A_{exit}(p_j - p_o)]$

▪ Specific Thrust $F_s = \frac{F_N}{\dot{m}_a}$

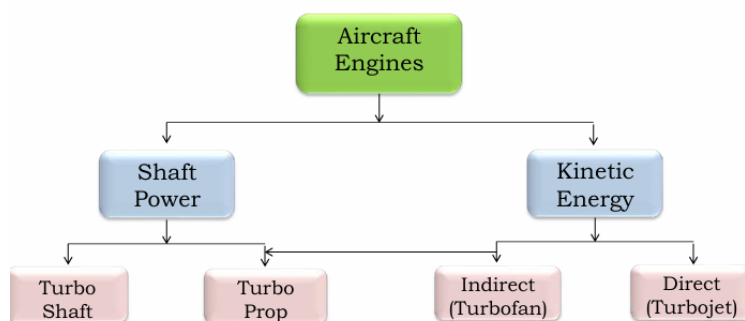
▪ Thrust specific fuel consumption (TSFC)

$$TSFC = \frac{\dot{m}_f}{F_N} \propto \frac{v_0}{\eta_{tot}}$$

Specific thrust allows to quantify how efficiently an engine can generate thrust w.r.t. its size by dividing the net thrust by the mass flow rate. The thrust specific fuel consumption is instead given by the fuel flow ratio divided by the net thrust and it is inversely proportional to the global efficiency.

Aero engine classification

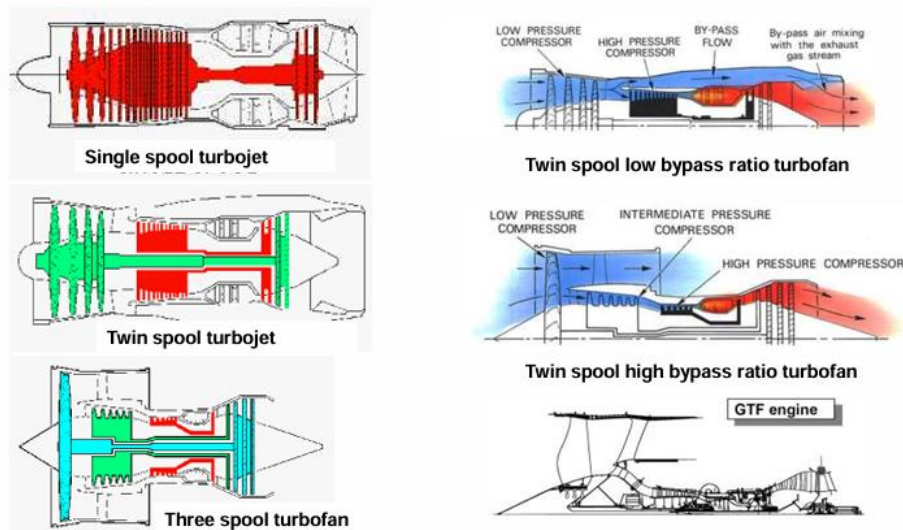
Aero engines can be classified in many ways. One is to distinguish them based on how the gas power is used.



The gas power can either be converted into shaft power or kinetic energy. Both the turbopan and the turboprop actually lie between the two categories since they use the gas power to both generate shaft power and thrust.

The other classification is based on the architecture of the engine. The simplest architecture is the single spool turbojet, which has been phased out for twin spool turbojets/turbopans (which have a high/low pressure compressor and a high/low pressure turbine) or even three spool turbopans.

This is because in the single spool turbojet, as the pressure increases the compressor stage loading is unbalanced.

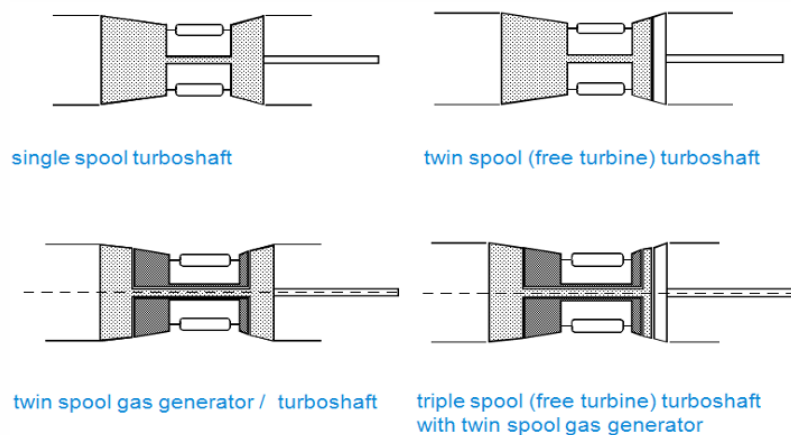


The different spools allow each part of the compressor to run at their optimal speed.

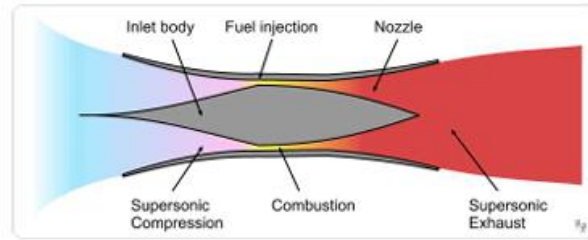
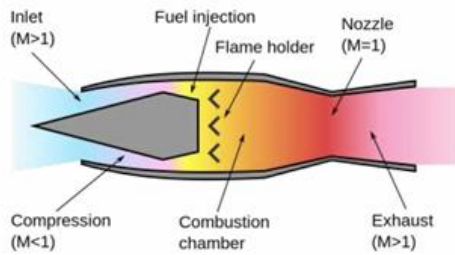
Turbofan engine can also be classified based on how large their bypass ratio (ratio between the mass flow rate in the fan and in the core) is.

An additional engine that is outside of this classification is the Gear TurboFan engine GTF, a two spool turbofan with a gearbox which decouples the fan and the low pressure compressor. This engine has the advantage of a three spool turbofan while effectively only having two spools.

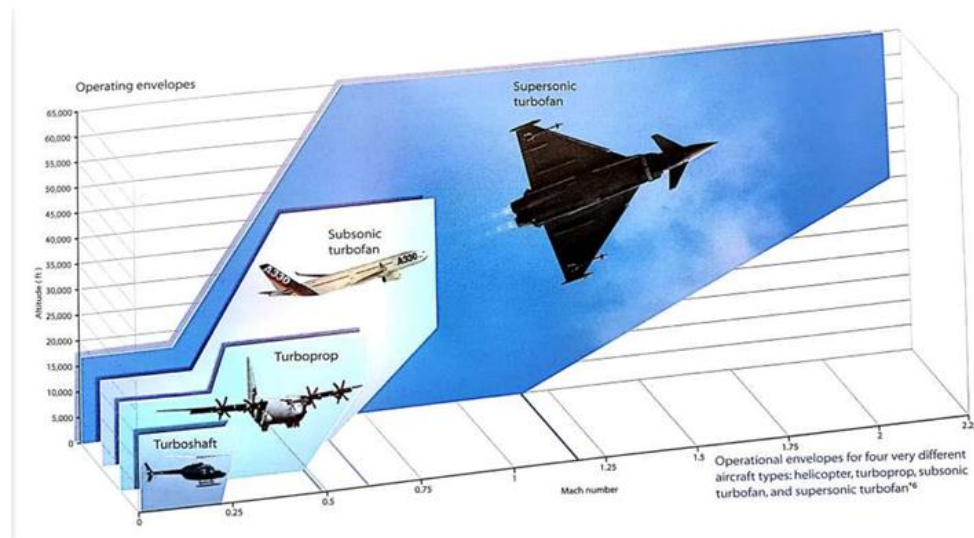
A similar classification can also be applied to turboshaft/prop engines.



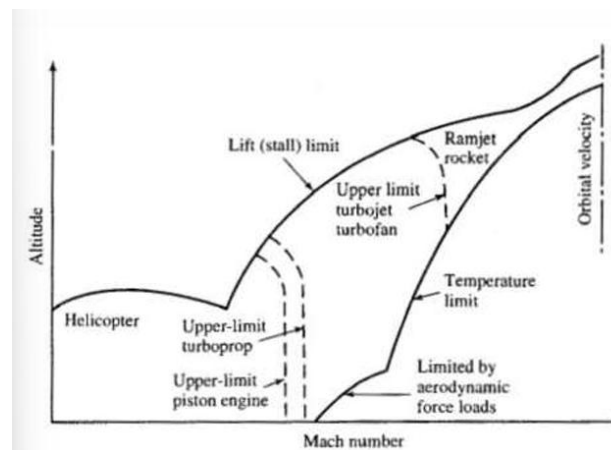
There are two other types of engines which follow the Brayton cycle that do not fit into these classifications : the ramjet and scramjet. These two types of engines do not have any mechanical components since they compress the air through dynamic compression while flying at supersonic speeds. Therefore the compressor stage is not needed and neither the fan stage which drives it. These engines only work at high supersonic speeds at which air compresses dynamically in virtue of the high speeds.



Different engine types have different operating envelopes as shown in the figure below.



The limits of this flight envelope are defined by different phenomena. At high flying altitudes and low speeds the limiting factor is the stall of the wings, while at low altitudes and high speeds it is the airfram/engine loads limits.



Another thing that distinguishes different engine architectures is the behaviour of propulsive efficiency to airspeed. In absolute terms, the turboprop has the highest propulsive efficiency at low speeds, however this drops drastically at transonic flight speeds due to tip shock on the propeller blades. The turbofan has the highest propulsive efficiency in the transonic and supersonic regimes,

beating out the turbojet. We also observe that as the bypass ratio increases the propulsive efficiency also increases.

Engine efficiency trends

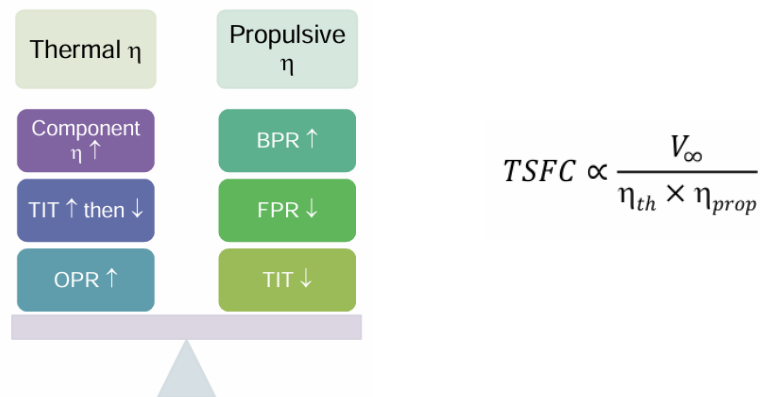
In general we see an increase of all the efficiencies with the development of new technologies. The efficiency of aero engines increase by 70% in the past decades, while the thrust specific fuel consumption of aero engines decreases by 50%.

The thrust specific fuel consumption reduction is due to the improvements of engine component efficiency, cycle efficiency and a large part being the propulsive efficiency.

Nowadays typical values for the thermal efficiency are around 50%, 70-75% propulsive efficiency (for high bypass ratio turbofan engines, for which the bypass ratio has also doubled in the past decades) and total efficiencies of 40%.

Design considerations

To minimize the specific fuel consumption of new engines we want to maximize both the thermal efficiency and the propulsive efficiency, which are dependent on the following parameters:



To increase the thermal efficiency we need to increase the efficiency of all engine components as well as the Overall Pressure Ratio OPR. For the Turbine Inlet Temperature TIT there is a specific value for which the thermal efficiency is maximized.

Meanwhile for the propulsive efficiency we want to increase the ByPass Ratio BPR with a combined reduction of the Fan Pressure Ratio FPR. Also we want the TIT to decrease.

We can see that we have opposing design goals, such as the increase in TIT to increase the thermal efficiency which leads to a decrease of propulsive efficiency.

Therefore design choices need to be made.

We can see how the design choices have lead to improvements over the years by comparing an engine from the '50s and a modern engine.

We observe an increase in thrust generated, combined with a decrease in weight and in specific fuel consumption. This is due to the higher TIT and OPR while having fewer compressor stages. We also can observe an increase in diameter together with the BPR, while the overall length decreased.

Performance F104 Starfighter ↔ F16

Engine Parameter	Unit	J79	F100
Thrust (wet)	kN	81	108
Thrust (dry)	kN	53.5	68
Gas generator airflow	kg/s	77	61.5
Sp. Thrust	kN/kg/s	0.69	1.1
Sp. Fuel Cons, SFC	g/Nh	84	68
Bypass Ratio	-	0	0.7
Turbine Inlet Temp	°C	990	1395
Comp pr. Ratio	-	13.5	25
No. of stages Comp	-	17	3+10
No. of stages of Turbine	-	3	2 + 2
Rotor speed	Rpm	7690	9600/11900
Weight	kg	1750	1360
Diameter	cm	77.8	100
Diameter	cm	530	488

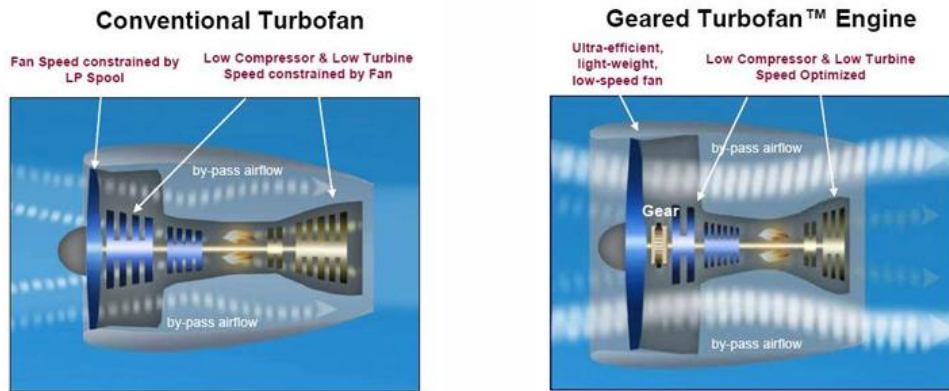
This analysis is valid for military engines, however similar trends can be observed for civil engines.

Parameters	JT 8D	CFM56	Leap-1A
Bypass Ratio	-	1.62	5.2
Core Mass Flow Rate kg/s	81.2	52.2	41.5
Fan Pressure Ratio	1.9	1.6	1.4
HPC Pressure Ratio	3.5	8	13
Turbine Inlet Temperature °C	1030	1270	1450
Overall Pressure Ratio	16.9	32.5	33
Fan, LPC & HPC Isentropic Efficiency	0.83	0.88	0.92
LPT and HPT Isentropic Efficiency	0.88	0.91	0.92
Combustor Efficiency	0.985	0.995	0.995
NR. OF STAGES IN COMPRESSOR / TURBINE	2+6+7 / 1+3	1+3+9 / 1+4	1+3+10 / 2+5
DIAMETER m	1.25	1.55	1.75

One small thing is that some newer engines are actually heavier than the old ones. This is due to the fact that fuel consumption has become one of the most important driving design parameters, while the thrust requirements have pretty much stayed the same.'

Property:	CFM56-5B	LEAP-1A26
Entry in service:	1996	2016
Take-off thrust [kN]:	≈120	≈120
Weight [kg]:	2381	3008

A new introduction is the GTF engine which has a few advantages over conventional turbofans.

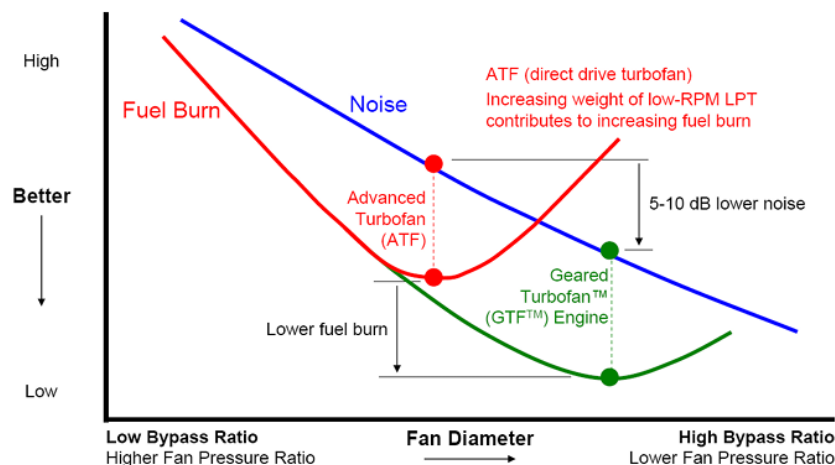


- De-couple the fan from low pressure spool by reduction gear system.
- High bypass ratio turbofan engine
- Low fan speed
- Higher low pressure spool speed

An ultra-high bypass ratio can be achieved this way, while the fan speed can be drastically reduced thanks to the gearbox, thus reducing the fan noise and increasing the propulsive efficiency.

At the same time the LPT can rotate at much higher speeds, greatly increasing its efficiency while decreasing the number of stages needed and thus the weight.

This reduction of fuel burn can only be achieved with this GTF architecture, since for a normal turbofan an increase in BPR eventually leads to higher fuel burn due to the increased weight of the LPT which has to spin at low speeds since it's coupled with the fan.

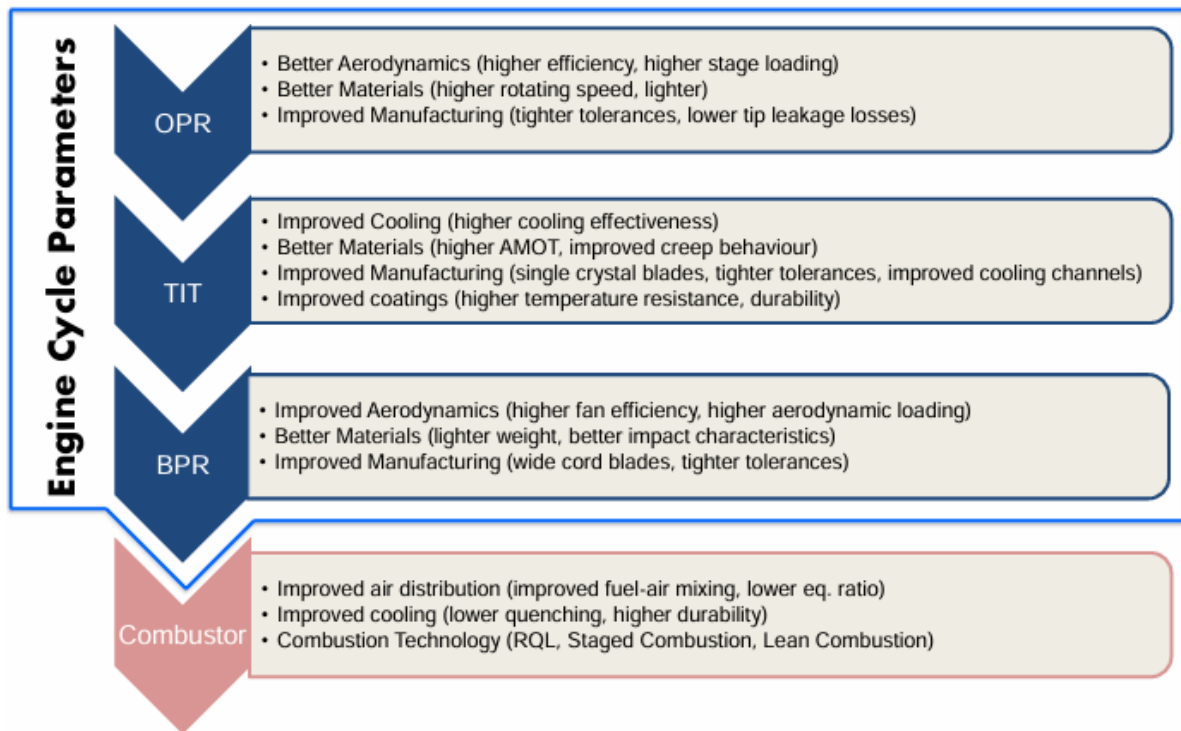


To increase the thermal efficiency we can increase the OPR. If we look at the trend over the years for the OPR this is exactly what we see, with modern engines having an OPR of around 50, almost double the OPR of engine from the '70s.

The TIT also shows a general trend to increase over the years with modern TIT reaching values of 2000 K. This is actually much higher than the melting point of the materials used for the turbine blades, however this gap is bridged by using advanced blade cooling techniques.

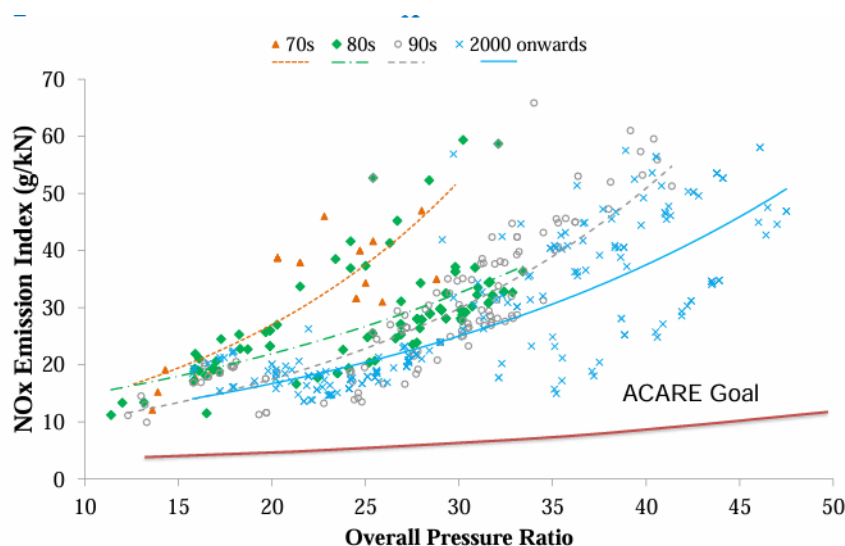
The BPR has had a much steeper increase over the last decades reaching values of 12-13, compared to BPRs of around 2-5 in the '70s and '80s. At the same time the FPR has been decreasing.

The single components efficiencies have also been increasing, with HPC efficiencies reaching values of almost 90% in the last couple of years, while also having higher BPR and lower number of stages.



Future design trends

While older designs were mainly driven by increased thrust and lower fuel consumption, modern goals also include the reduction of NO_x emissions, which tend to increase with OPR. We see that over the years the correlation between NO_x emissions and OPR is decreasing thanks to better combustion.

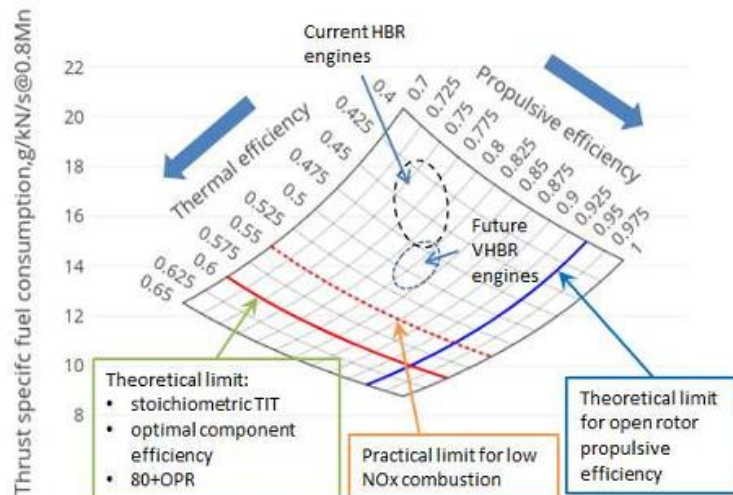


The ACARE goal is the goal that has been set to be achieved by 2050.

The main effects which have environmental effects are the NO_x and CO₂ emissions, as well as contrails left behind by the engine.

As we increase the design parameters of the engine (TIT, BPR and OPR) the fuel consumption and therefore the CO₂ emissions decrease, while contrails formation and NO_x emissions increase.

Other limitations exist for the future engines. The fuel consumption is driven by thermal and propulsive efficiency. The former has a maximum theoretical limit due to limitations on the TIT (stoichiometric TIT, above which combustion becomes less effective), on the efficiency of single components and on the OPR due to structural reasons. However, this limit is never going to be reached if we want to keep low NO_x emission, as that will become the limiting factor. The latter has a theoretical limit that is imposed for open rotor engines.



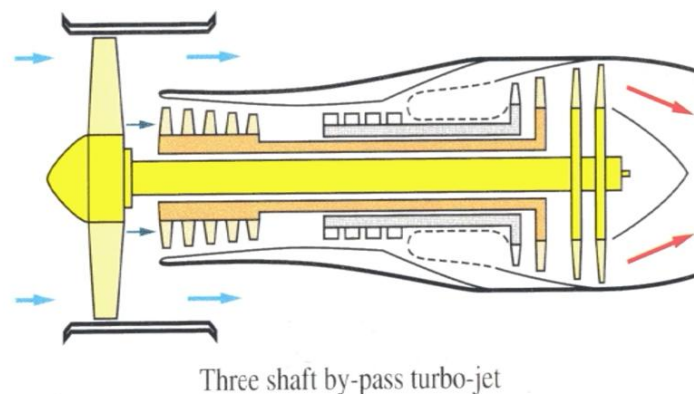
Overall however there is still room for improvement.

The main design drivers for future development are going to be emissions, efficiency, energy sources and noise.

Lecture 4

3 spool engine

As we can see from the image below the three spool turbofan engine has three different shafts. The fan is connected to the low pressure turbine since it wants to rotate more slowly, the booster (LPC) is connected to the intermediate pressure turbine and the HPT is connected to the HPC.

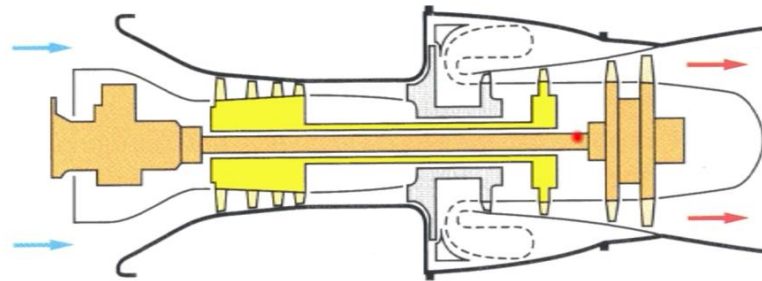


This architecture is not very common and is mostly used by Rolls-Royce engines. The main advantage of this architecture is that the intermediate pressure turbine is decoupled from the fan. In the design phase the fan is the most constricted component since the tip speed can easily reach critical speeds, increasing the noise produced and introducing shock losses in the fan exhaust (decreasing the overall efficiency of the engine). Therefore its rotational speed is strictly limited, which also limits the

LPC in a two spool turbofan (due to the low pressure increase it can provide it's typically called a "booster" stage).

With three spools, the LPC is free to rotate faster and allow for higher OPR.

Three spool architecture is also common for turboprops.



Twin spool free turbine turbo-engine, with front power drive

In this architecture the gearbox that drives the propeller is connected to a power turbine, which is decoupled from the other turbines. The other two shafts connect the HPC to the HPT and the LPC to the LPT.

The main advantage of having three spools for a turboprop is that it's really easy to start the engine: the core is easier to start than the propeller, so by starting it, it can provide power to the power turbine, which drives the propeller.

The other general advantage is the same that makes two spool engine more popular than one spool engines nowadays. The LPC and HPC have different optimal rotating speeds, so by decoupling them we can obtain a higher overall efficiency.

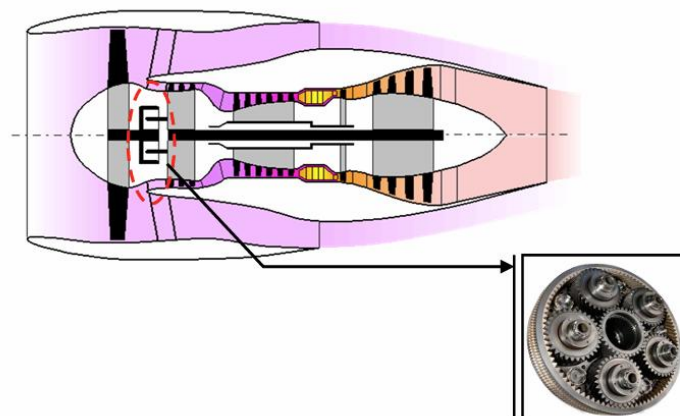
The main disadvantage is the higher system complexity (thinner shafts, vibrations,).

An example of a three spool turboprop is the TP 400 europrop which powers the A400M.

Gear turbofan engine

As we have seen before the GTF is a turbofan which has two main spools that connect the booster and the LPT, as well as the HPT and the HPC. However, the GTF is different than conventional twin spool turbofans, because the fan and the LPC are decoupled by a gearbox. This solves the same issue that is solved by the three spool turbofan with a lower system complexity (although mechanical losses are introduced in the gearbox). This is why the GTF is called a 2.5 spool engine.

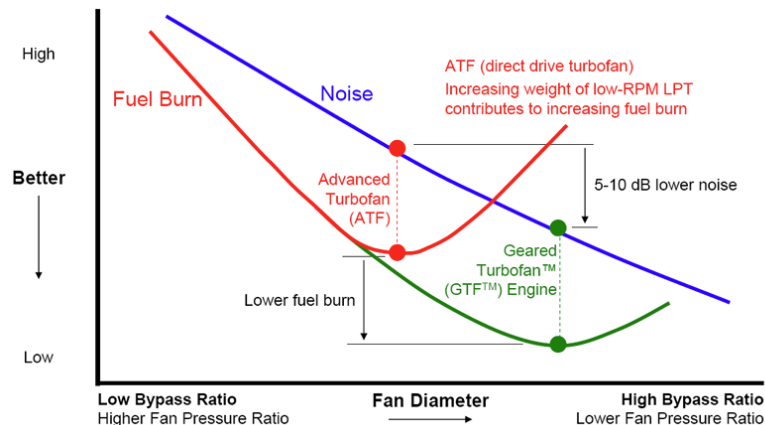
The reduction ratio of the gearbox is around 2-2.5, which means that the fan is rotating at around half the speed of the LPC. This allows for higher OPR while avoiding tip shock at the fan.



GTF also has less LPT stages (around 3) compared to the 5-6 stages that normal twin spool turbofans have. This is possible thanks to the fact that the LPT can now rotate much faster, extracting the same amount of power from fewer turbine stages. Additionally the number of LPC is the same.

Revisiting the graph from the previous lecture, for a twin spool turbofan, after a certain BPR (around 10-12) the fuel consumption actually starts to increase. The positive effect of the increased BPR is negated by the decrease in efficiency of the fan and the LPC.

This limit can be overcome by introducing a gearbox between the fan and the LPC. This allows to increase the BPR even more, while maintaining the gains on the fuel consumption.



Let's now summarize the advantages of decoupling the fan and the LPC:

- Low speed of the fan tips: leads to great noise reduction (as a rule of thumb the noise produced in the inlet is about proportional to the 4th power of the tip speed, while the noise produced by the nozzle is proportional to the 8th power of the nozzle speed. By increasing the BPR we can choose which contribution to reduce)
- Optimum speed for the fan and the LPC: this increases the efficiency and lower the fuel consumption
- High speed LPC and LPT: this allows the reduction in LPT stages (to about half) while increasing the pressure ratio of the LPC at the same time.

These benefits come at the price of increased weight and system complexity. Also the larger BPR and engine diameter might lead to an increase in installation penalty (aerodynamic losses caused by the interaction between the nacelle-pylon-wing).

Due to the higher LPT rotation the turbine blades have higher requirements, since the centripetal forces acting on them are increased.

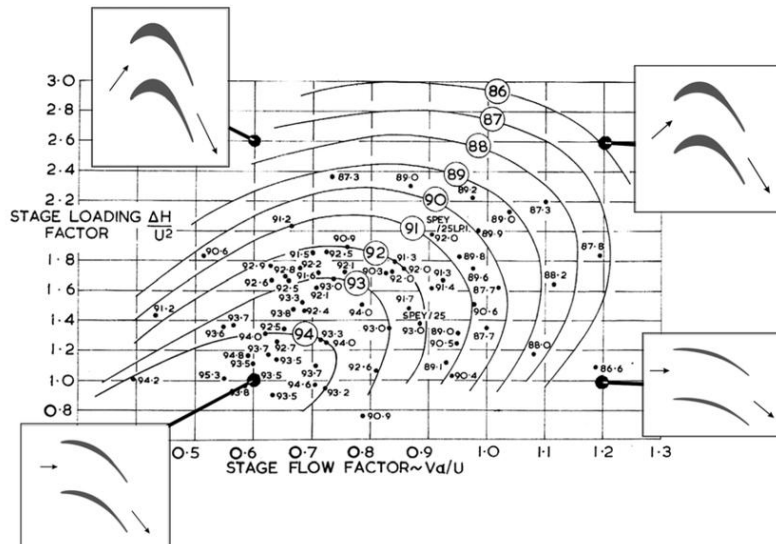
Finally the heat rejection of the engine becomes complicated (increased cooling system complexity and requirements).

A big challenge in designing the gearbox is maximizing the mechanical efficiency. This is not only too reduce power losses but also to reduce the cooling requirements.

Note: GTF engine have the lowest specific fuel consumption between all the different architectures.

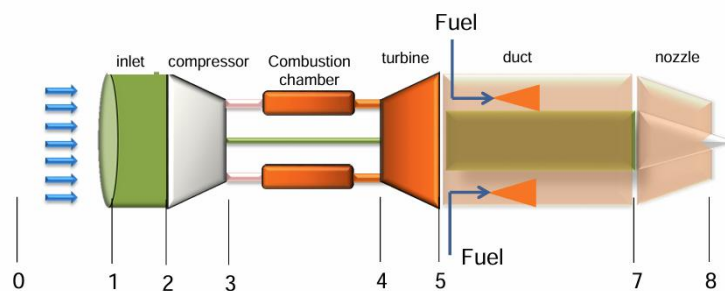
A three spool engine is slightly less efficiency since the turbine coupled with the fan still has to rotate at a slower velocity, which is not the case for a GTF.

From a smith chart we can see that a reduction in rotational velocity increases the stage loading (total enthalpy jump divided by the square of the velocity) and flow coefficient (proportional to the flow velocity), thus reducing the efficiency. Typically we want high velocity to reduce the stage loading.

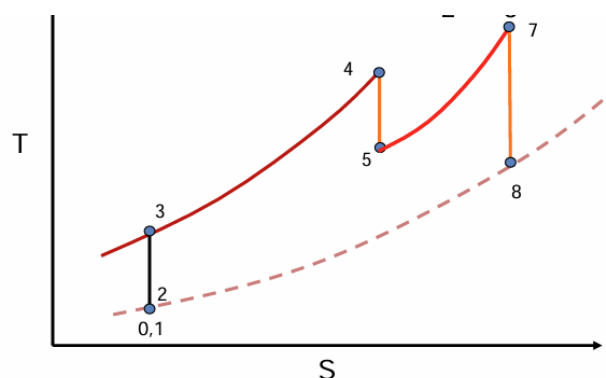


Afterburner

The afterburner is a long secondary combustion chamber, which is added to military turbojets/turbofan engines to reheat the exhaust gases that come out of the turbine to generate more thrust in the subsequent nozzle (typically a convergent-divergent nozzle).



Since we are reheating the gasses, the pressure slightly drops due to pressure losses, while the temperature increases rapidly to a temperature that is higher than the one reached in the primary combustion chamber. This is possible because the TIT is limited mainly by the heat resistance of the turbine blades, however there are no turbomachinery components behind the afterburner.



From the T-S diagram of the ideal cycle with afterburner we can quickly see that the power generated (and thus thrust, which is typically increased by 40-50%) is much higher since the expansion process from 7 to 8 is much longer than from 4 to 5. This comes at a high cost of an enormous reduction to the efficiency of engine and increased length.

If we consider afterburning turbofans, we see that they typically have a lower BPR than conventional turbofans. The bypass flow is mixed with the exhaust gasses coming from the turbine before the reheating process occurs.

To summarize the features of an afterburning engine are:

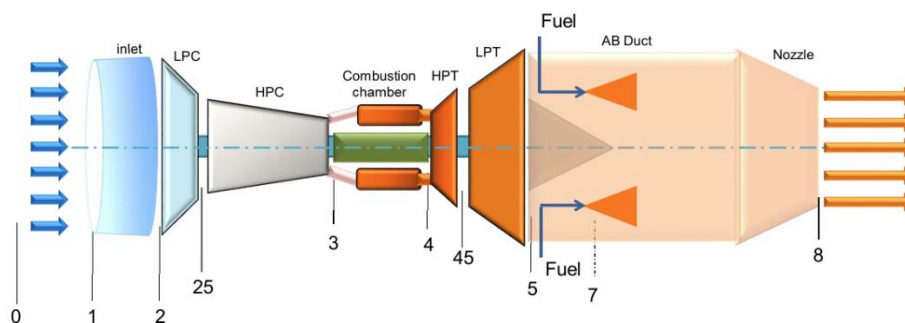
- The specific thrust can be increased instantaneously
- The thrust to weight ratio is very high
- The off design performance can be significantly increased without penalizing the design point performance (e.g. short take off, evading manoeuvres)
- The diameter of the engine is not increased

These come at some disadvantages:

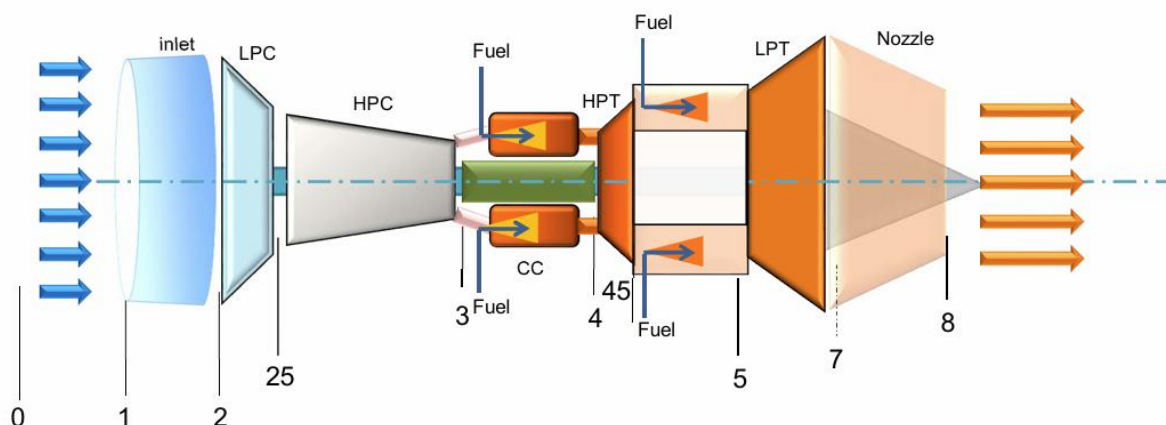
- The specific fuel consumption is increased significantly (up to 300%) when using after burner
- The length of the engine is more than a conventional engine
- The weight of the engine is more than a conventional engine
- The infrared signature is increased immensely when using afterburner
- There is a need for variable area nozzle with using afterburner

Inter-turbine burner engine

Let's consider a twin spool turbojet with afterburner.

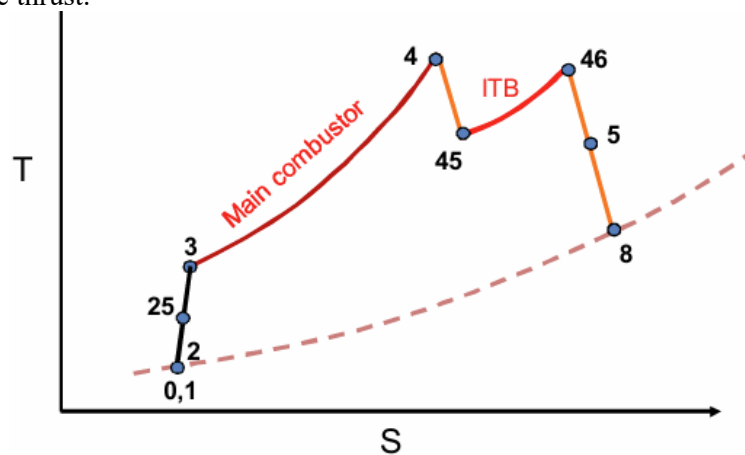


The idea at the basis of the inter-turbine burner engine is that we can insert the secondary combustion chamber ahead of the LPT instead of after it.

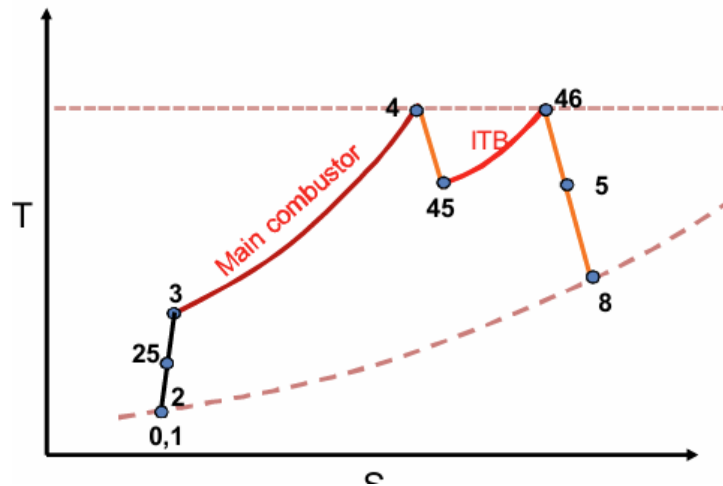


Just like the afterburner the inter-turbine burner is burning fuel at a higher temperature, however in virtue of the fact that it is doing it at a higher pressure, the thermodynamic efficiency of the process is much higher. This engine does produce more thrust than a conventional turbojet at a lower fuel "cost"

than a conventional afterburner, however the process is still less efficient than a normal turbojet which produces the same thrust.

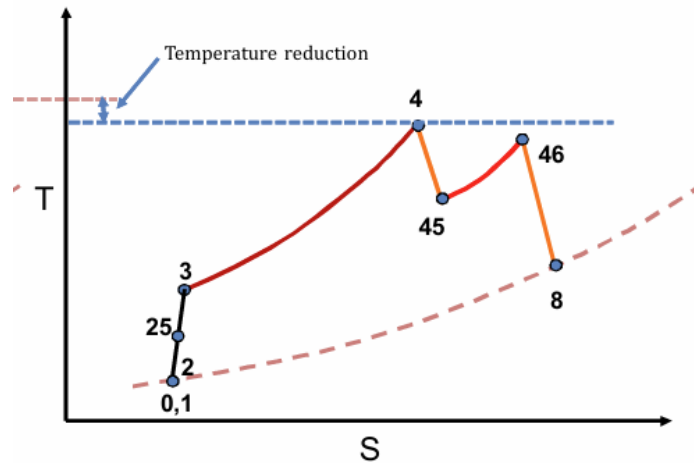


Thanks to the presence of an additional burner, the cycle can be optimized in many ways. If the goal is to maximize the specific thrust produced, then the thermodynamic cycle would probably look like the following:



The amount of fuel added to the ITB has to be so that the end temperature is the same as the TIT. By doing so the specific power of the system is maximized. This is because the temperature at the exit of the ITB is still limited by the LPT blades thermal resistance. By going as close as possible to this limit we maximize the power produced in the nozzle.

However, if we want to minimize the NO_x emissions, then the cycle would look like:



To reduce the NOx emissions we want to lower the TIT (NOx production increases almost exponentially with the combustion temperature after a certain threshold). To compensate for the lower TIT, we rise the temperature in the ITB.

Another effect which leads to NOx emissions reduction is the dissociation of NOx produced by the first combustor in the second combustor.

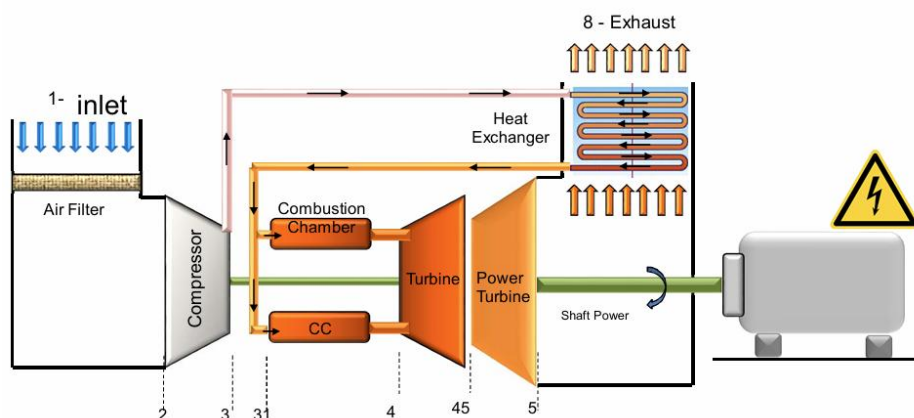
The volume of the ITB has to be higher than the volume of the first combustion chamber because the gasses are at a lower pressure, so to obtain the similar residence time (amount of time the gas stays in the combustion chamber to absorb heat) we need a larger volume.

Another advantage of the ITB is the possibility of lowering the TIT. This allows for a lower amount of cooling air, which is typically needed since the HPT blades are operating at higher temperature than their melting point. This air is taken from the compressor, effectively reducing the overall efficiency of the engine. By reducing the amount of air taken from the compressor, the overall efficiency is further increased.

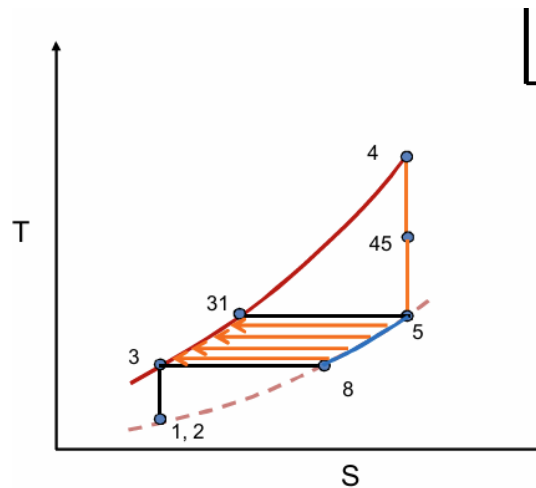
Obviously the disadvantages of this architecture is the increased size, weight and system complexity.

Recuperated gas turbine engine

Let's first look at an industrial example of a gas turbine.



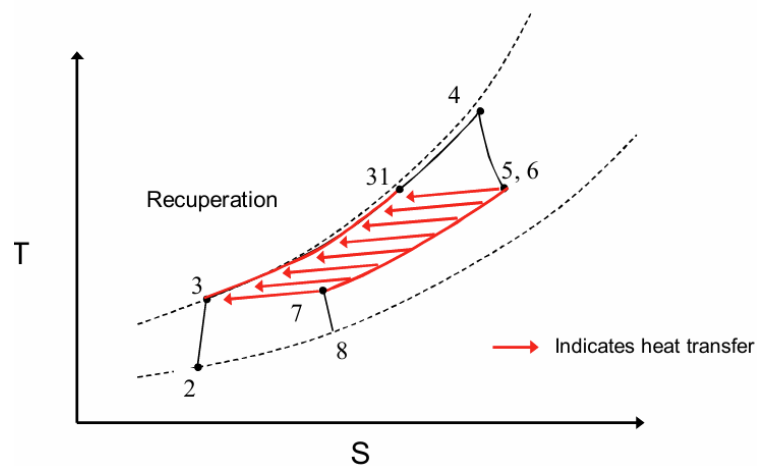
To increase the thermodynamic efficiency we can take the exhaust gasses, which are typically at a higher temperature than the gasses passing from the compressor to the combustion chamber, to heat the gas before it enters the combustor. This reduces the required heat that the combustion chamber has to provide, thus increasing the efficiency. This process is called recuperation.



The recuperation is most effective when the pressure ratio is low, since more of the heat can be extracted from the exhaust gasses.

In an aero engine the recuperation can only occur before the expansion in the nozzle. This is not as efficient as the recuperation in the industrial gas turbine, however recuperation from the exhaust gasses of the nozzle is not possible.

Recuperation has never been implemented in an aero engine also because it disturbs the air before it comes into the nozzle, reducing thrust generated.

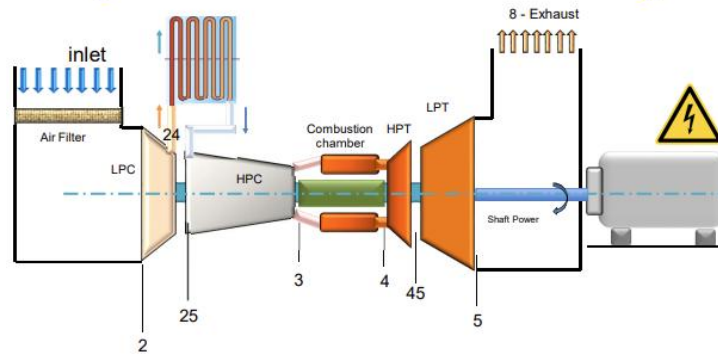


Lecture 5

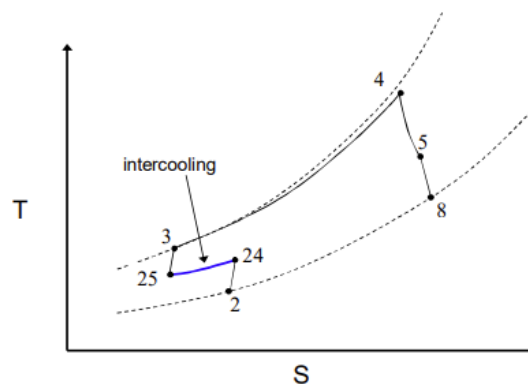
Intercooling and recuperation

Let's consider the concept of recuperation when applied to a 2 spool gas turbine.

2 Spool Gas Turbine with Intercooling

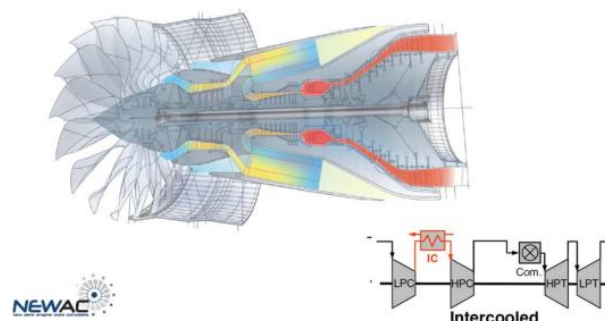


In order to reduce the amount of work required by the HPC is by intercooling. The air after the LPC is taken to an intercooler, where it is cooled through a heat exchanger filled with either water or surrounding air. Once the temperature of the air has reduced, it is then fed to the HPC. This significantly reduces the work required for the same pressure jump, since it is “easier” (it cost less power) to compress a fluid a lower temperature.



A downside to intercooling can be directly seen from the T-S diagram. Since the temperature of the fluid is now colder when it enters the combustion chamber more heat is going to have to be added into the system to get the same TIT, therefore more fuel is going to be consumed. In the following example the core air after the LPC is intercooled by using the bypass air.

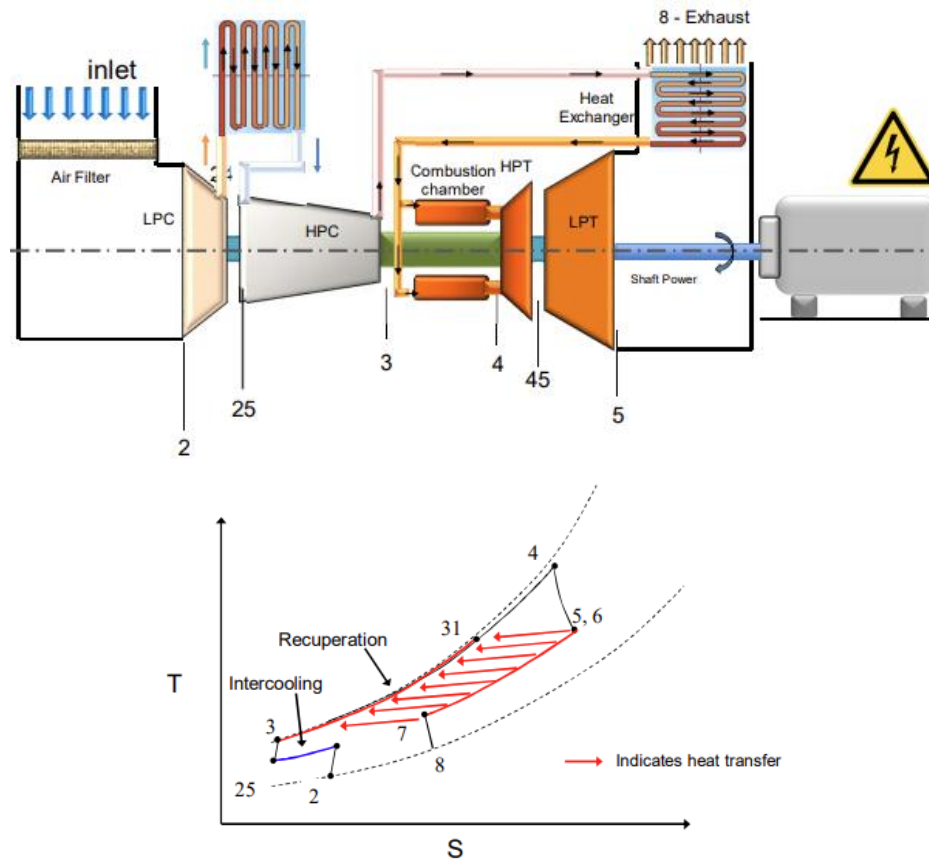
Intercooled Turbofan



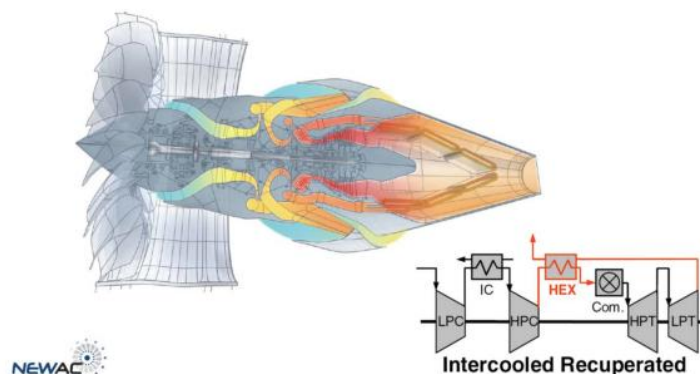
Since as we have previously seen the amount of NO_x produced in the combustion chamber is proportional to the temperature at which the combustion process occurs, by lowering the temperature the emissions of NO_x can also be reduced.

Intercooling however it's not used in aero engine due to the increased engine weight and [].

Intercooling can actually be combined with recuperation, since the second process is actually limited by the temperature difference between the exit air at the HPC and the exhaust coming out of the LPT. The intercooling reduces the first temperature, increasing the temperature difference and making recuperation possible and more efficient.



In a way recuperation also negates the disadvantage of intercooling since it reduces the amount of heat that needs to be transferred to the gas. If we applied these concepts to a turbofan:



From conceptual designs we know that the expected temperature rise of air after the HPC in the recuperator would be in the order of 200K, while the temperature drop in the intercooler would be around 70K. Also the heat exchanged with the bypass air adds energy to it and gets accelerated.

This third flow of cooling air could also be discharged by a third nozzle increasing overall thrust.

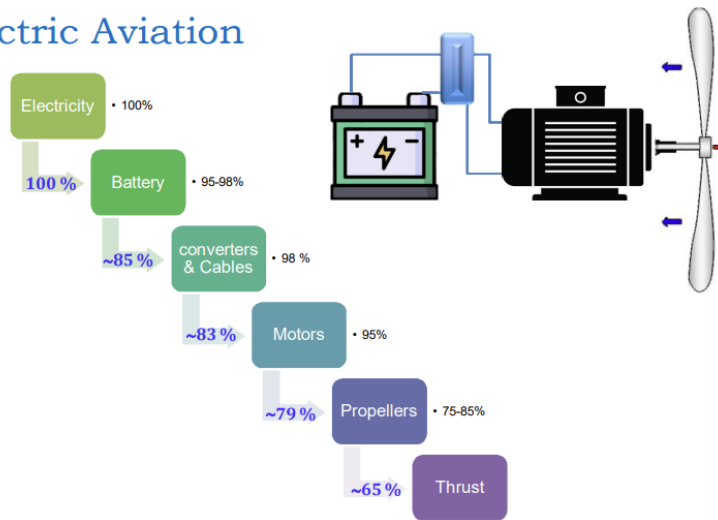
In the heat exchangers there are significant pressure losses, which can actually be offset by the temperature rise in the recuperator. However, pressure losses are a major limiting factor. Also system weight is increased, while reliability is reduced.

Since recuperation is typically more efficient than intercooling (the temperature jumps are different) the TIT can actually become higher and therefore the NO_x emissions are raised.

Hybrid electric propulsion

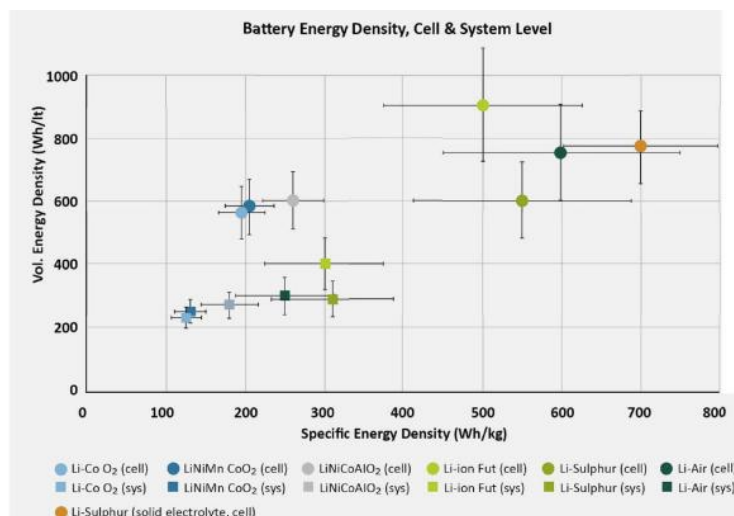
As we have seen previously one of the main components of a propulsion system is the energy source. One of the many possibilities are batteries, which carry electric energy. If we look at the efficiency of such a system we have that:

Electric Aviation



Electric energy is stored in batteries, which then power electric motors through converters and cables. Every step of the process introduces small losses (which lead to a global efficiency of around 60%), which overall are much lower than the losses that occur inside a conventional aero engine.

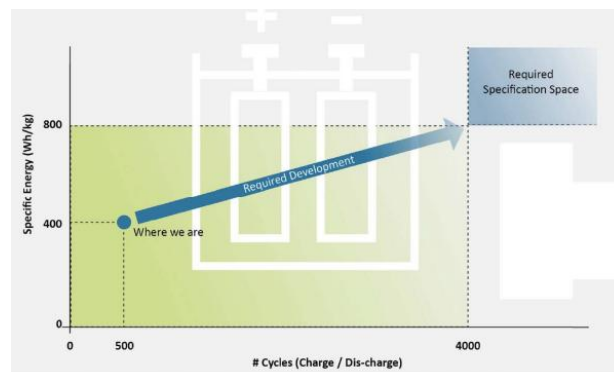
One of the main constraints when it comes to electric energy propulsion is the energy density of batteries.



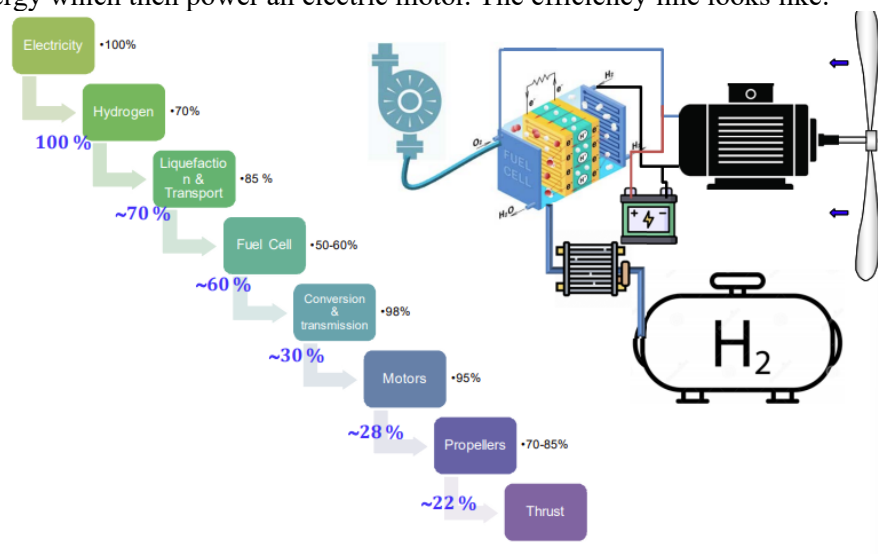
At the cell level the specific energy density and the volumetric energy density are actually quite high, however, at the system level (along with cooling systems and other systems needed) the energy

density drops. Nevertheless at the cell level, the energy density of kerosene is still around 40 times that of the best batteries.

Another major limitation is the battery life of current batteries. If we were to use completely a battery each recharge cycle (meaning we use their full capacity) we see that the number of cycles for which they can be used is greatly reduced and not satisfactory.



The implementation of electric energy can also be done together with another form of energy. Fuel cell electric systems use hydrogen, which is passed through a condenser and fed into a fuel cell, together with air coming from a compressor. Inside the fuel cell chemical energy is transformed into electrical energy which then power an electric motor. The efficiency line looks like:



In summary, electric propulsion systems have the following advantages:

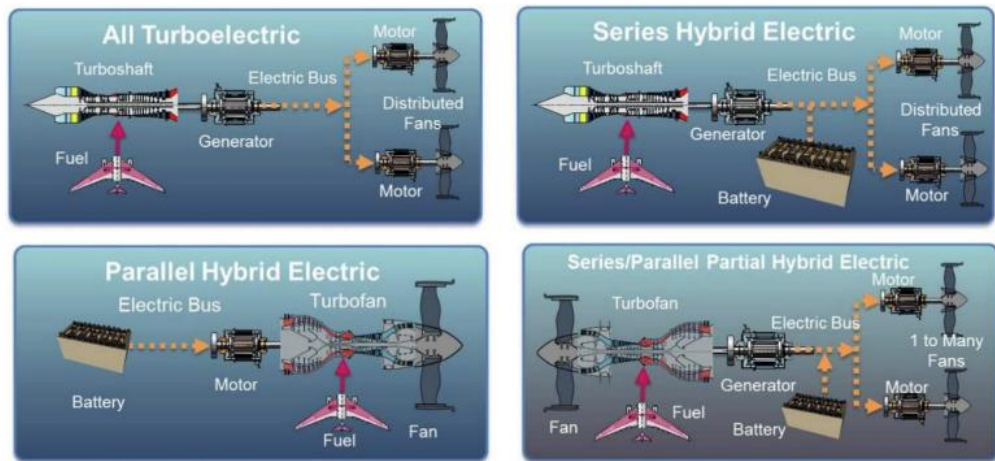
- Higher efficiency, specially for systems which use batteries
- Low emissions
- Lower noise
- Architectural flexibility
- Reduction in the control surface area
- One engine out condition is not a constraint anymore
- Possibility of thrust vectoring
- Suited for urban mobility

However, there are a few disadvantages:

- The energy density of electrical components is not great
- The propulsion system weight is high
- The integration of propulsion systems with aircraft become complex

- The reliability of such systems is not known yet
- Electromagnetic interference is an open issue
- Lightning strikes and other issues need to be investigated

Yet another option is to integrate electrical power systems in conventional aero engines. Hybrid electric architecture can be classified as follows:

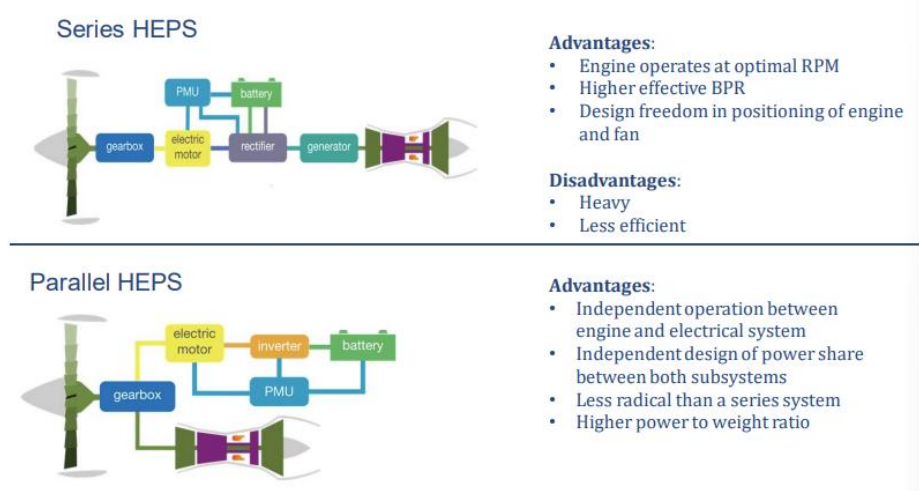


In the turboprop engine, the gas turbine is not producing any thrust, but only electric energy which is then used to drive propellers. This is essentially a turboprop.

A battery can be added in series to the turboshaft to aid in energy generation to drive the propellers. Depending on the flight phase the power generated by the turboshaft can be used to recharge the batteries. The batteries in this case act as buffers to smoothen out transition phases and can help to reduce the total emissions.

If the batteries are added in parallel then typically we have a turboprop instead of a turboshaft. In this case the battery is driving an electrical motor, which is assisting a gas turbine (it can boost the efficiency in many ways by providing energy to all components).

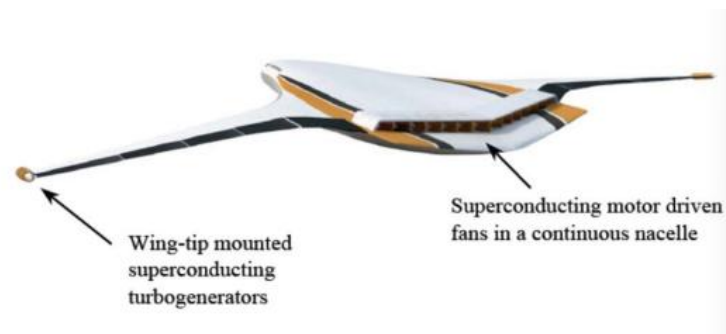
Finally, a combination of the two previous system can be obtained. Batteries in parallel are assisting a turboprop, which is producing thrust and electrical energy which is used to drive propellers or recharging the batteries, depending on the flight phase.



Note: for the HEPS the fan is decoupled from the compressors, therefore the overall efficiency of the engine is greatly increased.

In the series architecture the fan and the core are completely separated.

An example of HEPS is the NASA N3-X, a concept aircraft with a blended wing-body, distributed propulsion and hybrid electric power systems.



The gas turbine engines are at the tips of the wings and they only produce power, which is then used to run smaller electric engine that are distributed over the body.

This distributed propulsion systems have quite a few advantages, such as:

- Higher effective bypass ratio
- Lower noise
- Better suited for boundary layer ingestion
- Architectural flexibility
- Reduction in the control surface area
- One engine out condition is not a constraint anymore
- Possibility of thrust vectoring

As we can see the advantages match up with the advantages of HEPS. The disadvantages are:

- More complex system architecture
- The integration with the aircraft becomes more complex
- The reliability of such systems is not known
- The energy density of electrical components is not good
- Electromagnetic interference is an open issue
- Lightning strikes and other issues need to be investigated

An example of parallel HEP engines are electrically assisted turbofans, which feature an electric motor that drives the fan. This moto brings great advantages during take-off, climb, descent and taxiing. Thus for short range aircraft the efficiency is increased by quite a bit, meanwhile for longer ranges no particular benefit are obtained.

Open rotor propulsion

An open rotor engine, also known as a propfan is an aero engine which combines features of turbofans and turboprops to obtain the thrust capabilities of a turbofan in transonic regimes, while maintaining a turboprop efficiency.

Although an interesting concept propfan engines never caught on mainly due to the great cabin noise they produce when compared to normal turbofans.

The main advantages of such an aero engine architecture are:

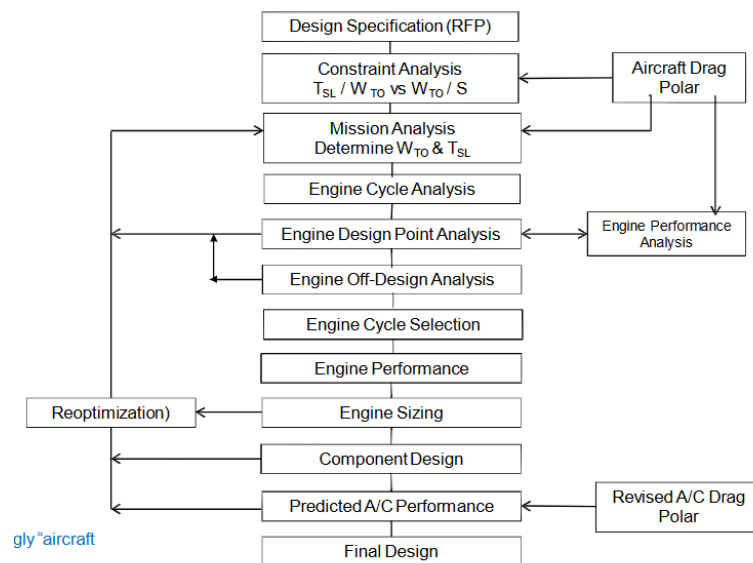
- Large savings in fuel burn due to the large BPR
- Very low FPR, no duct and low downstream swirl when compare to conventional turboprops
- Reduced nacelle drag

These come at a cost:

- Noise emission is greatly increased when compared to a normal turbofan
- New engine installation needed with unprecedented issues
- Balde containment issues

- Power transmission to rotors (introduced mechanical losses)

Note: engine design cycle

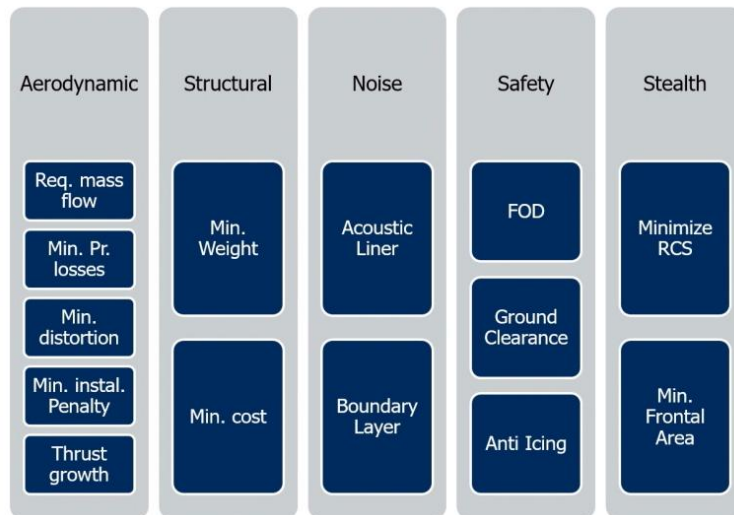


Lecture 6

Engine intake

The design criteria of the inlets can be classified into 5 main types:

- **Aerodynamics:** one of the main requirements is that the inlet has to provide the adequate mass flow rate to the engine with minimal pressure loss and minimum distortion. The installation penalty (interaction between nacelle, pylon and wing) has to be reduced. Finally, the thrust growth is also an important factor. When designing a family of aircrafts (e.g. A320) the thrust growth has to be taken into account, since future aircraft will require more powerful engines and ideally we'd like to keep the same intake.
- **Structural:** the objective is to minimize the weight and the cost.
- **Noise:** important elements are the acoustic liner inside the intake and how the boundary layer behaves.
- **Safety:** foreign object damage is particularly important for military aircrafts. Ground clearance and anti icing systems drive the design of the intake. An example of the importance of ground clearance is the intake of the Boeing 737 NG, which has an oval shape to maximize the mass flow rate coming into the engine, while maintaining a certain clearance.
- **Stealth** (only important for military aircrafts): we want to minimize the radar cross section, the infrared emissions and the frontal area.



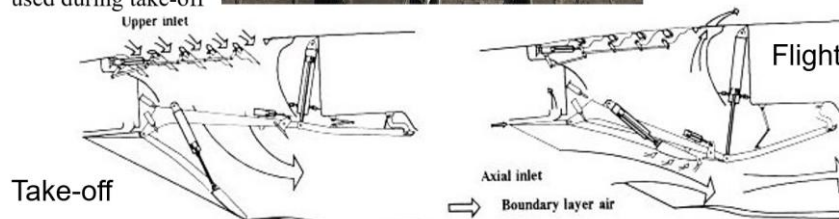
Note: foreign object damage refers to object present on the runway that get accidentally ingested by the engine and lead to great damage. The intake has to be designed to minimize the risk of foreign objects. An extreme example of an inlet designed to minimize the risk of FOD, are the top inlets of the Mig 29, which can provide enough mass flow rate for the aircraft to take off with the main inlets closed. This feature can also be used during high speed manoeuvring to increase the stall margin of the intakes.



Inlet door (down) with porous surfaces for boundary layer bleed



Auxiliary inlet doors used during take-off

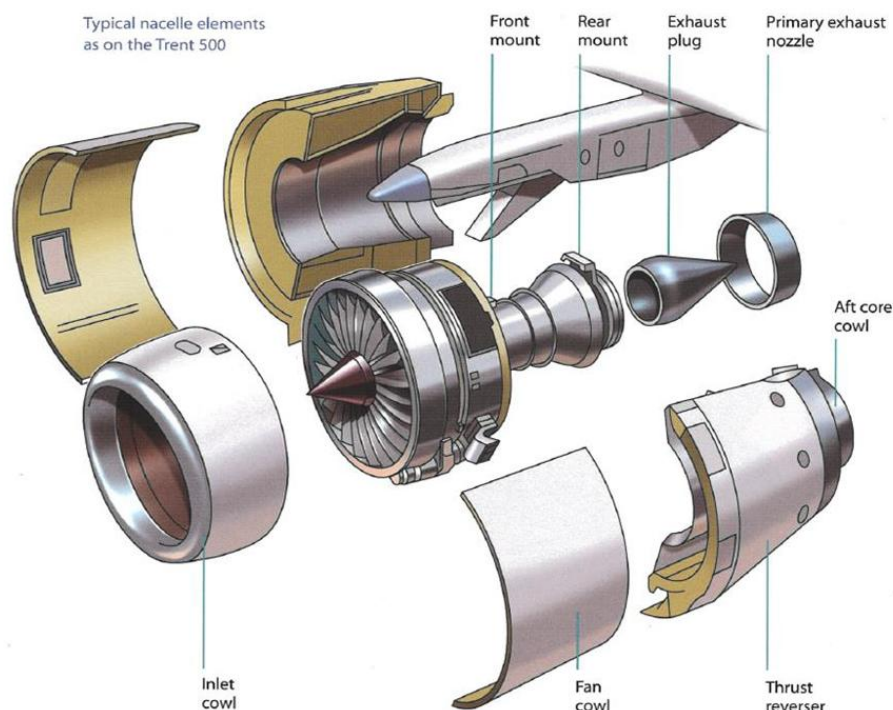


Some important considerations for inlets:

- The intake has to always provide the required mass flow rate in every condition. Therefore the main flight conditions which drive the inlet design are the take-off condition and the cruise condition.
- During cruise one important factor to take into consideration is the ram recovery, or dynamic compression of the fluid. Since the inlet slows down the entering air to around $M=0.6$ at the fan face, some kinetic energy is transformed into potential energy through the ram effect. This translates into an increase in pressure.
- Intake can be distinguished into subsonic and supersonic intakes. For the former boundary layer behaviour is of the utmost importance, while for the latter the shock structure through which the flow is slowed down to subsonic conditions is much more important.

Another example of inlet design mainly driven by one discipline is the inlets of the F117 nighthawk, which are mounted over the wing and are protected by a mesh. All of these measures are taken to improve the stealth of the aircraft.

The classic schematic of an engine nacelle can be seen in the figure below.



We see that the thrust reverser system is integrated with the engine nacelle. These systems (which can be of bucket, cascade or plume type) reverse part of (or all of) the flow to create reverse thrust and aid the breaking phase on icy/wet runways. These systems cannot however be used to certify landing distance on dry runways due to their low reliability.

Nevertheless, they are used to reduce break wear. Most of the modern thrust reverses on civil a/c are of the cascade type and reverse only the fan flow.

Another important part of the nacelle is the inlet cowl. This component is lined on the inside with a material that absorbs acoustic waves to reduce the noise produced. Also inside the inlet cowl a heating system, which is part of the pneumatic system, prevents the formation of ice in the intake which could get ingested by the engine.

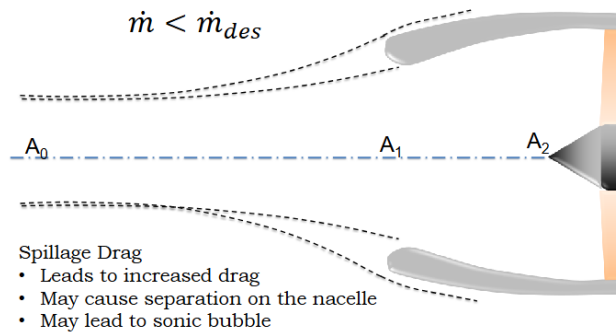
The geometry of the cowl can be defined total diameter of the nacelle, the diameter of the highlight (which is the “leading edge” of the airfoil that makes up the cowl) and the diameter of the throat.

The last two parameters define the mass flow rate ratio of the intake MFR:

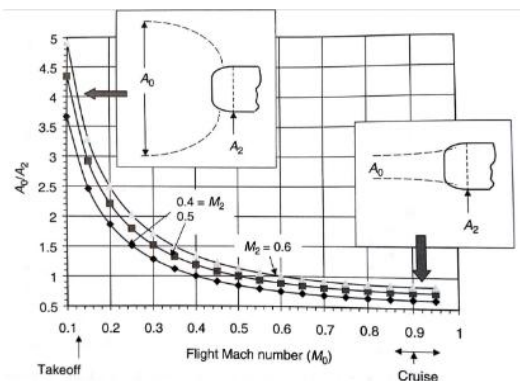
$$MFR = \frac{A_{\infty}}{A_h}$$

Where A_{∞} is the cross section of the stream tube that gets ingested by the intake. In the two conditions that drive the design of the inlet we have:

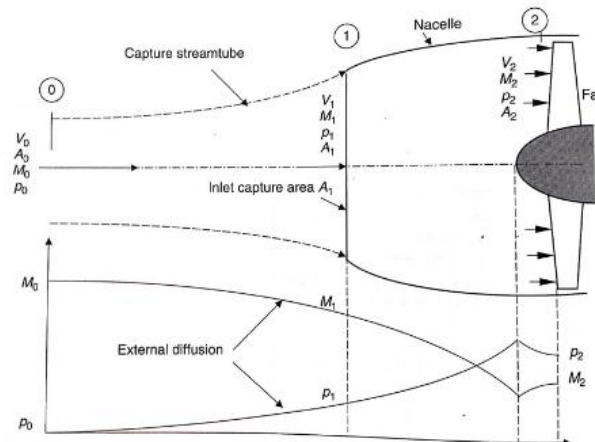
- **Cruise:** low MFR, which means that the stream tube expands over the cowling, leading to an increased risk of shock outside the cowling. The design for this phase focuses mainly on the external geometry. For mass flow rates different than the design value (lower values for example), the stream tube ingested might be slightly larger, leading to spillage drag and possible separation on the nacelle.



- **Take-off:** high MFR, which means that the stream tube converges into the intake and separation can occur on the inside of the cowling. The design of this phase focuses mainly on the internal geometry.



The pressure and Mach number variations in the inlet are as follows:

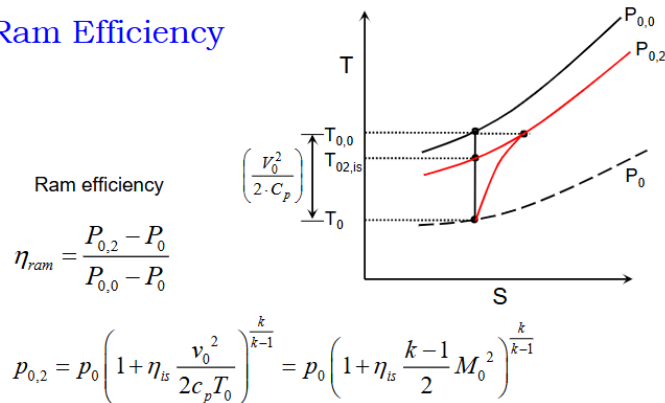


The flow is slowed down and the pressure increases. Especially at high Mach numbers, the incoming air into the inlet is slowed down and in the process some of the kinetic energy is converted into potential, increasing the static pressure. This phenomena is called ram compression.

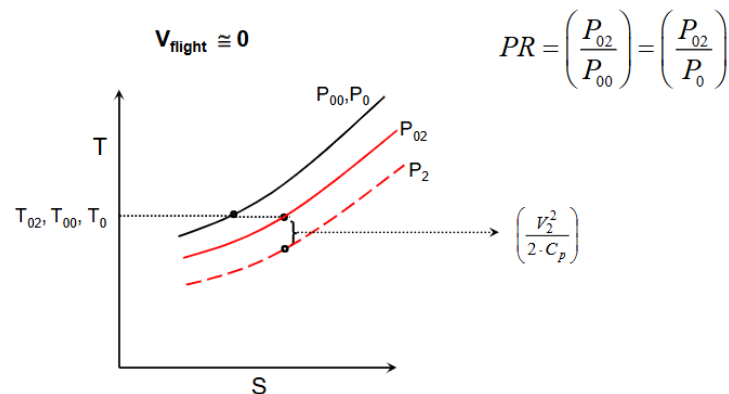
A ram efficiency can be defined to take into account pressure losses that occur during this process, while an isentropic efficiency can be defined to take into account losses in total energy. Up until $M = 0.8$ the two are pretty much equivalent, however the introduction of shocks at higher Mach numbers changes things and a ram recovery factor is used (ratio between the total pressure after the inlet and before). Empirical relations can be used to evaluate the ram recovery factor (typical values of 0.6-0.7):

$$P_{02}/P_{00} = 1.0 - 0.075 (M-1)^{1.35}$$

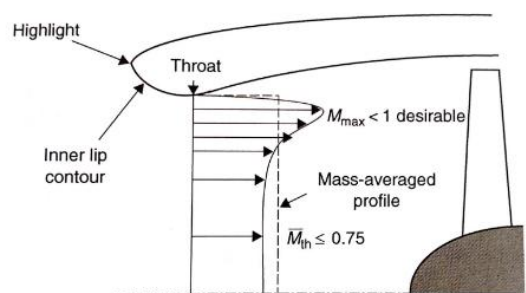
Ram Efficiency



At take-off since the flight speed can be approximated to zero, the total pressure is equal to the ambient static pressure and there is almost no ram effect.



Note: the curvature of the inside of the inlet throat causes a non uniform velocity profile. This is why we typically use an average value for the Mach number in the inlet.

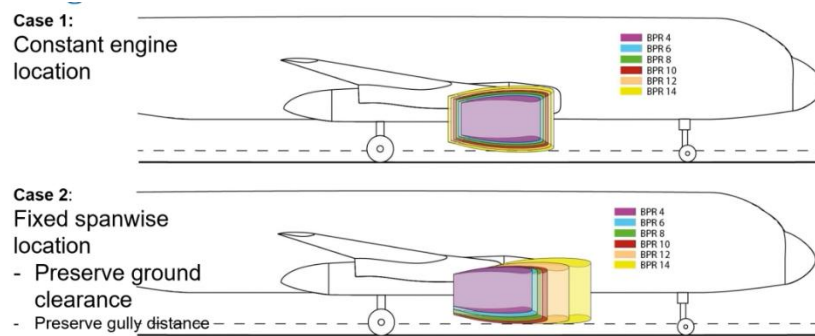


Flow curvature at the throat causes a non uniform velocity profile

Installation effects

As aviation has evolved we can clearly see a tendency for the engine sizes to increase. This is to accommodate for higher and higher BPR.

Engine can be mounted on the wing in two different ways:

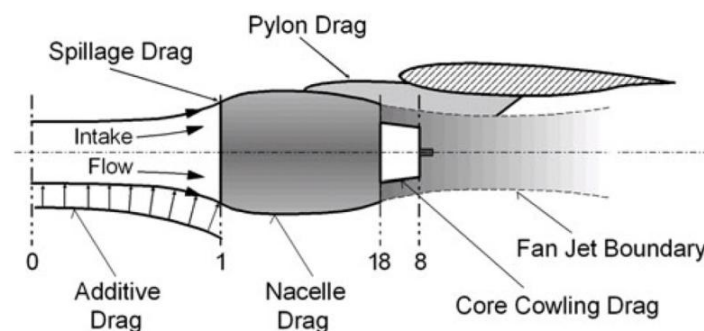


In the second case, to maintain the required ground clearance the position of the engine is shifted forward.

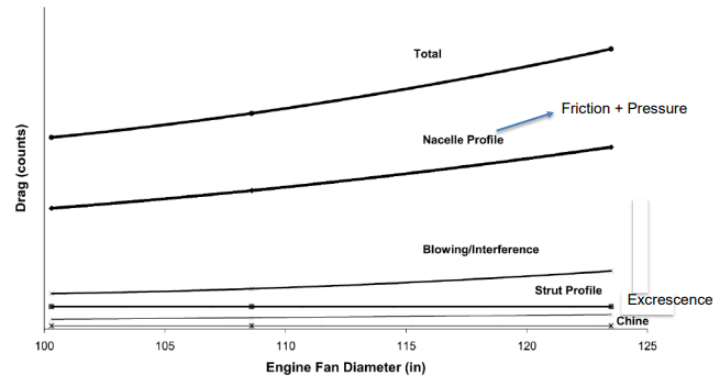
When we talk about installation penalties we are referring to the interactions between all of the components that are in close proximity to the installation point of the engine. Interference effects between the nacelle-pylon-wing/fuselage can lead to great super-velocities and therefore increased drag and risk of shock-induced separation.

However, this is not the only drag component we need to take into account. The drag generated by a nacelle can be divided into 4 components:

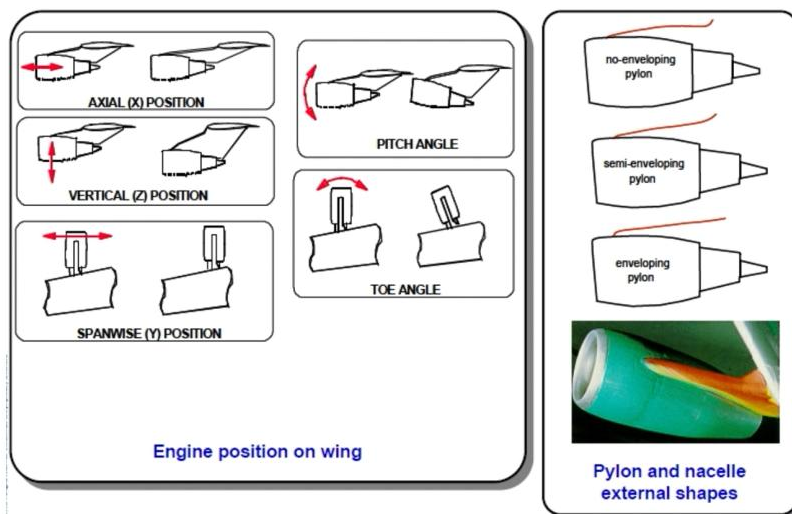
- Profile drag: the air flow has to move around the nacelle. This creates pressure drag known as profile drag.
- Spillage drag: when the mass flow rate is different than the design mass flow rate, the stream tube coming into the engine has to converge/diverge more. This can lead to mass flow rate spillage over the nacelle which increases drag.
- Wave drag: supersonic velocities over the nacelle can lead to shocks and shock-induced separation.
- Windmilling drag: this drag component is only relevant when the engine is inoperative and the mass flow rate through it is drastically lower than in normal operating conditions. The windmilling drag component takes into account the increase in drag from the OEI condition.
- Additive drag: is the pressure drag generated from the pressure difference between the air inside the stream tube and the air around it.



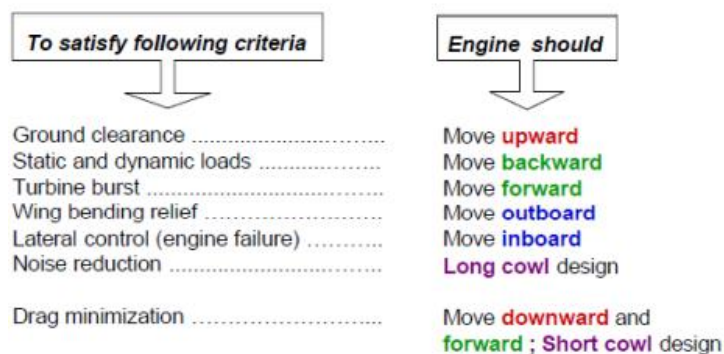
The main design parameter that affects drag generation of the nacelle is obviously the nacelle diameter. As BPR increases, all of the drag components tend to increase, so the gains brought on the TSFC are slightly offset by this increase in drag. This is what determines the optimal BPR of an engine.



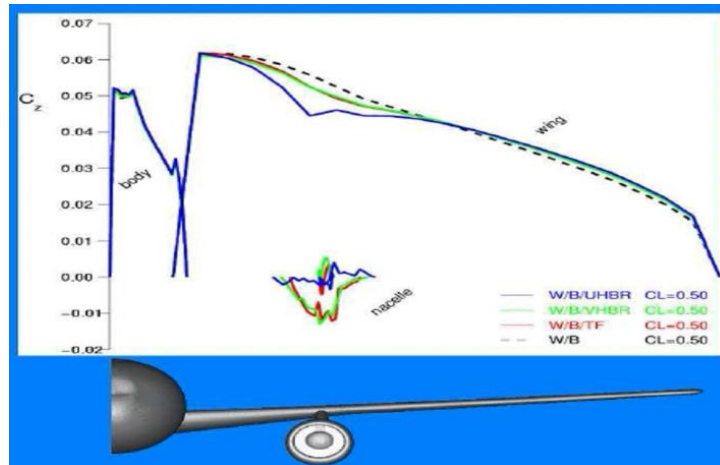
In civil aircrafts the installation of an engine on the wing can vary greatly:



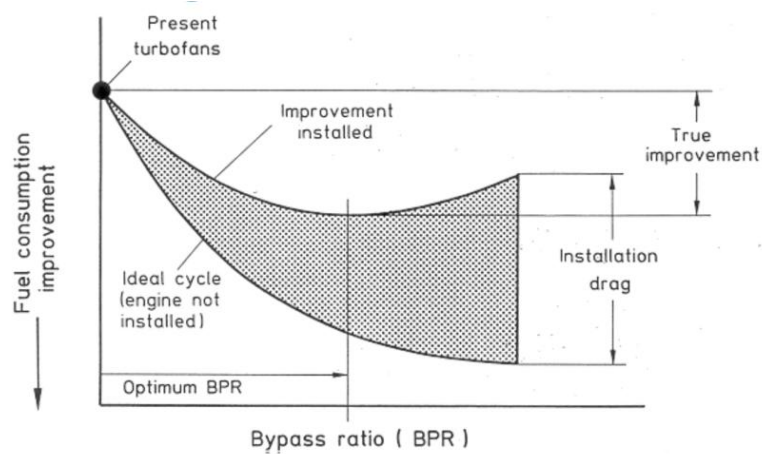
These movements are obviously constrained. For example, the vertical position effects the ground clearance, the spanwise position is determined by the tail size for the OEI condition and so on. The general design criteria for determining the engine installation position are:



Another important aspect that need to be taken into account when choosing the engine installation position is the fact that the presence of the nacelle “spoils” the flow behind it that goes over the wing. This reduces the local lift produced and can be observed from the spanwise distribution of lift. This can be offset by adding fins on the engine nacelle which generate stable vortices.

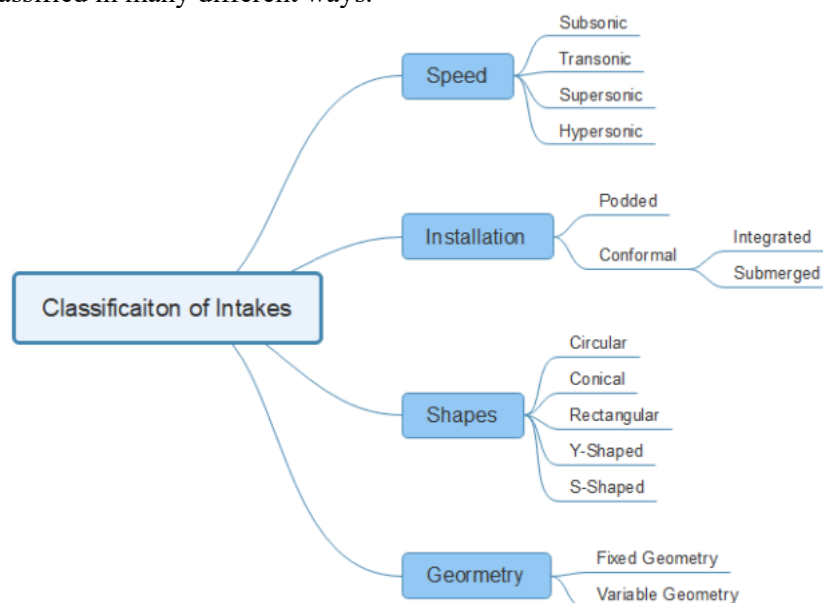


In conclusion, while for an isolated turbofan an increase in BPR to very high values can lead to great fuel consumption improvements, in reality the installation penalty determines a much lower optimal BPR to minimize fuel consumption.



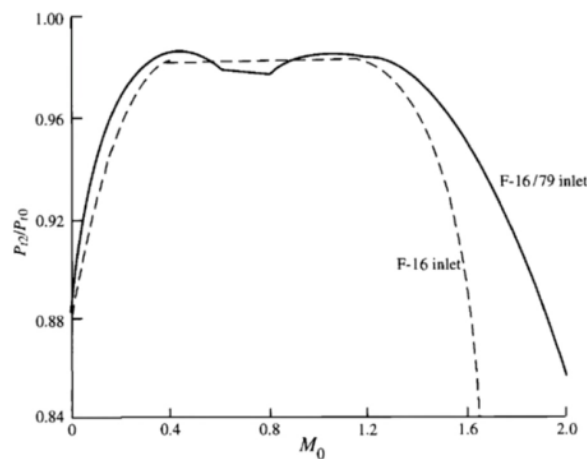
Intake classification

Inlets can be classified in many different ways.



Military aircrafts typically use supersonic, conformal (meaning that the inlet is flush with the aircraft geometry), rectangular/ Y-shaped and of variable geometry. For civil aircrafts the inlets are instead podded, circular and have a fixed geometry.

The inlet efficiency (the losses of pressure that occur in the inlet) vary over the mission profile. The inlet can be designed to optimize one of the phases of a typical mission, for example supersonic inlets tend to have very thin lips to avoid shocks over the cowling during cruise. This however greatly reduces the inlet performance at lower Mach numbers since separation can occur inside is when the stream tube converges. If we look at two different inlets developed for the F-16 (on with variable geometry and one not), we see that the variable geometry inlet is able to maximize the pressure ratio over a larger operational envelope.



Despite the better performance the F-16 was eventually fitted with the fixed geometry inlet due to the lower system complexity-cost-maintenance.

Inlet efficiency is extremely important, since for every percentage loss we have an overall decrease in the performance of the engine of about .5%.

Inlet distortion

Let's consider a Y-shaped inlet, in other words an inlet which has two intakes, which join together to provide the engine with only one mass flow rate.

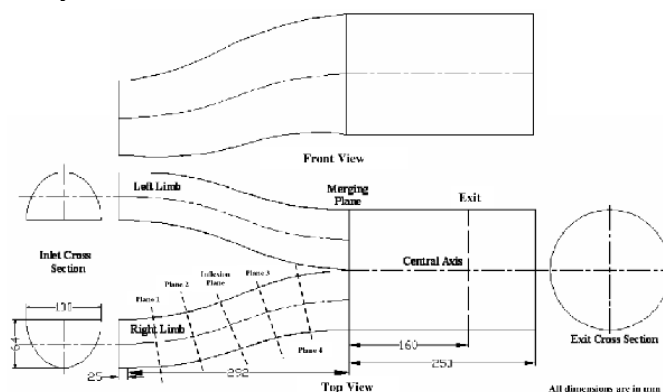
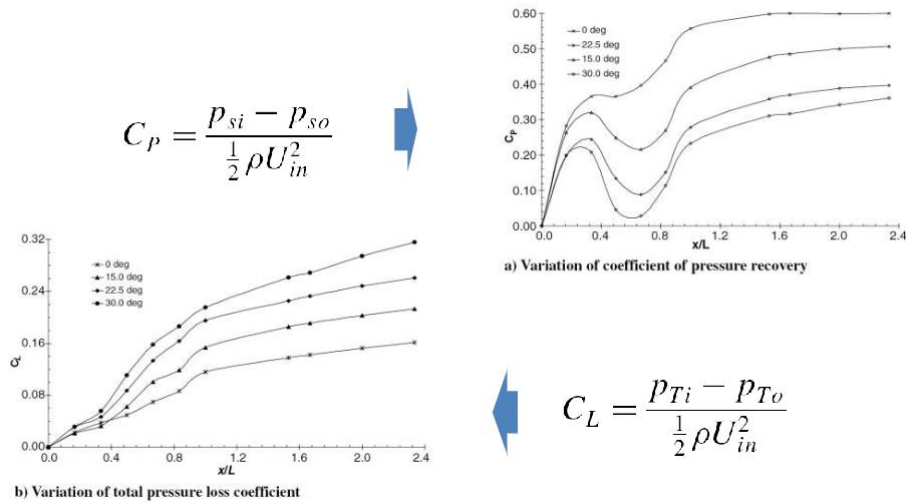


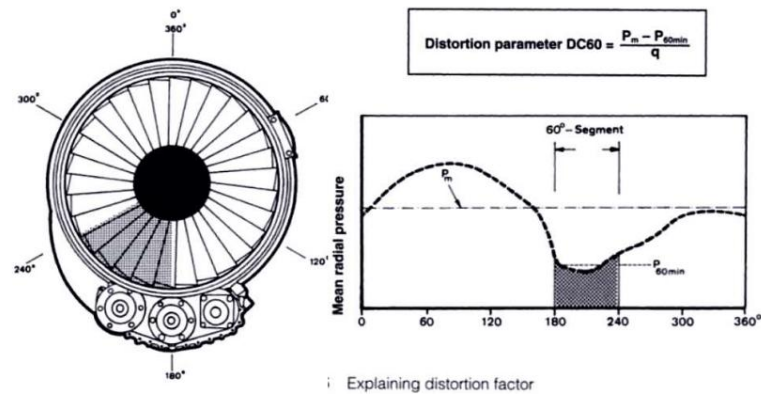
Fig. 3 Schematic layout of the Y duct showing double offsets.

In this kind of duct a prominent boundary layer forms in both the branches and does not disappear when they merge. This, together with the peculiar geometry, creates a non uniform pressure and velocity distribution.

If we take a look at the total pressure loss diagram and the pressure recovery diagram at different yaw angles we see a non linear behaviour.



This means that the engine is facing non uniform flow at the inlet. If we look at the pressure profile along the inlet walls we see high and low pressure regions. A distortion parameter can be defined to quantify the difference between the max pressure and min pressure in a 60 degree section.



When the engine is facing non uniform flow at the inlet, the compressor characteristics change. In particular the surge line of the compressor map comes down, which reduces the operational margin of the engine (increasing the risk of surge in the engine and catastrophic accidents).

The way distortion travels around the inlet can be tested with a fan test rig by putting a distortion screen over a certain 60 degree region in front of the fan. This screen creates located distortion, that allows us to evaluate the effects of the distortion on the duct, fan and compressor.

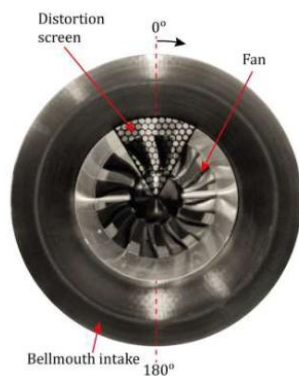
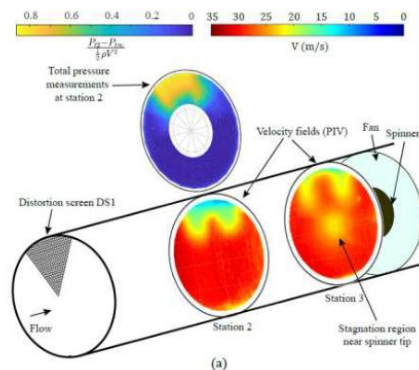


FIGURE 2. Front view of the test rig



Experimental flow field measurements using Particle Image Velocimetry

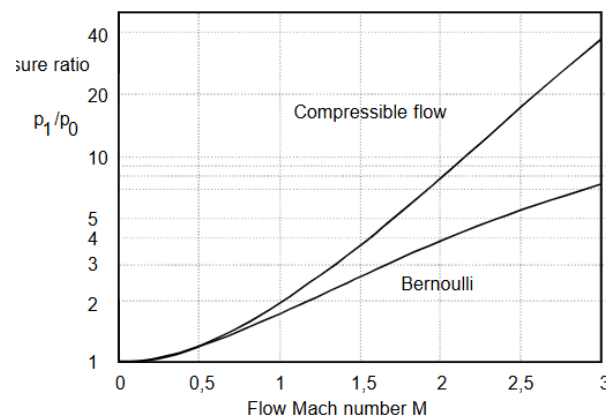
From the image on the right we can clearly see that distortion in the inlet makes the flow in the duct non-uniform and this introduces losses in all the downstream components.

Distortion can come from many sources. On novel aircraft architectures distortion at the inlet can come from boundary layer ingestion (e.g. BWB prototypes).

One big issue created by distortion is the fact that once the flow is not uniform anymore, asymmetrical loads on the fan appear. This can lead to cyclical loads that promote fatigue. A way to reduce the distortion effects is to increase the inlet length. This allows the flow to become more stable by the time it reaches the fan.

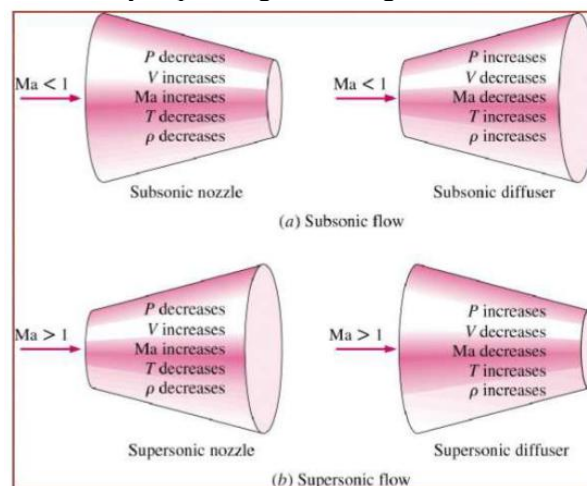
Supersonic inlets

For supersonic inlets the difference between the static pressure and the total pressure increases as the Mach number increases. As Mach increases also this difference cannot be evaluated with Bernoulli's equation since compressibility and shocks invalidate the assumptions. We can see the Bernoulli prediction deviate more and more as Mach number increases above $M = 1$.



For very high Mach numbers, this difference becomes so big, that when the flow is slowed down to subsonic velocities in the inlet, the static pressure increases drastically and the compressor has to do little to no work (ram compression).

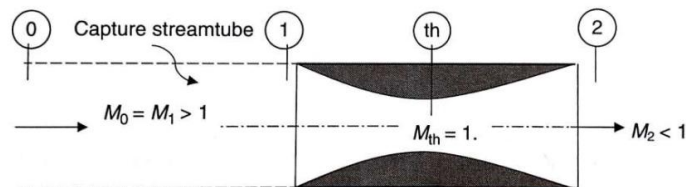
The geometry of the duct is the thing that has the most influence on the flow behaviour in an inlet. We see that the flow behaves differently depending on the regime:



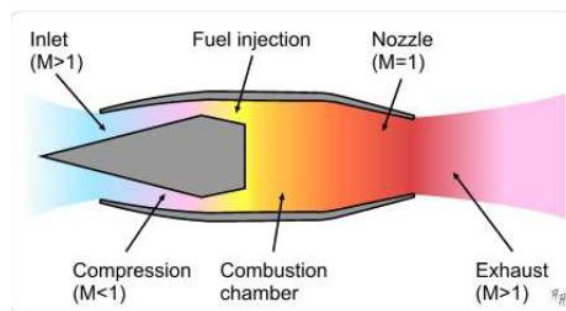
$$dA/A = (M^2 - 1) * (dV/V)$$

We see that supersonic flow behaves in an opposite manner to subsonic flow. Thermodynamic properties always change in the opposite way to the velocity.

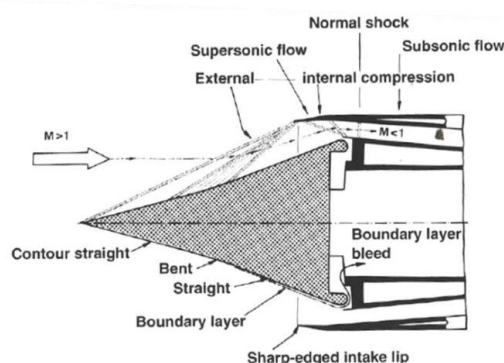
As an example let's consider a supersonic convergent-divergent inlet. The supersonic flow slows down in the convergent part of the inlet and reaches the throat in sonic conditions. Now since it's subsonic, in the divergent part the flow will continue to diffuse, exiting the duct in subsonic conditions. This geometry is what is typically used for supersonic inlets.



One of the applications of this kind of inlet is the ramjet engine, in which all of the compression is done by the ram compression. The compressor is therefore taken out, together with the turbine which drives it. In high supersonic regime these kinds of engines are the most efficient, however they cannot work in sonic/subsonic conditions. Both the inlet and the nozzle have a convergent-divergent geometry, one to slow down supersonic flow to subsonic conditions and the other to speed back up to supersonic conditions.



One of the possible geometries of such an inlet is the spike geometry. In supersonic conditions, the shock waves attach to the tip of the spike. Then the position of the spike is determined so that these shockwaves all attach to the lip of the cowling ensuring maximum mass flow rate with very little spillage. A system of oblique shocks gradually slows down the flow until a last normal shock inside the inlet brings it in subsonic conditions.



One of the main issues of fixed convergent-divergent inlets is the unstart. For a fixed geometry inlet, the C-D duct chokes at a specific Mach number and the position of the last normal shock changes as the Mach number increases. As Mach number increases even more small perturbations can actually make it so that the normal shock comes out of the inlet forming a bow shock in front of it. This

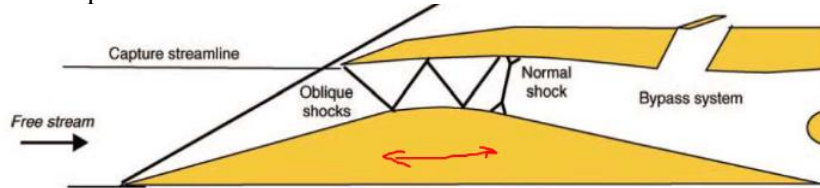
increases the spillage drag greatly and reduces the incoming mass flow rate to the engine decreasing overall thrust.

For this kind of inlet we can define three operational regimes:

- Subcritical: the Mach number is lower than the design value. The normal shock occurs ahead of the cowl lip increasing the spillage drag.
- Critical: the Mach number is equal to the design value. The normal shock is located at the cowl lip.
- Supercritical: the Mach number is higher than the operational Mach number. The shock moves aft and gets stronger, but the total mass flow rate into the engine stays the same.

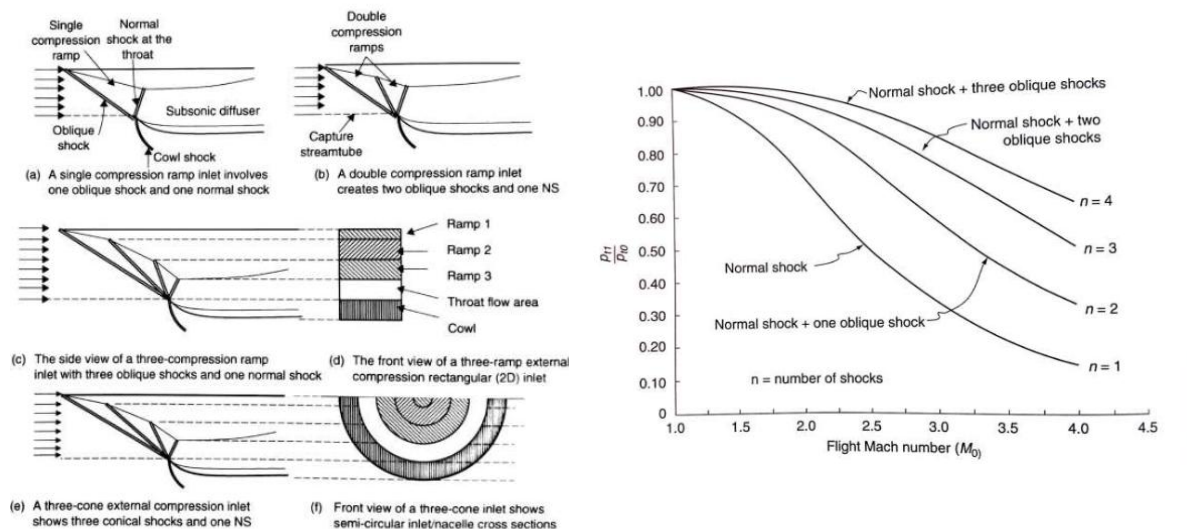
To avoid the increase in spillage drag that occurs in the subcritical condition due to the normal shock being outside of the inlet we can have a variable geometry inlet, for which the spike position can be moved forward or aft.

This type of systems typically have a bypass system that allows for spillage of the mass flow rate in the inlet to control the position of the normal shock.

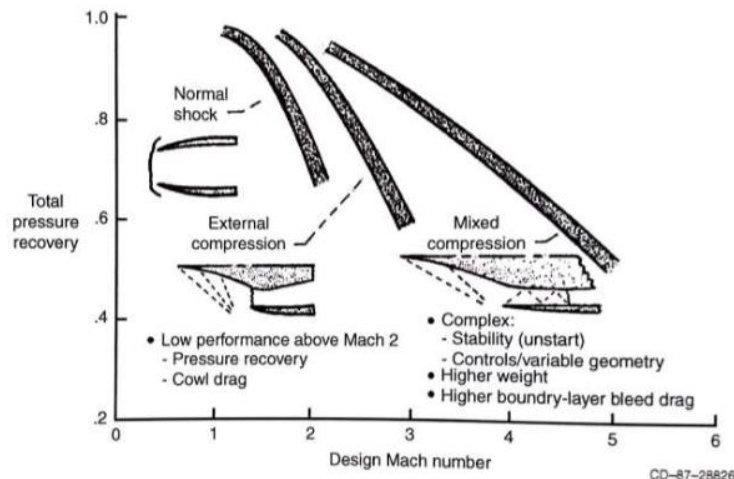


Another way to control the normal shock position is through active flow control (providing suction through the skin of the inlet).

Having a strong normal shock in the inlet creates great pressure losses since the compression that occurs across it is not isentropic but is instead associated with a great increase in entropy. Therefore to reduce pressure losses we prefer to have a system of oblique shocks followed by a weak normal shock. To have this we can have an inlet which has different ramps with different ramp angles, each creating an oblique shock. This is called a mixed compression inlet.



To summarize the total pressure recovery for different inlets:



Lecture 7

Engine exhaust

The main purpose of the nozzle is to convert the potential energy that is available after the turbine into kinetic energy at a minimum pressure loss. Ideally the exit pressure at the nozzle is equal to the ambient pressure to reduce losses in the generation of thrust. When this occurs the expansion is called complete, however this is typically not the case. Other design objectives of the nozzle are: suppress jet noise, permit the control of the engine operating characteristics (by changing the nozzle area the mass flow rate can be changed. Only for military a/c).

For some low bypass ratio engines the core and bypass flow need to be mixed in the nozzle before expansion, therefore this is in some cases a requirement for the nozzle.

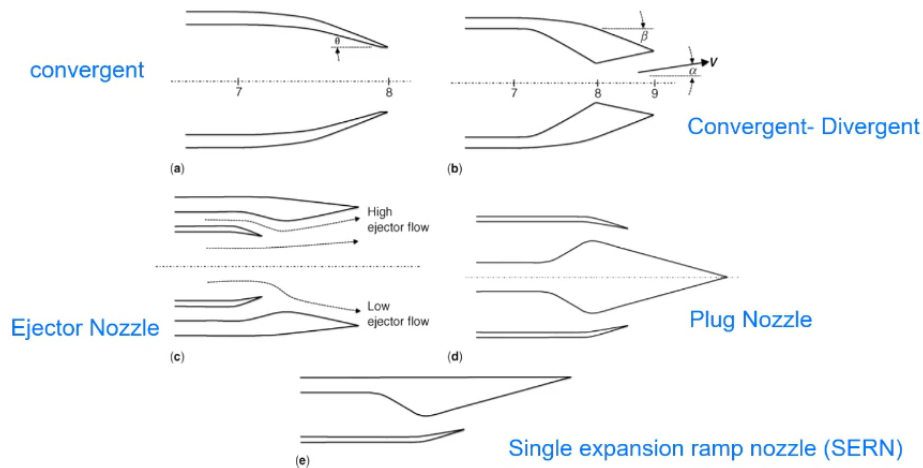
For most engines the thrust reverser system need to be considered. This is a part of the exhaust system, which has to be designed with this in mind.

Note: for military aircrafts the exhaust system has some additional requirements such as: permit afterburner operations, allow for thrust vectoring and suppress radar cross section.

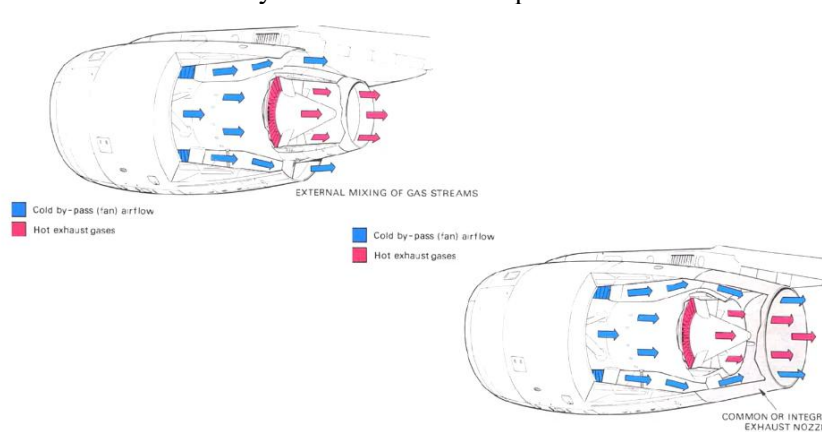
All of these requirements have to be met against minimum cost, weight and drag while meeting the reliability goals.

Nozzles can be classified as follows:

- **Convergent:** this is the most common type of exhaust system used for most civil engines. Typically at design condition in the converging section, also known as the throat, the flow is sonic. This geometry accelerates the gasses converting potential energy into kinetic.
- **Convergent-Divergent:** the convergent section is followed by a divergent nozzle. This type of nozzle is used for military aircrafts and rocket propulsion. The flow reaches sonic conditions at the throat and keeps expanding in the divergent section reaching even higher supersonic velocities.
- **Ejector:** used for supersonic aircrafts. This kind of nozzle entrains ambient air in order to reduce the jet velocity and therefore the jet noise.
- **Plug:** not used anymore. This is a kind of convergent-divergent nozzle that allows for the variation of the geometry
- **Single Expansion Ramp Nozzle SERN:** used only for hypersonic aircrafts. The upper surface of the nozzle is actually the body of the aircraft. The geometry is designed so that the expansion continues outside the nozzle and onto the body.

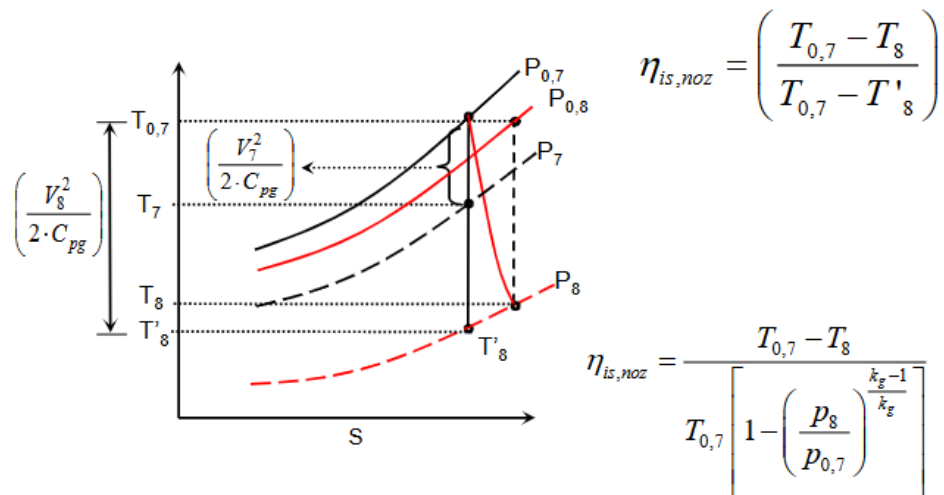


For commercial engines an unmixed convergent nozzle is typically used. The bypass and core have separate nozzles that allow for separate expansion. As we have already said, on some older jets or business jets, the two flows are actually mixed before the expansion.



From a thermodynamic point of view mixing lowers the overall exhaust velocity for the same thrust, improving propulsive efficiency. It can also reduce kinetic energy losses in the jet and jet noise. However, mixing incurs pressure losses, reducing total pressure available at the nozzle and slightly lowering ideal thrust. Also the mixer itself is heavy and requires space.

To take into account all the losses that occur in a real nozzle we can introduce the nozzle efficiency, which is given as the ratio between the enthalpy jump of the real expansion (between the total pressure and the ambient static pressure) and the ideal isentropic expansion:



Since no work or heat is being added to the flow, the total temperature stays constant, while the total pressure slightly decreases due to friction with the nozzle walls.

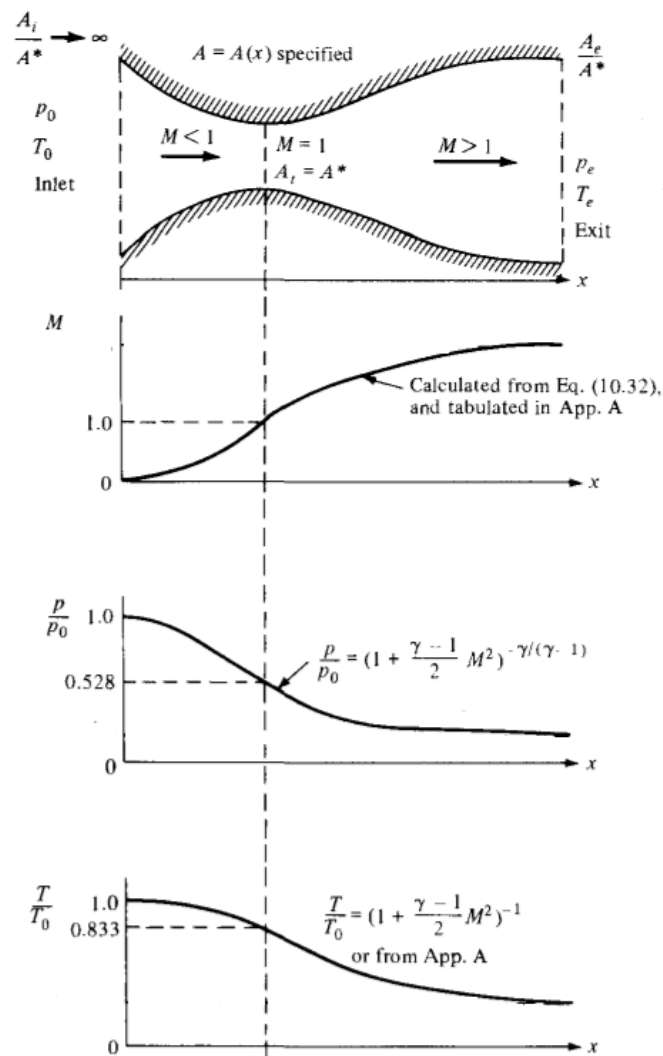
The sources of nozzle losses are:

- Exhaust velocity angularity: due to the swirl introduced in the turbine or the nozzle itself the flow is not perfectly axial and this component cannot contribute to thrust generation. This is considered a loss
- Flow leakage: relevant only for variable geometry nozzles. These nozzles are made up of petals that overlap with each other and airflow can leak through the gaps leading to losses
- Friction inside the boundary layer

In engineering practice the isentropic efficiency is hardly ever used, instead the following coefficients are defined to account for losses:

- Discharge coefficient: given as the ratio of the real mass flow rate and the ideal one. Can be expressed as the ratio between the effective area of the nozzle and the geometric area
- Velocity coefficient: ratio between the real jet velocity and the ideal one
- Thrust coefficient: given as the product of the two previous ones.

If we look at the static quantities throughout the nozzle we have:

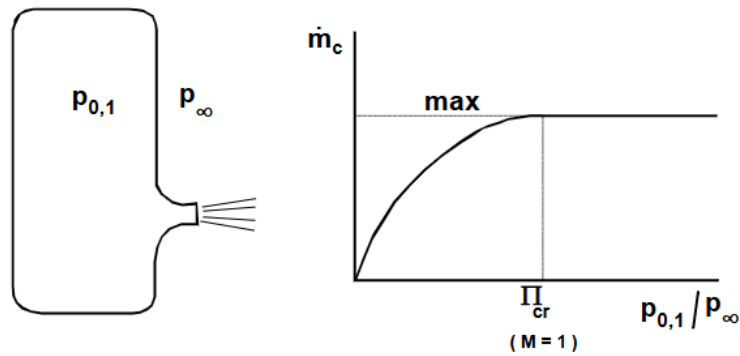


In the convergent section we have that pressure and temperature are both decreasing since the flow is accelerating. For a complete expansion the Mach number reaches one at the throat and then the expansion continues in the divergent part since the flow is now supersonic.

$$\frac{dV}{V} = - \left(\frac{1}{1 - M^2} \right) \frac{dA}{A}$$

Changes in velocity are related to changes in section are by the above formula, which states that subsonic and supersonic flow behave in the opposite way when encountering a nozzle. The thermodynamic quantities are inversely correlated to the velocity of the flow.

To determine whether or not a nozzle is choked we have to look at the pressure ratio across the whole nozzle. If the pressure ratio P_t/P_0 is over the critical pressure ratio then the nozzle is choked. This means that the flow reaches sonic conditions at the throat and no information can travel upstream anymore. This means that all of the flow properties remain at the choked condition even if the pressure ratio increases further. The change is that the expansion is now not complete anymore and therefore the pressure at the nozzle exit will be higher than the ambient pressure, so the expansion will continue outside of the nozzle.



$$TR_{cr} = \frac{T_{0,7}}{T_{throat}} = \frac{k_g + 1}{2} \quad \Pi_{cr} = \frac{p_{0,7}}{p_{throat}} = \left(\frac{k_g + 1}{2} \right)^{\frac{k_g}{k_g - 1}}$$

The equations above are used to evaluate the critical temperature and pressure ratios. Depending on the nozzle efficiency it can increase slightly since higher pressure ratios are needed to reach sonic conditions.

Here are some examples of calculation for an unchoked nozzle with complete expansion:

$$\frac{p_{0,7}}{p_\infty} \leq \Pi_{cr} \quad T_{0,7} - T_{jet} = T_{0,7} \cdot \eta_{is,jet} \left[1 - \left(\frac{p_\infty}{p_{0,7}} \right)^{\frac{k_g - 1}{k_g}} \right]$$

$$p_{jet} = p_\infty$$

$$V_{jet} = \sqrt{2c_{p,g}(T_{0,g} - T_{jet})}$$

$$FN = \dot{m} \cdot (V_{jet} - V_0)$$

$$\begin{aligned}\dot{m} &= \rho A v \\ p &= \rho R T \\ v &= M \sqrt{k_g R T}\end{aligned}\quad \begin{aligned}T &= T_0 / \left(1 + \frac{k_g - 1}{2} M^2\right) \\ p &= p_0 / \left(1 + \frac{k_g - 1}{2} M^2\right)^{\frac{k_g}{k_g - 1}}\end{aligned}$$

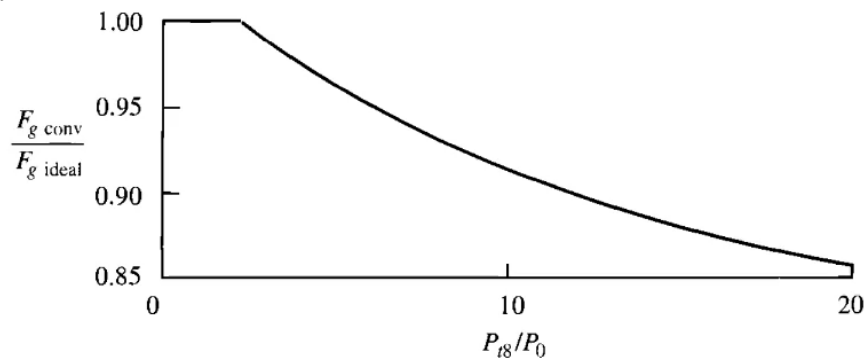
$$A = \frac{\dot{m} \sqrt{T_{0,g}}}{p_{0,g}} \sqrt{\frac{R}{k_g}} \frac{\left(1 + \frac{k_g - 1}{2} M^2\right)^{\frac{k_g + 1}{2(k_g - 1)}}}{M}$$

And a choked nozzle with incomplete expansion:

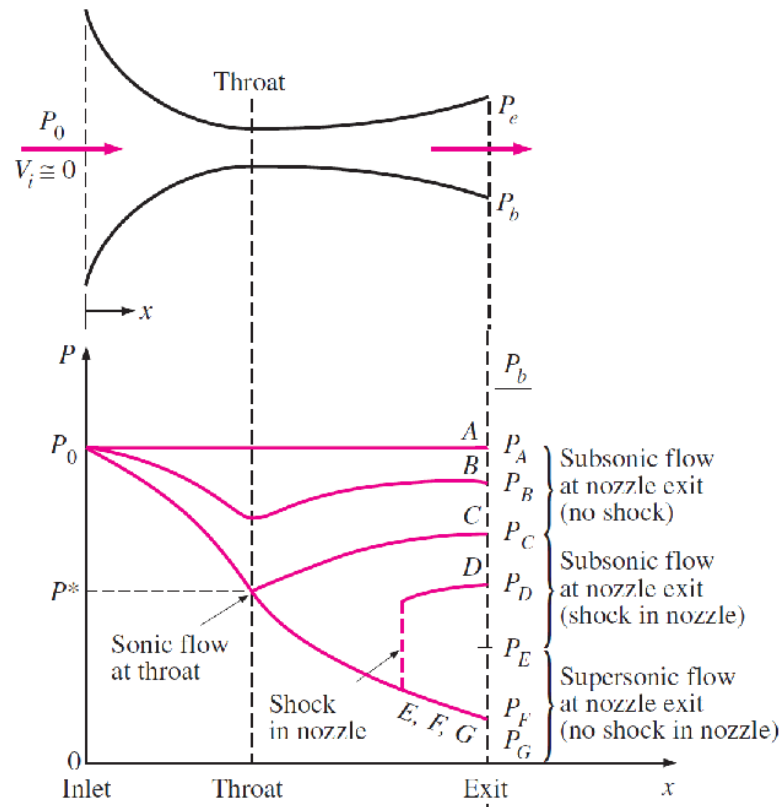
$$\begin{aligned}p_8 &= p_{0,7} / \Pi_{cr} \\ T_8 &= T_{0,7} / TR_{cr} \\ v_{jet} &= \sqrt{k_g R \cdot T_8} \\ A_{Nz} &= \frac{\dot{m} \cdot R \cdot T_8}{p_8 \cdot V_{jet}}\end{aligned}\quad \begin{aligned}\frac{p_{0,7}}{p_0} &> \Pi_{cr} = \frac{1}{\left[1 - \frac{k_g - 1}{\eta_{is,jet} \cdot (k_g + 1)}\right]^{\frac{k_g}{k_g - 1}}} \\ TR_{cr} &= \frac{k_g + 1}{2}\end{aligned}$$

$$\begin{aligned}F_N &= \dot{m} \cdot (V_{jet} - V_0) + A_{Nz} \cdot (p_8 - p_0) \\ F_N &= \dot{m} \cdot (V_{eff} - V_0) \quad \leftarrow \text{solve for } v_{eff}\end{aligned}$$

For choked nozzles if the pressure ratio continues to increase we see that we start getting losses of thrust that are accounted by the thrust coefficient. This is due to the fact that the expansion that occurs outside of the nozzle does not contribute to thrust generation. Some of this energy is still recuperated from the pressure thrust term, however it is not as efficient as directly converting that energy into kinetic energy.



Let's now look at how the flow behaves in a convergent divergent nozzle:



When the pressure ratio across the nozzle is 1 there is no mass flow rate and the flow is still. Once we start rising it the flow starts flowing through the nozzle, it reaches a maximum velocity at the throat and slows down again in the divergent part.

This situation persists for all pressure ratios under the critical pressure ratio. At the critical pressure ratio, the flow reaches sonic conditions at the throat and then slows down again in the divergent part. For higher pressure ratios there is one design pressure ratio for which the expansion is complete. In this situation the flow reaches the throat at sonic conditions and keeps expanding in the divergent part. In this case we have that the pressure at the exit of the nozzle is equal to the ambient pressure.

For every pressure ratio between the critical one and P_E however we have that a normal shock occurs in the nozzle. The flow keeps expanding in the divergent part of the nozzle until it cannot manage the pressure difference between the flow itself and the ambient pressure. Therefore a rapid compression occurs in the nozzle that brings the flow to subsonic conditions and the pressure to the ambient pressure. This condition has to be avoided at all cost since the normal shock introduces a lot of noise, structural damages and losses.

In this region, as the pressure ratios keep increasing the normal shock moves downstream, until we reach P_E , at which point the normal shock is on the exit section of the nozzle. For pressure ratios in between the ideal pressure ratio P_F and P_E we have that the flow will follow the ideal expansion curve and reach the nozzle exit with a pressure that is lower than the ambient pressure. This is why, for this and the previous regime the nozzle is said to be **overexpanded**. The compression of the flow occurs outside of the nozzle through a system of oblique shocks.

Finally if the pressure ratio is higher than the ideal pressure ratio the nozzle is **underexpanded**, since the flow will still follow the ideal expansion curve and have a higher pressure than the ambient one. Therefore, the flow will continue to expand outside the nozzle through expansion fans which flare the plume. This is then followed by the formation of shock diamonds (Mach disks) further downstream to realign the flow.

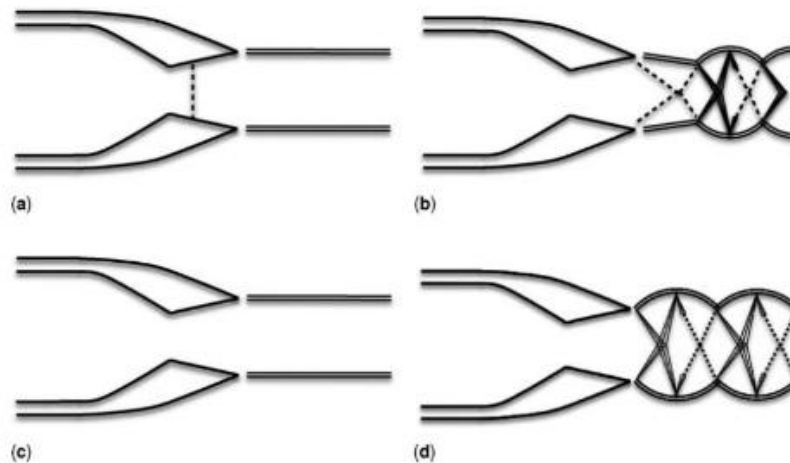


Figure 4. Nozzle expansion. (a) Overexpanded - normal shock inside, $NPR \ll NPR_{design}$; (b) overexpanded - oblique shocks at exit, $NPR < NPR_{design}$; (c) perfectly expanded, $NPR = NPR_{design}$; (d) underexpanded, $NPR > NPR_{design}$.

If we look in detail the sequence of compression and expansion waves for an over and under expanded nozzle we see:

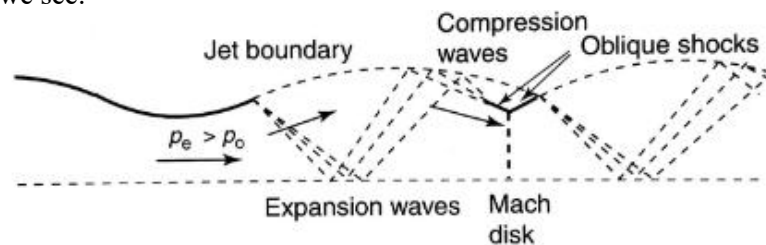


Figure 4.14 Flow pattern in under-expanded nozzle jets.

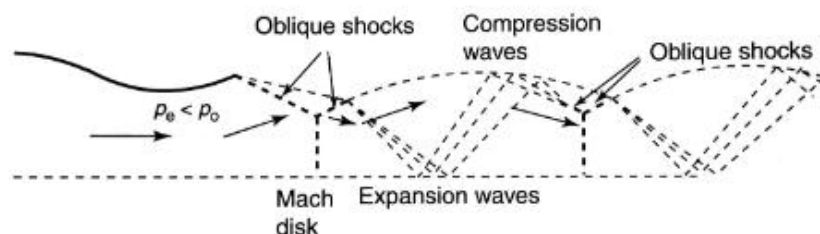
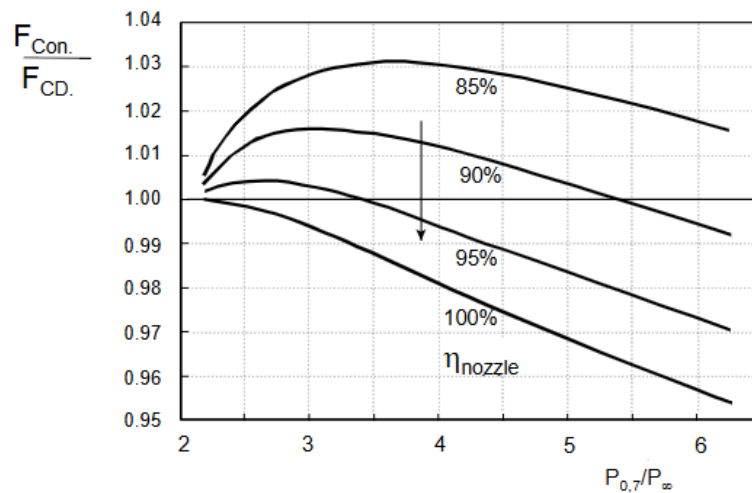


Figure 4.15 Flow pattern in over-expanded nozzle jets.

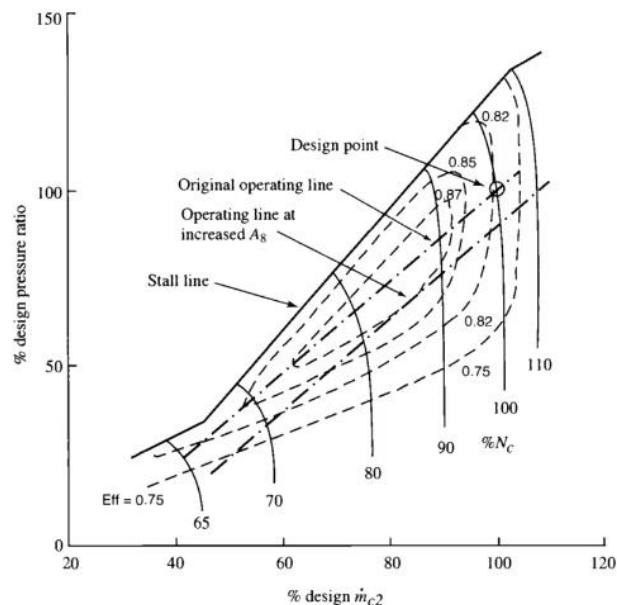
The main difference is that for an over expanded nozzle the first thing that occurs outside the nozzle are oblique shocks that attach to the nozzle lip and converge into a Mach disk. This is then followed by a succession of expansion and compression fans.

For an under expanded nozzle the first thing the flow encounters is an expansion fan which can further down coalesce into oblique shocks and Mach disks.

From the graph below we can see that convergent nozzles have better thrust coefficients at low pressure ratios (typically for subsonic aircrafts). However, their efficiency drops once the pressure ratios surpass pressure ratios that are a bit higher than the critical pressure ratio (pressure ratios of around 5). That is supersonic aircrafts and fighters use convergent-divergent nozzles.



Note: a variable area nozzle, as we have already said, can be used to control the mass flow rate through the engine. We can see this from the compressor map:



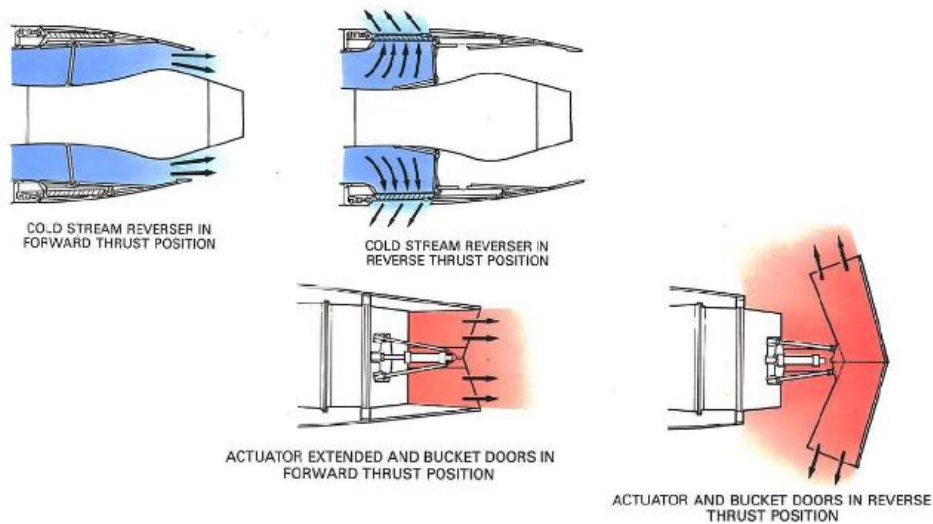
By closing the area we reduce the allowed mass flow rate through the engine and we move closer to the surge line.

Thrust reversers

Thrust reversers are used to reduce break wear during landing and add a layer of safety especially for wet and icy runways. They cannot be used for the certification of the landing distance in dry conditions due to their low reliability.

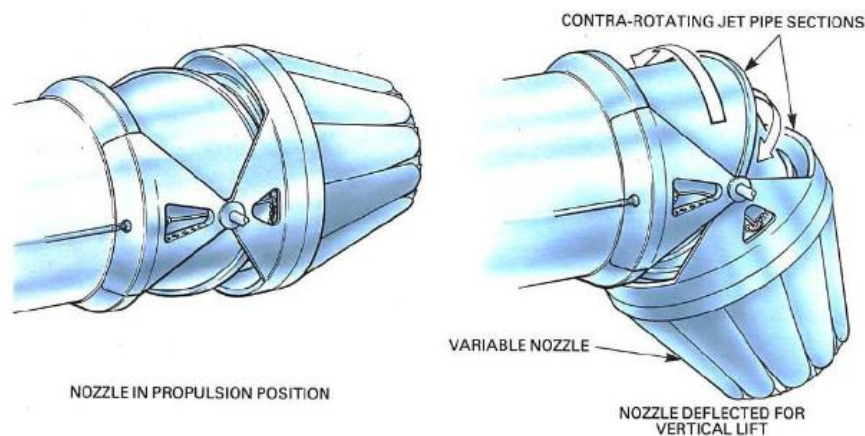
The basic idea is to redirect the exhaust gasses forward in order to produce negative thrust. To do this there are three different types of mechanisms:

- Bucket type reversers: two panels close over the exhaust and reverse both the core and bypass flows. Not used anymore if not only for business jets.
- Cascade type reversers: the engine nacelle opens up revealing cascade grids which redirect the gasses forward. Blocker doors block the flow from going out the nozzle.
- Petal type: doors open all over the nacelle.



Thrust vectoring

Thrust vectoring means that the axis along which the thrust is applied is changed. This is only used for military aircrafts and can be used for short take-off or even vertical take-off.



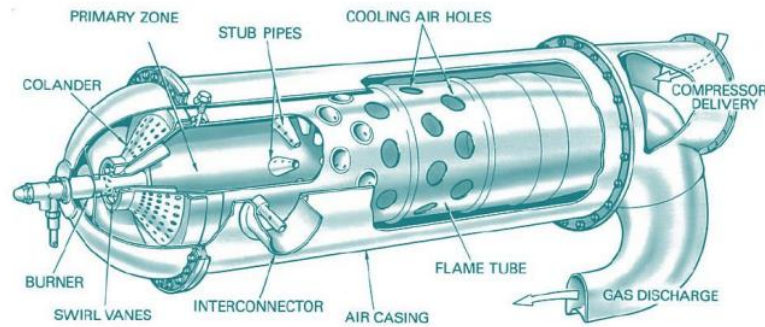
Lecture 8

Combustion chamber

The combustion chamber is located between the compressor and turbine in a gas turbine and required to convert the chemical energy of the fuel into thermal energy with the smallest possible pressure loss and with the least emission of undesirable chemicals. In other words, the combustor provides the heat input into the gas turbine cycle. It receives air from the compressor, introduces a stream of fuel into it, creates the conditions for the fuel and air to mix and react and, eventually, delivers a mixture of hot post-combustion gases to the turbine. Such a process is commonly referred to as **internal combustion**.

For the sake of analysis, combustion systems are generally classified as either those of constant pressure or constant volume.

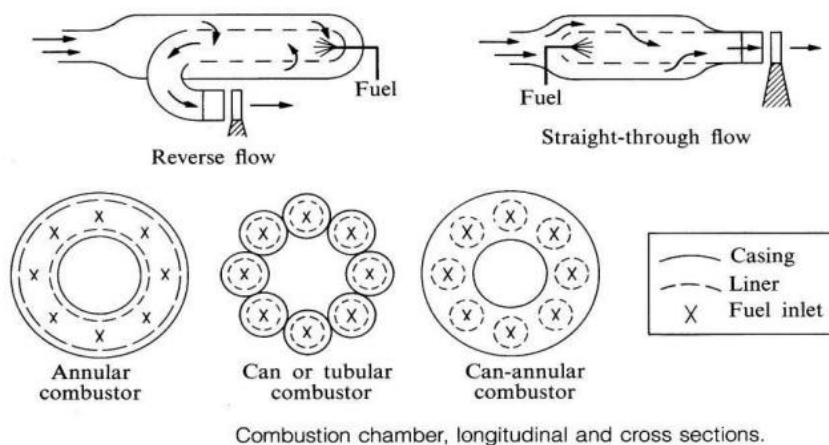
One of the first combustors for a gas turbine engine was the whittle combustor



In the figure fuel injectors, swirl vanes (which introduce swirl in the flow to stabilize the combustion) and a colander (which separates the incoming flow into two components that are then mixed once the combustion is concluded) can be seen in the figure above. For the first combustors the only objective was to introduce heat, without worry about emissions. Nowadays the combustion chamber design is not only focused on burning the fuel efficiently but also introducing the minimum amount of pollutants as possible.

Combustion chambers can be classified depending on:

- Direction of the flow:
 - **Reverse:** the Whittle combustor is a reverse flow combustion chamber, where the flow comes in from one end, heat is added and the flow then reverses its direction and goes out the same way
 - **Straight:** flow comes in one end, heat is added in the middle and then exhaust gasses come out the other end:
- Combustor type: the geometry of the combustor
 - **Annular:** used in modern engines
 - Can of tubular combustors: used in the first generations of engines
 - Can-annular combustor: in between the two.



The early gas turbine engines used can type combustors, which had holes in the sides to connect all of the cans that were positioned in a circle. This was to equalize the pressure in all of the cans in case of an issue in one of them or to reignite one of the flames.

In the 70's the design shifted towards an annular architecture. This has the advantage that flame propagation is much easier and less cooling air is required. The liner (material that "lines" the combustion chamber walls) has holes that allow air spilled from the compressor to form a thin film

which takes away heat from the liner allowing temperatures that are higher than the melting point of the material.

Enthalpy

Enthalpy is one of the most used concepts to study the process of combustion. There are three types of enthalpy we can use:

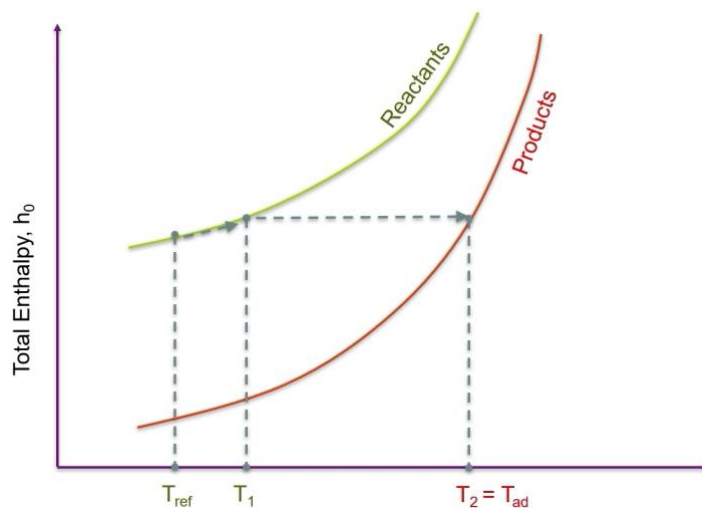
- Sensible: measures the increase of temperature of an object to which heat is transferred to
- Latent: energy required for phase transformation
- Reaction enthalpy: energy that is obtained from an exothermic chemical process or energy required for an endothermic process. For the study of combustion processes, reaction enthalpy is the property of interest. In this type of enthalpy the enthalpy of formation can be defined, which is the amount of energy released as heat at a given temperature when 1 mole of an element is produced (for reference elements like H_2 , O_2 and N_2 the formation enthalpy is zero). This is used to evaluate the amount of heat released from a chemical reaction.

On a molecular level, as we go from elemental molecules to gaseous atoms we need to give an amount of heat that is equal to the energy of dissociation.

The enthalpy of reactant molecules can be higher or lower than the enthalpy of product molecules depending on the process we are considering. In the figure the enthalpy of products is less than that of the reactant molecules, therefore the reaction is an exothermic one. This jump is what we care about in combustion, the amount of heat released by the process is equal to the difference in enthalpy formation of the reactants and the product molecules.

The jump from the elemental molecules and the reactant molecules instead is quantified by the enthalpy of formation of the reactants.

If we chart the total enthalpy of both the products and the reactants we see that they both increase with temperature (sensible enthalpy). If we were to have an adiabatic combustion, no heat would leave the system, therefore the total enthalpy would remain constant. The new thermodynamic state would be on the reactants curve at a higher temperature.



The total enthalpy at a given temperature can be defined as the sum of the formation enthalpy and the sensible enthalpy. The heat of the reaction can be used to evaluate the final enthalpy of the products, starting from the enthalpy of the reactants.

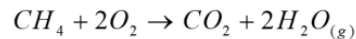
$$h_i(T) = h_{f,i}^0(T_{ref}) + \Delta h_{s,i}(T),$$

or

$$\Delta h_{s,i}(T) = h_i(T) - h_{f,i}^0(T_{ref}) = \int_{T_{ref}}^T c_{p,i} dT;$$

h_i - mass or molar specific enthalpy of species i.

If we consider a process of combustion involving methane and oxygen we would get CO₂ and water vapor. The heat of reaction of this process is just the difference in formation enthalpy of the products and the reactants.



$$(Q_P)_{NTP} = (\Delta H_f)_{CO_2} + 2(\Delta H_f)_{H_2O} - (\Delta H_f)_{CH_4} - 2(\Delta H_f)_{O_2}$$

$$(Q_P)_{NTP} = (-393522) + (-241827) - (-74897) - 2(0)$$

$$(Q_P)_{NTP} = -802279 \text{ kJ/k mol}$$

From this the heating value of a fuel can be evaluated.

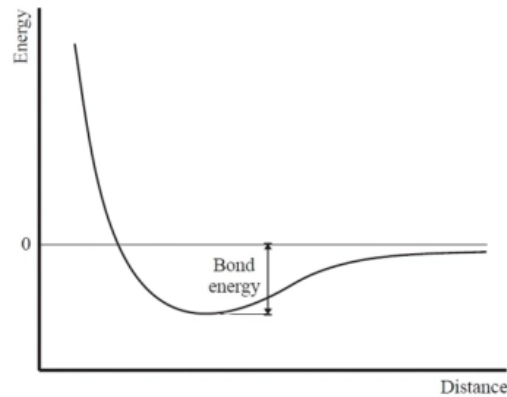
Bond energy

We're considering the transformation of chemical energy into heat. Chemical energy is the energy that is conserved in chemical bonds: when a bond is broken energy is released and vice versa.

Table 11.1 Some Atomisation, Dissociation and Resonance Energies (Based on 25 °C)

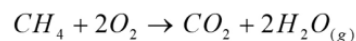
Bond Atomisation (ΔH_a)	Energy MJ/kmol	Bond Dissociation ($\Delta H(X-Y)$)	Energy MJ/kmol	Resonance	Energy MJ/kmol
H-H	435.4	C-H	414.5	Benzene: C ₆ H ₆	150.4
C (graphite)	717.2	N-H	359.5	Naphthalene: C ₁₀ H ₈	255.4
(O = O) _{O₂}	498.2	O-H	428.7	Carbon dioxide: CO ₂	137.9
N≡N	946.2	H-OH	497.5	-COOH group	117.0
		C-O	351.7		
		C=O	698.1		
		C-C	347.5		
		C=C	615.5		
		C≡C	812.2		

The bond energy is defined as the energy required to separate two atoms, in other words the minimum potential energy relative to when the atoms are infinitely separated. The minima coincides with the equilibrium state. To move away from this equilibrium, energy needs to be provided; this is the bond energy.



If a isobaric or isochoric reaction is carried out in stages, the algebraic sum of the heat evolved in separate stages is equal to the total amount of heat released when the reaction occurs directly, which means that the heat liberated in a reaction is independent of the path of the reaction between the initial and the final states. This is known as the **Hess Law**.

Let's consider the oxidation of methane (combustion). The simple reaction is:



However, in reality there are approximately 53 species involved and 325 reactions occur. Thanks to the Hess law we can still evaluate the enthalpy of the reaction from the initial and final states, independently of the stages in between (since combustion can be assumed to be occurring at constant pressure or at constant volume).

The fuel used in aviation are all different types of hydrocarbons. From the following table the heating values and the stoichiometric air to fuel ratio of these fuels can be computed. The stoichiometric air to fuel ratio is the ideal ratio between air and fuel mass to have a complete combustion, where all the reactants are transformed into products, with no leftover reactants. In other words this value is the amount of air needed to burn a unit mass of fuel.

Fuel	Chemical Formula	Molecular Weight (m_u) (integrals)	Enthalpy of Formation, Δh_f (kJ/kmol)	Higher Enthalpy of Reaction, Δh_R (MJ/kmol)	Lower Enthalpy of Reaction, Δh_R		Stoichiometric Air–Fuel Ratio, e_{st}
					(kJ/kg)	(MJ/kmol)	
<i>Alkanes</i>							
Methane	CH_4	16	−74,898	−891	−50,047	−802.9	17.16
Ethane	C_2H_6	30	−84,724	−1561	−47,519	−1428.8	16.02
Propane	C_3H_8	44	−103,916	−2221	−46,387	−2045.4	15.60
Butane	C_4H_{10}	58	−124,817	−2880	−45,771	−2660.2	15.38
Pentane	C_5H_{12}	72	−146,538	−3538	−45,384	−3274.3	15.25
Hexane	C_6H_{14}	86	−167,305	−4197	−45,134	−3889.3	15.16
Heptane	C_7H_{16}	100	−187,946	−4856	−44,955	−4504.4	15.10
Octane	C_8H_{18}	114	−208,586	−5515	−44,820	−5119.5	15.05
Cetane	$C_{16}H_{34}$	226	−454.3	−10,711	−44,000	−9963.4	14.88
<i>Aromatics</i>							
Benzene	C_6H_6	78	+82,982	−3304	−40,605	−3172	13.20
Toluene	C_7H_8	92	+50,032	−3950	−40,967	−3774	13.43
Xylene	C_8H_{10}	106	+18,059	−4598	−41,276	−4378	13.60
Naphthalene	$C_{10}H_8$	128	+150,934	−5233	−39,455	−5057	12.87
<i>Alcohols</i>							
	$C_nH_{2n+1}OH$	$C_nH_{2n+2}O$					
Methanol	CH_3OH	32	−201,301	−764	−21,114	−676	6.44
Ethanol	C_2H_5OH	46	−235,466	−1410	−27,742	−1278	8.85
Propanol	C_3H_7OH	60	−235,107	−2069	−31,504	−1893	10.30

Some observations:

- The enthalpies of reaction for many hydrocarbon fuels on mass basis is similar, between 40-50 MJ/kg.
- The stoichiometric air-fuel ratio for many hydrocarbons lie between 13:1 to 17:1

- Some fuels have positive formation of enthalpies
- All fuels have negative enthalpies of reaction (they release heat when they participate in a reaction)
- The enthalpies of reaction for alcohols is less than non-oxygenated fuels (as oxygen does not provide any energy of reaction)
- Commonly used fossil fuels are a mixture of several hydrocarbon compounds

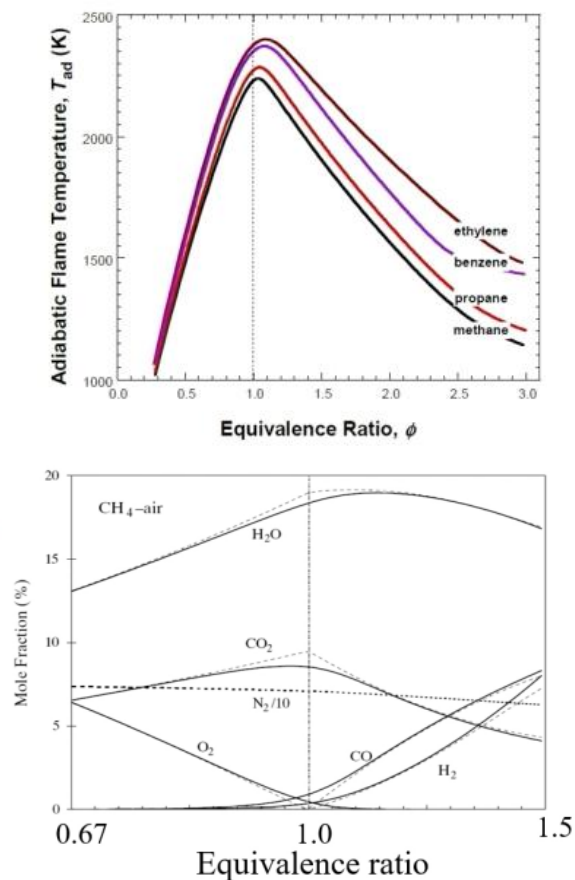
When a mixture of air and fuel burns with an air to fuel ratio higher than the stoichiometric ratio, the combustion is called lean combustion (or weak) since there is a lower than ideal quantity of fuel in the mixture and some air will not participate in the reaction and will instead absorb some of the heat released during the reaction. Thus the final temperature of the exhaust gasses will be lower.

If the mixture of air and fuel burns with an air to fuel ratio lower than the stoichiometric ratio, the combustion is called rich combustion since there is a higher than ideal quantity of fuel in the mixture and some of this fuel will not be burned during the reaction.

If a fuel-air mixture burns adiabatically at constant pressure, the absolute enthalpy of the reactants at the initial state equals the absolute enthalpy of the products at the final state. The final-state temperature is defined as the constant-pressure adiabatic flame temperature.

$$H_{prod}(T_{ad}) - H_{reac}(T_0) = 0$$

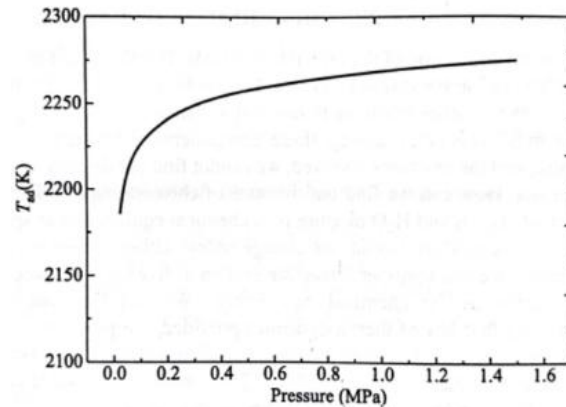
The adiabatic flame temperature (the temperature the mixture will reach if there are no heat losses in the system) will be highest when the air to fuel ratio is close to the stoichiometric one, since there won't be any unused reactants absorbing heat.



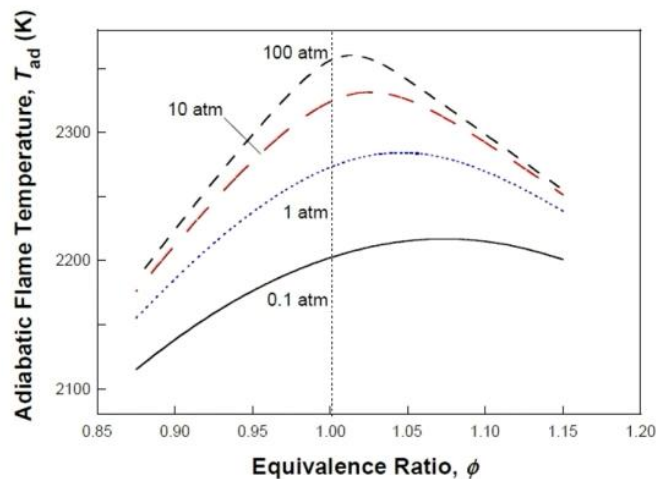
In these graphs the equivalence ratio appears. This is just the ratio between the stoichiometric air to fuel ratio and the real air to fuel ratio. When it's lower than one, the mixture is lean, while when it's higher than one the mixture is rich.

We see that the peak in adiabatic flame temperature is slightly shifted to the right of the equivalence ratio of 1. This shift is due mainly to dissociation which occurs at high temperatures. From the second graph we see that the CO_2 dissociates at around an equivalence ratio of 1 and this means that some of the fuel reactants will dissociate. Thus a slightly higher air to fuel ratio is required.

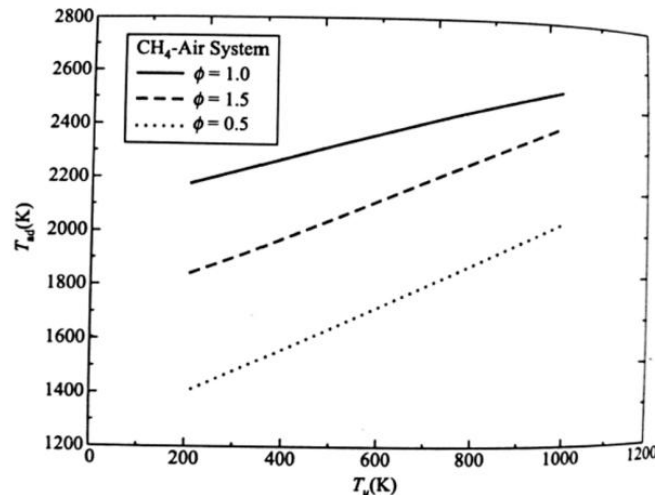
Let's now look at how the pressure in the combustor effect the adiabatic flame temperature.



We see that the adiabatic flame temperature increases with pressure until a plateau is reached. Higher pressure shifts chemical equilibrium toward more complete combustion and more dissociation suppression, so less energy is spent breaking molecules apart. This raises the flame temperature. As pressure continues to increase, dissociation becomes minimal and the reactants/products are already near their equilibrium limits, so further pressure increases add little change, producing a plateau.



If we instead consider the effects of the inlet temperature we see that as the inlet temperature increases, the adiabatic flame temperature increases but not 1 to 1.



This is because part of the added inlet energy is consumed by greater product dissociation at the higher flame temperature. Dissociation absorbs energy, reducing how much of the inlet-temperature rise shows up as temperature gain in the adiabatic flame.

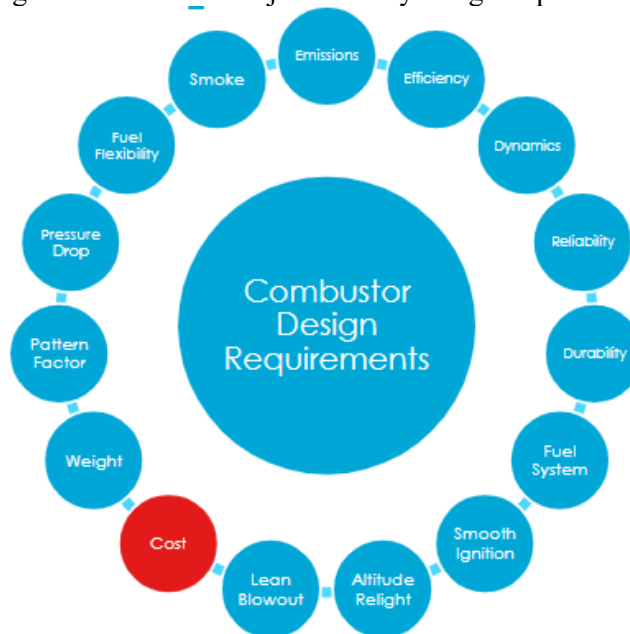
Combustion can be classified in different ways:

- Based on the mixing of fuel and the oxidizer:
 - Pre-mixed: fuel and oxidizer are mixed before the reaction. Burns at a selected equivalence ratio and are therefore prone to flashback. They are prone to instabilities and have a small operating envelope. Since there are no strong temperature gradients the NOx emissions are also contained as well as soot production. CO production becomes hard to control.
 - Non pre-mixed: also called diffusion burning. It is flashback free and also very stable. They allow for a broad operating envelope and have reliable ignition. However, these flames produce high amounts of NOx and soot due to the high temperature gradients that form on the boundary between the fuel and the air. Here dissociation of nitrogen and oxygen occur, creating different types of nitrogen oxides. Since the NOx emissions are high, the CO emissions are easy to control
 - Partially pre-mixed: most combustor are of this type
- Nature of flow:
 - Laminar
 - Turbulent
- Phase of reactants:
 - Homogeneous: the reactants are all in the same phase
 - Non homogeneous: the reactants are in different phases such as kerosene (liquid) and air (gas)
- Temporal variation:
 - Steady: typical of laminar flow flames
 - Non-steady: typical of turbulent flows
- Speed of the combustion:
 - Deflagration: slower reaction in which the pressure tends to increase
 - Detonation: much quicker reaction in which the pressure increases
- Speed of the flow:
 - Compressible
 - Non compressible
- Heat transfer:
 - Adiabatic: gas turbine combustion is adiabatic since heat is not being extracted from the system, but only used to increase the temperature of the products
 - Non adiabatic
- Convection:

- Natural
 - Forced
- Spatial dependence (strongly dependent on the geometry of the combustor):
 - 1-D flame
 - 2-D flame
 - 3-D flame
- Reaction rate:
 - Equilibrium
 - Finite rate

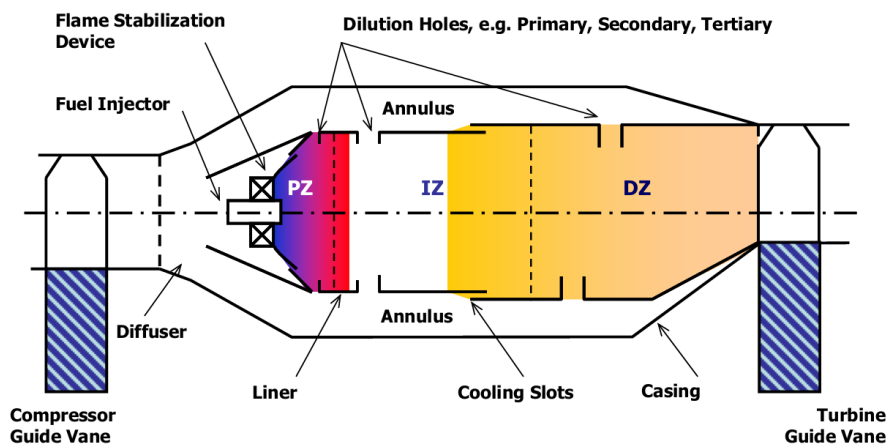
Combustor components

The design of an aero engine combustor is subject to many design requirements.



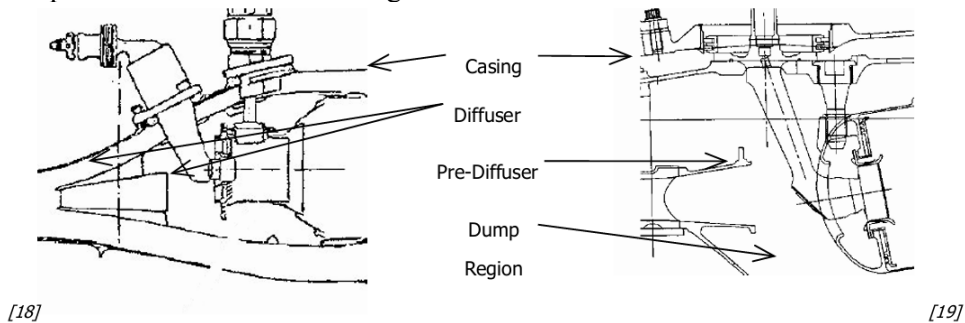
Note: pattern factor is the temperature profile at the combustor exit.

All combustors incorporate a list of the main components which are always responsible for the same functions despite the diversity in their design. These components are a diffuser, casing, liner, fuel injector, some type of a flame stabilization device and a cooling arrangement.



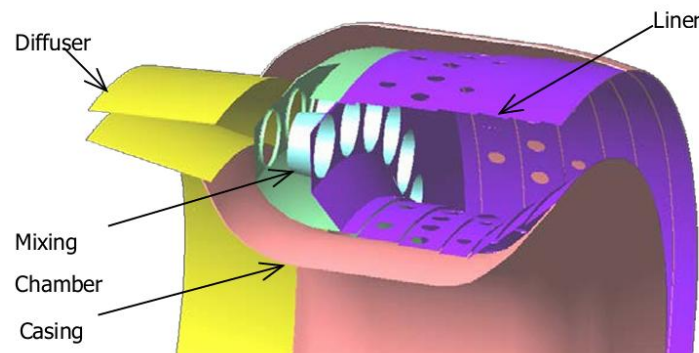
Quantitatively, the drop in total pressure associated with combustion is proportional to the dynamic pressure of the flow. Therefore, the compressor discharge velocity should be reduced to minimize the loss. It is customary to do so by incorporating a diffuser at the combustor inlet. Additional functions of the diffuser are to recover the flow dynamic pressure by raising its static pressure and smoothen flow instabilities.

There are two main design philosophies for the diffuser. One is to employ a relatively long duct to achieve a gradual flow deceleration. The other is the so-called “dump” diffuser, which consists of a short smooth pre-diffuser where the air velocity is reduced to about half its inlet value. At exit, the air is literally dumped into the combustor casing.



The aerodynamic diffuser traditionally has smaller pressure losses but results in a considerable length. The dump diffuser is exactly opposite. Its shorter length made this type the automatic design choice for modern aircraft engines.

Next are the casing and liner. The simplest possible form of combustor would be a straight-walled duct connecting the compressor to the turbine. Unfortunately, this arrangement is impractical due to the fact the pressure loss incurred would be excessive. Simply fitting a diffuser, however, is not sufficient to obtain a viable combustion system. First of all, the air velocity remains too high for a flame to sit. Secondly, poring fuel directly into the available airflow would result in no flame, as the air to-fuel ratios are too high for the mixture to ignite. The airflow should therefore be partitioned and the flame stabilized. Arrangements for that are provided inside the liner located inside the casing and downstream the diffuser. The space between the liner and casing is called annulus.



The liner is divided into divided into a primary (PZ), secondary (SZ) and tertiary or dilution (DZ) zones.

A fuel injector and a flame stabilization device mark the front boundary of the primary zone. In some cases the PZ may also be preceded by an additional mixture preparation device: a mixing chamber. The main function of the PZ is to anchor the flame and provide sufficient time, temperature and mixing to achieve complete combustion of the incoming fuel air mixture. If the PZ temperature is around or above than 2000 K, dissociation reactions will result in significant amounts of CO in the gases. If the residence time of the fuel-air mixture in the PZ is too short, appreciable amounts of UHC and CO will be present due to incomplete combustion. CO and UHC are first of all polluting species. Secondly, their presence in post-combustion gases equals wasting the heat.

To provide additional time and space for the burnout of CO and UHC, the secondary zone succeeds the PZ. The right conditions are created in the SZ by adding an extra air to, on one hand, reduce the temperature and, on the other hand, not to quench combustion reactions.

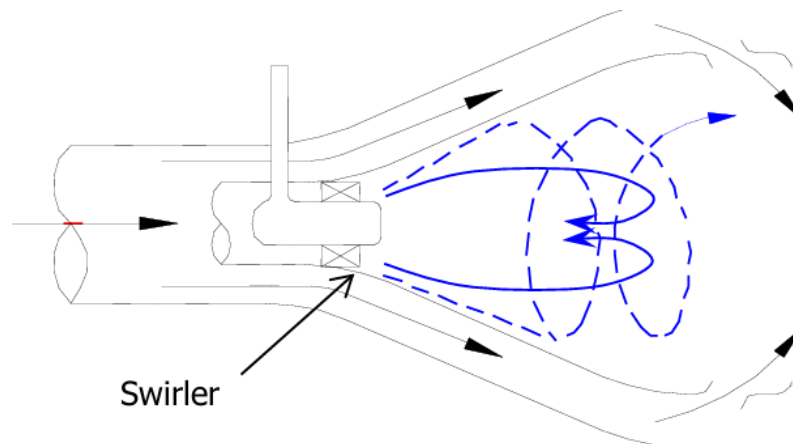
At Low altitude operation (higher pressures in combustor), the SZ needs to recover dissociation losses (primarily $\text{CO} \rightarrow \text{CO}_2$) and complete the burning of any leftovers from the PZ. Some of the soot that is produced in the PZ is also oxidized in the SZ.

At high altitude operation (lower pressures in combustor) the reaction rate in the PZ is lower due to the lower pressure and so the SZ needs to act as an extension to the PZ allowing for higher residence time.

The remaining air is admitted into the dilution zone, mixing it with the core air. This is done to reduce the temperature of the outlet stream and make its temperature pattern acceptable to the turbine.

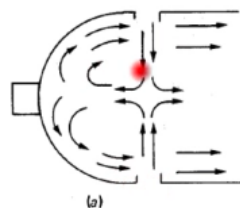
Another important element is the fuel injector. Fuel can be in gaseous form or in liquid form. A common method of supplying a gaseous fuel into combustion chamber is forcing it through a specially designed orifice. For liquid fuels it is more complicated since the fuel needs to be vaporized through atomization, otherwise the combustion will occur heterogeneously around the big droplets of fuel. The fuel injectors used for liquid fuels like kerosene are therefore called atomizers.

Air leaves the compressor at a speed on the order of 150 to 200 [m/s] for the case of an industrial gas turbine or a large aircraft engine. The velocity is reduced about twice in the combustor diffuser. It is further decelerated down to 25 to 35 [m/s] in the primary zone of the liner. However the flow velocity still remains far greater than the speed of the flame. To prevent the flame from being blown away, it is therefore necessary to set up local regions with much smaller velocity. The most common solution is to generate a flow reversal in the liner PZ. An efficient way to reverse the flow is to impart a swirl to it. Recirculation is created in the core region of the swirl where the amount of rotation is high.

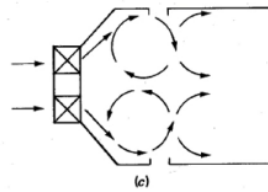


Somewhere near that boundary, the velocity assumes the value that matches the value of the flame speed. This is the region, where the flame “anchors”. Swirlers also contribute to fuel-air mixing and provide effective ways to control the stability and intensity of combustion and the size and shape of the flame region.

Other ways to supply flame stabilization are:



Stabilization by opposing jets



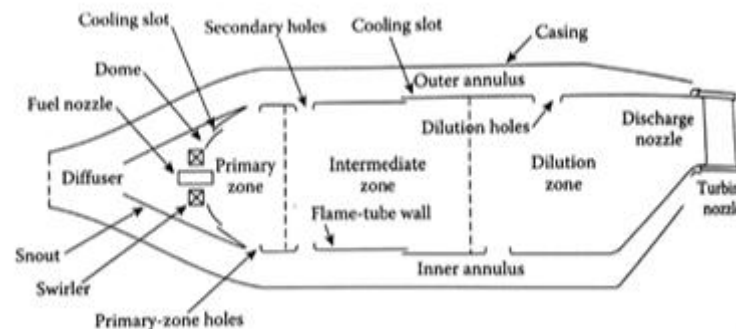
Stabilization by swirling flow assisted by secondary injection

The stabilization by swirling flow assisted by secondary flow injection is the most common solution and it is a mix of the two previous solutions.

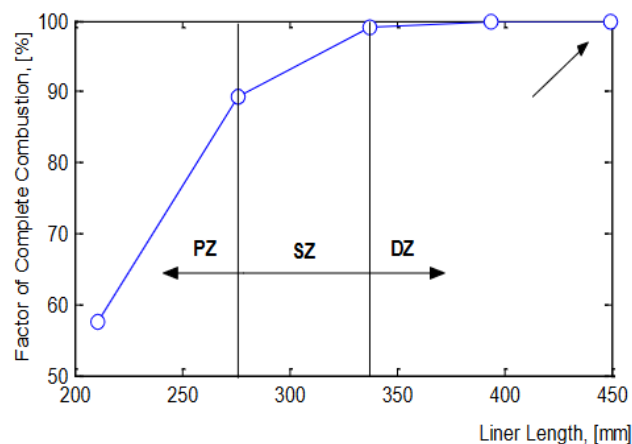
The combustor liner is exposed to very high temperatures during gas turbine operation. To ensure the life of the liner, it is necessary to remove the heat transferred to its walls and prevents contact with the hot combustion gases. An efficient way to cool and protect the liner is film cooling. This technique employs stacks of holes through which air is injected axially along the inner surface of the liner wall to provide a protective cooling film. This film is being gradually destroyed downstream. Therefore, normal practice is to provide a succession of cooling-hole bands or slots along the length of the liner.

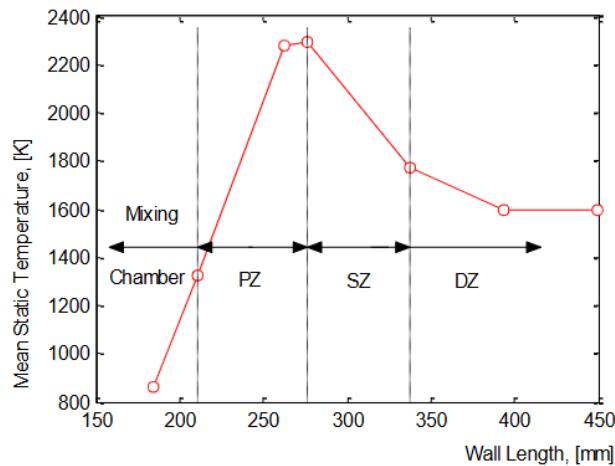
In summary, the incoming airflow is split into two flows since there is too much air for a complete combustion; 20-25% of the air goes into the core while the rest flow in the annulus. The core air then enters the diffuser where it is slowed down. Fuel is then injected into the core flow and the swirler aid with mixing. Another one of its functions is to introduce a swirl to the flow velocity to further slow down the air and allow a flame to be sustained. This all occurs in the primary zone. The flame stay in between the primary and intermediate zone. Meanwhile the liner walls are cooled by the secondary flow that enters through small holes and forms a protective film.

The in the dilution zone the rest of the air is mixed to ensure the correct temperature profile to the turbine inlet.



The evolution of combustion and the temperature variation in a typical combustion chamber are plotted in the graphs below.

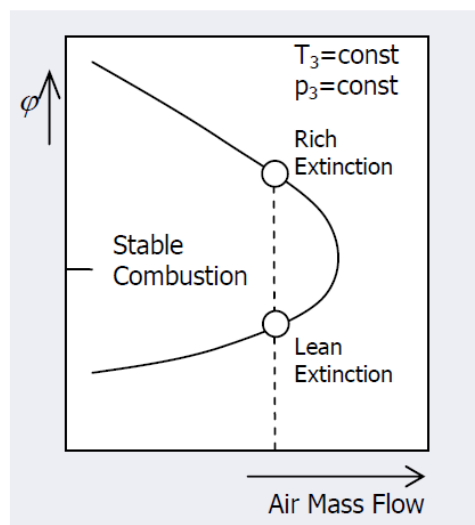




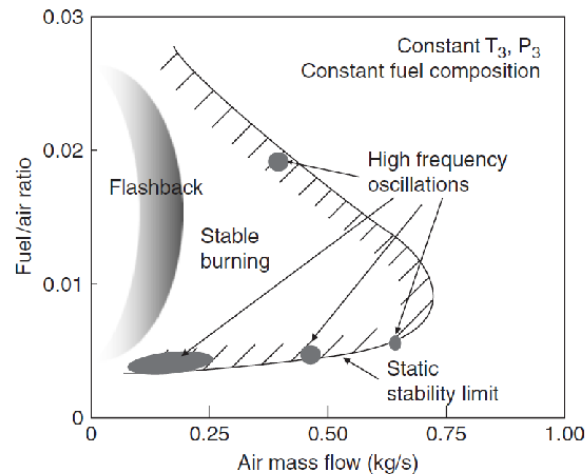
Most of the temperature rise occurs in the PZ, since that's where the flame sits. Then in the SZ and the DZ temperature decreases again since secondary flow is being mixed with the core flow.

Flammability limits

Gas turbine combustion is supposed to be a self-sustaining continuous process under broad operating conditions. We can define a region of equivalence ratio and air mass flow (or air velocity) outside of which combustion cannot take place.



We see that as the flow velocity increases the equivalence ratio range over which combustion can occur decreases. The curve is defined by a rich extinction limit and a lean extinction limit. However, a combustor must operate well within these limits, since there are many effects we want to avoid close to the boundaries.

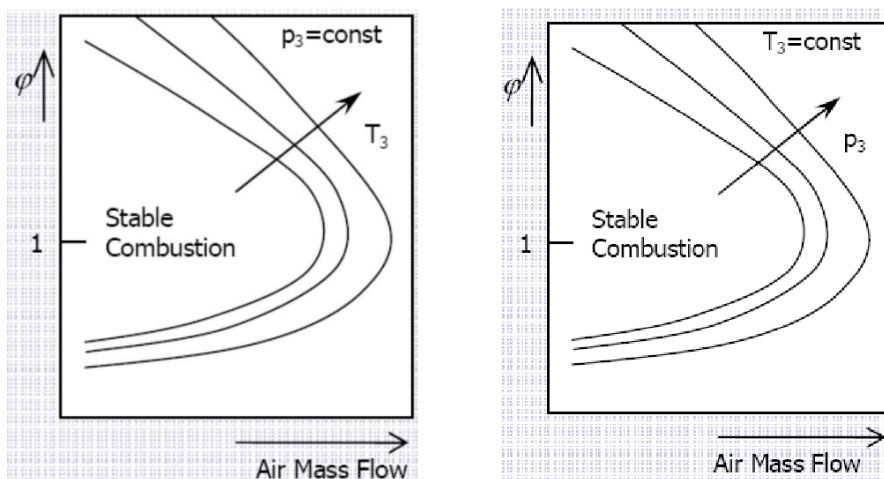


Flashback is a phenomena in which the flame propagates upstream and can only occur at low velocity flow. This can greatly damage the combustor. Near the limits high frequency oscillations in the flame occur, which can cause a flame-out. These acoustic oscillations then lead to unsteady heat release, which in turn causes flow perturbation that cause acoustic oscillations. This cycle continues and it creates pressure fluctuations which decrease the combustors efficiency and structural vibrations.

Just like for the adiabatic temperature, inlet temperature and pressure have an effect on these limits. As we increase these two properties the limits move to the right, increasing the possible operating range of the combustor.

Higher pressure and higher inlet temperature both make ignition and reaction kinetics faster and easier, so mixtures that were previously too lean or too rich can now sustain a flame.

This is particularly advantageous during take-off, when the pressure is much higher. The critical condition is relight at cruise, since pressure and temperature are both very low.



If we look at the below chart for example we see that engines tend to have overall equivalence ratios that are outside of these limits, especially at take-off. Therefore the design of the combustion chamber needs to endure higher local equivalence ratios so that we stay in the flammability limits.

Also bigger engines have higher overall equivalence ratios.

The stability of the flame is also a function of other combustion parameters such as:

- Ignition delay: the time between when a fuel–air mixture is exposed to ignition conditions and when rapid heat release (combustion) actually begins. It increases with temperature.
- Activation energy: the minimum energy the reactants must acquire for a chemical reaction to proceed at an appreciable rate. As the equivalence ratio increases and approaches 1, the activation energy decreases. Varies depending on the type of fuels.

Combustor efficiency

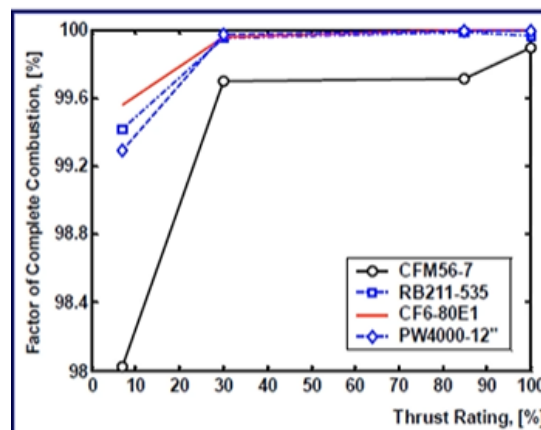
Since the purpose of the combustor is to provide heat to the flow, the combustor efficiency can be defined as the ratio between the actual enthalpy rise and the ideal enthalpy rise, where the enthalpy rise is equal to the heat added to the system since the combustion is isobaric.

$$\eta_b = \frac{\bar{c}_p \left[(\dot{m}_a + \dot{m}_f) T_{t4} - \dot{m}_a T_{t3} \right]}{\dot{m}_f \cdot h}$$

Note: The higher heat of combustion - also known as the higher heating value - is calculated assuming that all the water in the products has condensed to liquid. The lower heating value, LHV, corresponds to the case where none of the water is assumed to condense. The latent heat contained in the water vapor remains un-liberated in the last scenario

The heat released in a combustion chamber is always evaluated with the lower heating value of the fuel, which, for an aviation fuel such as kerosene, is always around 42.8-43 MJ.

Typically combustor efficiencies are very high and tend to increase with pressure and temperature due to the increased reaction rate (for the same residency time, combustion is more complete). The opposite is true for the Mach number, since this decreases residency time. Since the design point of the combustor is the cruise condition we see that the efficiency tends to increase as we increase the thrust generated.



The main losses in the combustor come from the pressure losses that occur due to friction, mixing, and heat addition as well as dissociation that occurs at high temperatures. Other factors are incomplete combustion and heat losses.

Pressure loss due friction is a major contributor especially in the DZ since the flow is accelerating back up again.

Let's now summarize all of the characteristics of an actual aero engine combustor:

- Primarily turbulent combustion since the velocity of the flow is still very high even though the flow is slowed down significantly
- External ignition is required to light the combustion, then combustion is self-sustained unlike in automotive engines

- Constant pressure distribution (very small pressure losses)
- Pressure losses penalize performance and have to be limited
- Low relative volume and weight
- Fast volume and heat dynamics
- Low residence time (in the order of a couple of milliseconds)
- Adiabatic combustion (only very small heat losses that penalize the performance)
- High pressure and inlet temperatures (40 (large engines)-60 (smaller engines) bar and $T > 900\text{K}$)

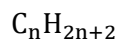
As fuels the aviation industry mainly uses liquid kerosene although different fuels are being studied both in liquid form and gaseous form.

We also observe that there are different types of combustor types:

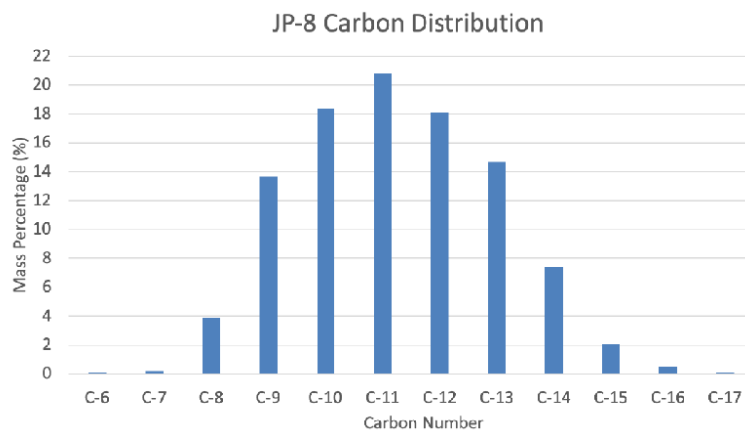
- Conventional
- RQL, lean burn, staged combustor, etc...
- Alternative technologies
- Use of catalysts and inhibitors (not currently used, active area of research)

Fuel

The fuel used by aviation is kerosene, which is a product of the distillation of crude oil together with gasoline, diesel and many other petrol derivatives. Kerosene is a mix of tens of different compounds, whose relative concentration vary depending on the use cases of the fuel. The compounds with the highest concentration are alkanes (paraffines) which have the following chemical formula:



Aromatics are also present with very small concentrations to reduce the soot formation in the combustion chamber. The most common carbon chains present in kerosene are the C-11 and C-10 chains as we can see from this distribution.

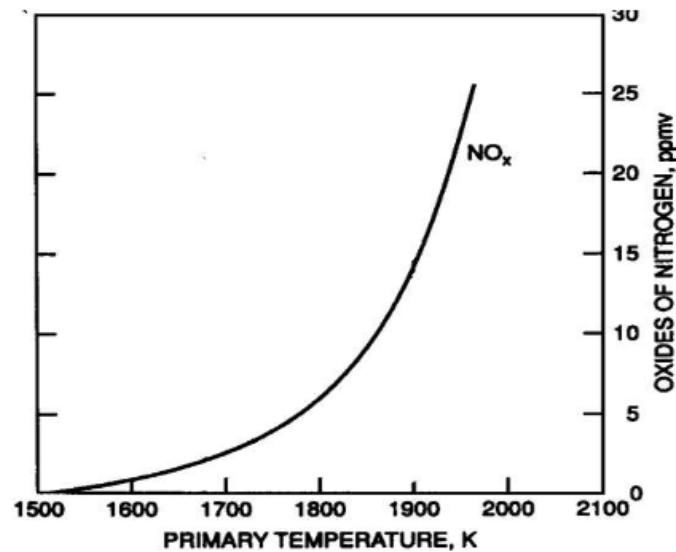


Engine emissions

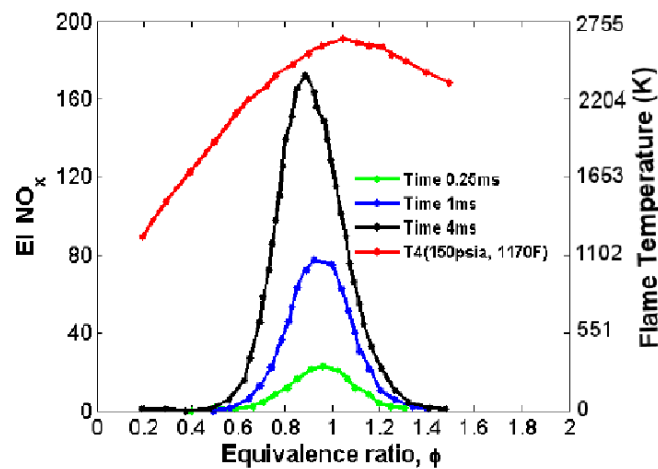
Engine emissions are strictly regulated for the Landing and Take-off cycle LTO. This cycle is made up of 4 phases: the approach, the taxi, the take-off and the climb. The cruise condition is not regulated. This is mainly to reduce local air pollution near airports. All of the products of the combustion, such as carbon monoxide, nitrogen oxides, smoke and unburned reactants are highly harmful for human beings.

The component of the emissions that is most regulated are the NO_x emissions since they can have the most dangerous effects.

There are many different ways that nitrogen oxides can form, however we focus on thermal NO_x, which is created inside the combustion chamber especially at high temperatures.

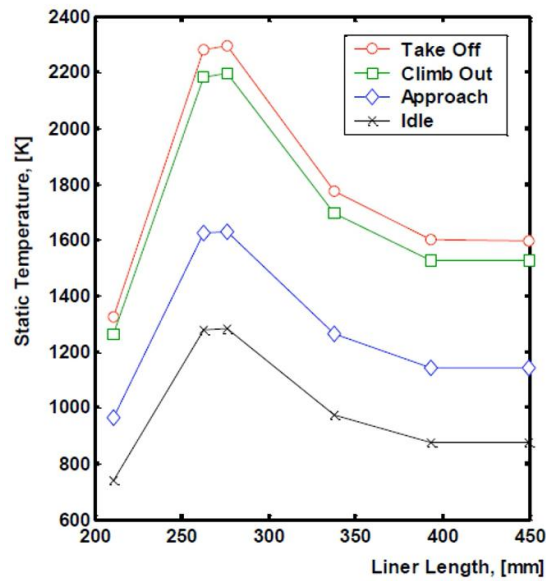


Factors that influence the NO_x emissions are all the factors that influence the adiabatic flame temperature, since the emissions increase exponentially after a certain temperature is surpassed. Also residence time plays a role (higher residence time corresponds to higher NO_x emissions).

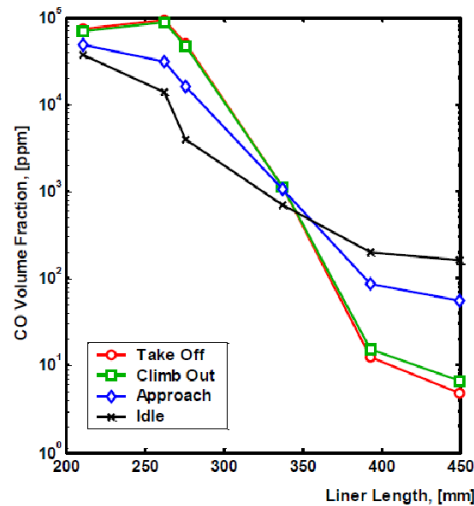


Even though the peak of the adiabatic flame temperature is slightly on the rich side, we see that peak emissions are slightly on the lean side. This is because on the rich side the availability of oxygen is less.

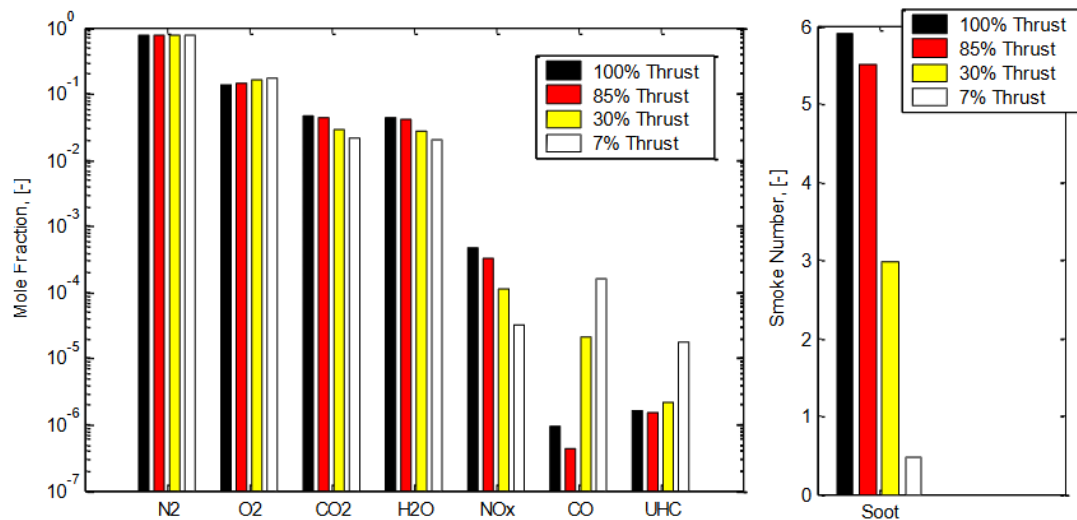
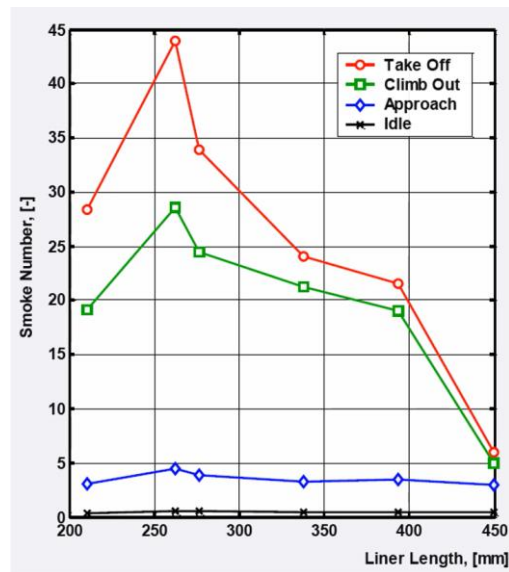
We can see that different temperatures in the combustor are reached for different phases of the mission:



The CO production is opposite of the NOx production: it is highest for idle conditions and lowest for take-off and climb.



Soot emissions seem to follow the same trend as NOx emissions. The controlling factors are the fuel to air ratio, pressure, temperature, quality of fuel injection and mode of the fuel injection. The equivalence ratio is particularly important for the soot emissions, therefore the injection quality and mode are important.



The main emissions from aero engines are molecular nitrogen and molecular oxygen since the combustion typically occurs in lean conditions, some of the air does not participate in the combustion process.

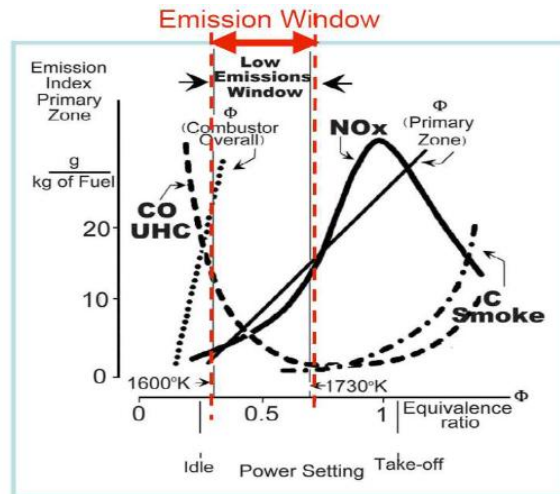
For higher thrust settings the amount of CO₂, H₂O, NO_x and soot produced are higher since more fuel is being burned. For the CO₂ and H₂O it's because the amount produced per kg of fuel burned stays pretty much constant.

CO emissions follow the opposite trend since it is formed at lower temperatures (it cannot be oxidized into CO₂) and is also a product of incomplete combustion.

Unburned hydrocarbons are highest when the thrust is minimum.

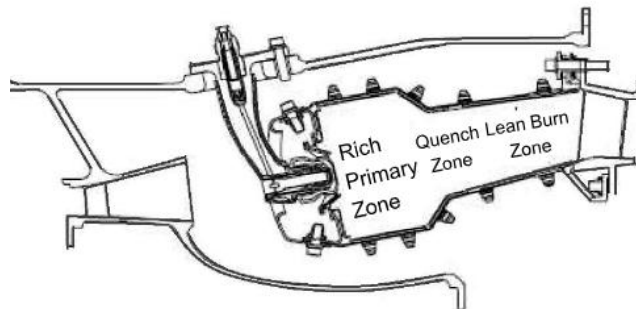
Note: UHC and CO behave similarly, as do NO_x and soot emissions.

By superimposing all of these trends we get a pretty small low emissions window around low equivalence ratios.



RQL

Rich burn-Quick quench-Lean burn engines are a combustor architecture used to reduce the risk of pressure and acoustic oscillations in the combustor when it's operating near its flame out limit. In the RQL the equivalence ratio is kept quite rich in the PZ (close to 2). This leads to relatively low temperatures in PZ and so low NOx emissions. This is followed by a quench zone where a lot of air is added from outside the liner, leading to a last zone of lean burn. This way the region where NOx emissions are very high is avoided by avoiding very high temperatures without sacrificing too much the efficiency.



Most of the modern engines used the RQL technology. A big amount of soot is produced in the rich burn zone since the equivalence ratio is high, however it is usually oxidized in the SZ and DZ.

TAPS

A Twin Annular Premixing Swirler is an alternative technology used by GE for their combustors. In this case the primary zone is already lean since almost 70% of the incoming airflow goes in the core of the combustor. This is a huge difference compared to the RQL combustor where only 25% of the air actually goes in the core. Since the amount of air for TAPS combustors is very high there is also a need for very complex swirlers to slow down the flow and stabilize the flame.

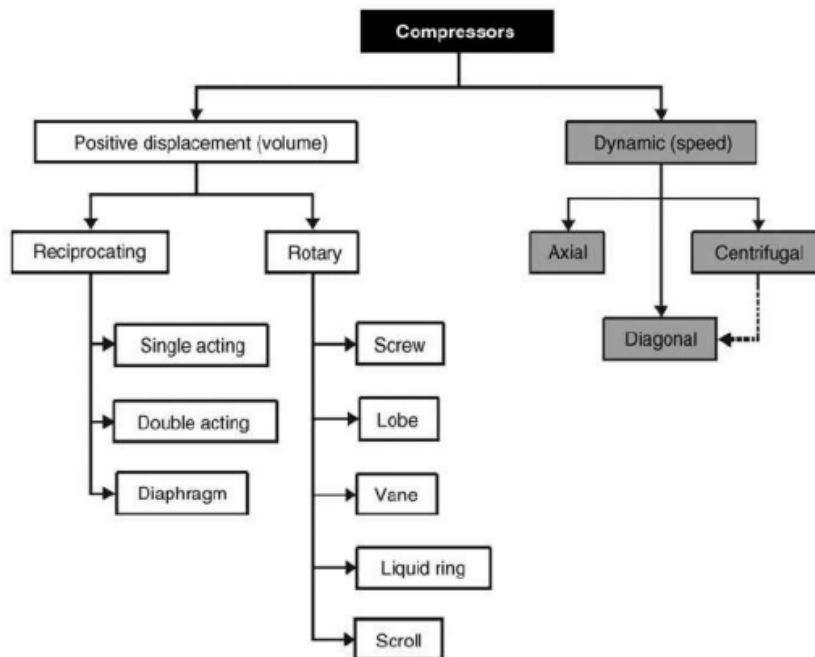
Lecture 10

Turbomachinery

An aero engine is mainly made up of turbomachinery: both turbines, both compressors and the fan are all examples of turbomachinery.

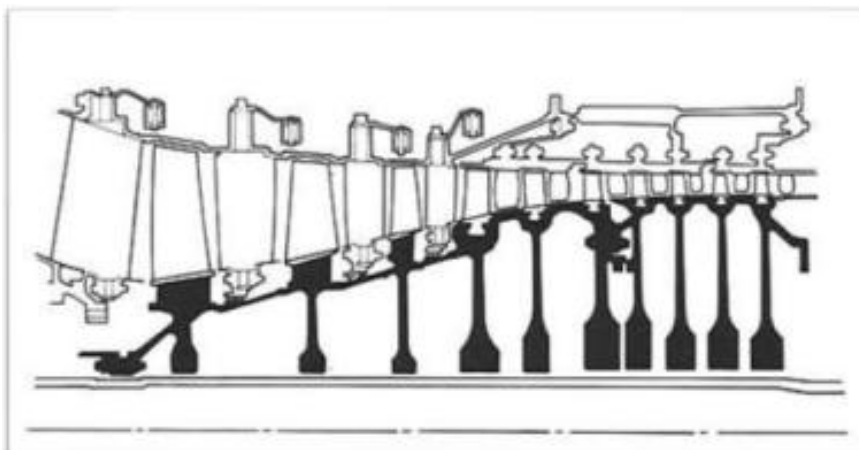
Let's first look at the compressor. There are many different types of compressors and they can be classified into:

- Mechanical compressors: they use moving mechanical parts to compress the air. Can be further classified into:
 - Positive displacement: compression of the fluid is achieved by compressing the fluid in a closed volume that reduces.
 - Dynamic: compression is achieved by doing work on a moving fluid without changing its volume. Can be then classified based on the direction of movement of the fluid:
 - Axial
 - Centrifugal
 - Diagonal: mix of the previous two



Axial compressor

The most used architecture for compressors in aero engines are axial compressors. In an axial compressor there are a lot of stages in order to achieve a high-pressure ratio in the most gradual way. Mechanical power of the rotor is used to increase the kinetic energy of the fluid. Passing through the stator, this kinetic energy is converted in pressure by a diffusion process. The stator (or diffuser) is also used to modify the direction of the fluid and make it as it was before entering the next rotor.

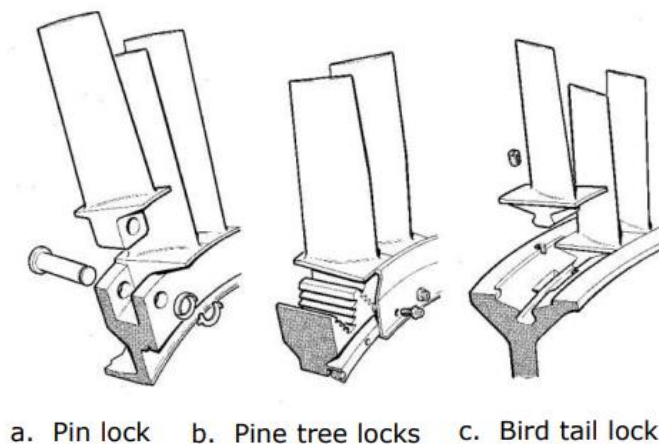


From the figure above we can see that the rotor stages are connected to the shafts that drive them using the power harvested by the turbine stages. Also we can see that the stator stages are also connected to a mechanism, which allow them to rotate around their axis. These are called variable stator blades and allow the compressor to function even in the harshest operating conditions.

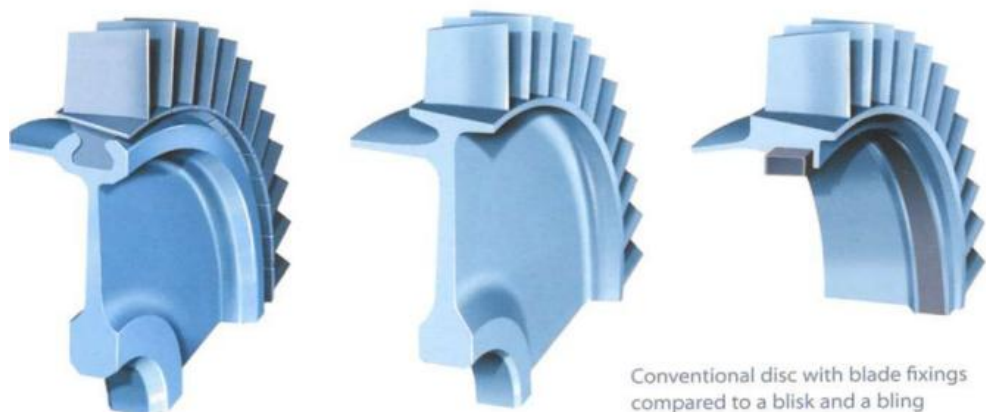
From the figure we can see that the size of the blades decrease progressively as we move through the compressor. This is done to maintain the mass flow rate constant. Since pressure is increasing, density is also increasing. Thus to maintain the mass flow rate constant, the cross-sectional area needs to decrease.

In some cases before the first stage of the compressor a IGV Inlet Guide Vane is present, which aligns the flow so that the absolute velocity the first rotor sees is equal to all the other rotors.

The root of the blades are made to slide into place around a disk that is connected to the shaft. They are then locked in place through different mechanisms, such as:

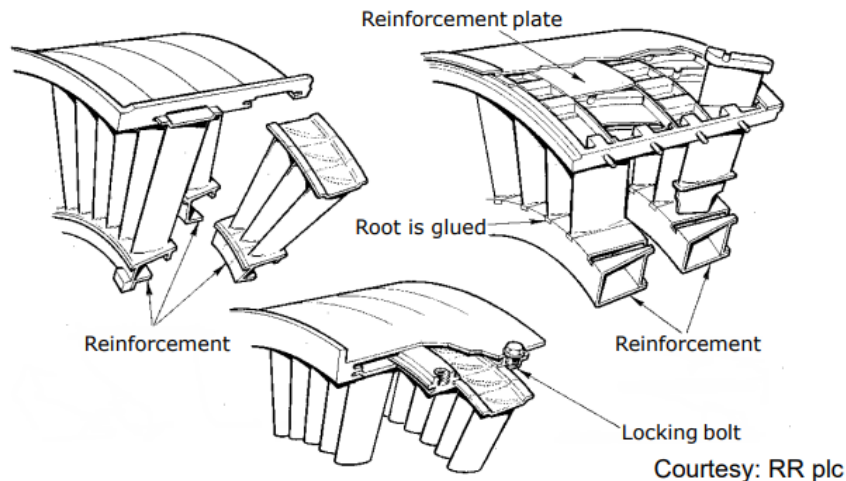


For certain modern engines, the blades are machined with the disk so to reduce mechanical complexity and weight. These kinds of disks are called bladed disks and are much harder to manufacture, much more expensive and are impossible to repair (they can only be replaced).



Bladed disks are typically never used in civil aircrafts and are only used in some military engines.

For non-variable stator blades, they are anchored both at the root and at the tip. In some cases to save on weight the stator blades are only connected to the outer shell of the compressor and are only glued to the core.

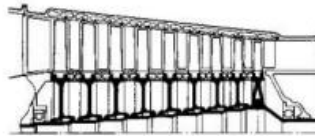


For smaller engines sometimes the saving in cost of only anchoring one side of the stator blades are enough to outweigh the increase in losses due to the presence of a gap.

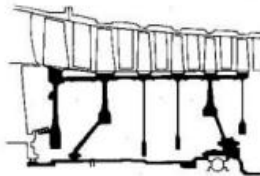
For the disks that connect blades to the shaft there are many different architecture.

- Disk rotor: used only for older engines. A series of disks connected together hold the blades of the rotor stages.
- Drum rotor: arms connect the blades hubs to the shaft
- Drum rotor with variable vanes

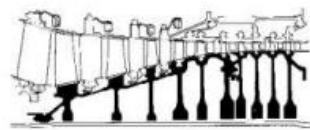
a. Disk rotor



b. Drum rotor

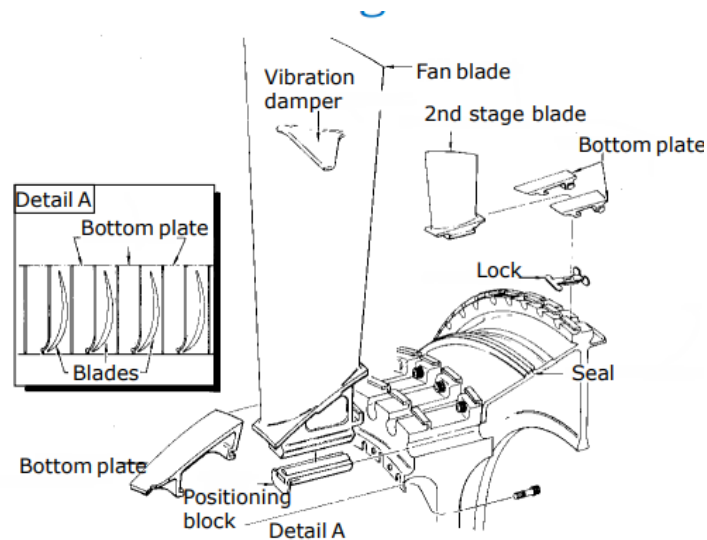


c. Drum rotor with variable vanes



The design of these disks are very important since they are subject to very high centrifugal stresses and can harmfully contribute to the propagation of vibrations that can lead to structural damage.

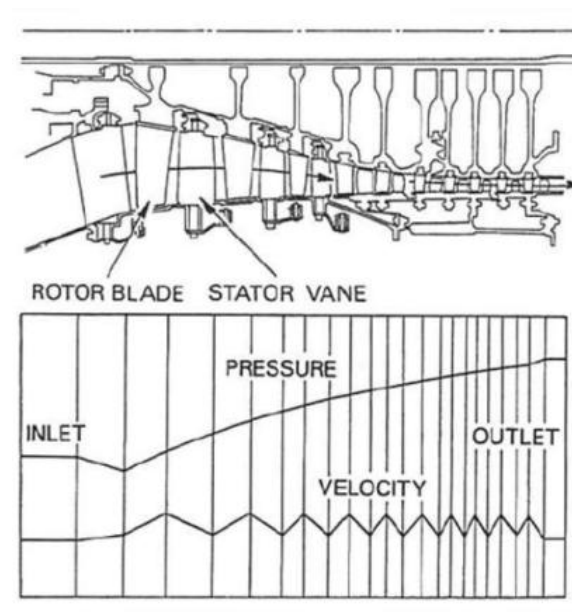
Let's now look at the fan blades. Unlike the compressor blades, the fan blades are hollow inside and have an internal honeycomb structure. Moreover fan blades typically have airfoil sections and supports towards the tip that allow them to "rest" on one another and greatly reduce vibrations.



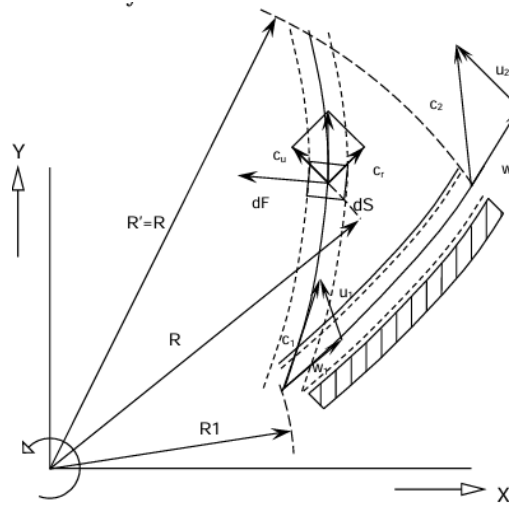
Modern fan blades are much more complex than oldern fan blades. They have more exotic 3D geometries and sometimes are made of carbon fiber composites with titanium tips.

Let's now consider more in depth how an axial compressor functions. For our considerations we assume that there are no 3D effects and that the flow is completely axial.

The stator imparts work on the flow, increasing it's velocity and pressure. Then in the stator no work or heat is being exchanged by the fluid, so kinetic energy is being directly transformed into an increase in pressure. It works as a diffuser. The variation of pressure and velocity along a generic axial compressor is as follows:



Because of the blading arrangements, the flow inside a turbo-machine rotor is both unsteady and asymmetric relative to a control surface fixed in space. However, the unsteadiness is periodic so that, on average, we may omit the time contribution of the velocity vector to the torque equation. We consider now a compressor and a fluid element along its path.



For the whole compressor holds:

$$\text{Power} = P = M \cdot \omega \quad (7.1)$$

$$F dt = d(m c) \quad (7.2)$$

Where c stands for the absolute velocity of a fluid particle. From this follows

$$F = \frac{d}{dt}(m c) = \dot{m} c + m \dot{c} \quad (7.3)$$

Assuming $\dot{c} = 0$ it follows

$$F = \dot{m} c \quad (7.4)$$

For the fluid element holds:

$$dF = \dot{m} dc \quad (7.5)$$

$$dF_r = \dot{m} dc_r \quad (7.6)$$

$$dF_u = \dot{m} dc_u \quad (7.7)$$

As the radial component gives no contribution to the torque M , we obtain:

$$dM = r dF_u \quad (7.8)$$

$$dM = r \dot{m} dc_u \quad (7.9)$$

The torque required for the compression is:

$$M = \dot{m} \int_1^2 r dc_u \quad (7.10)$$

From which follows that

$$M = \dot{m} (c_{u2} r_2 - c_{u1} r_1) \quad (7.11)$$

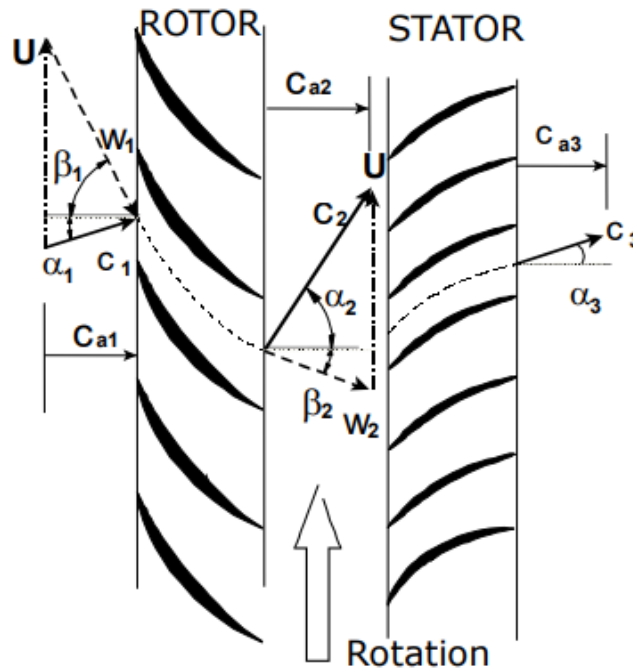
$$P = M\omega = \dot{m} (c_{u2} r_2 - c_{u1} r_1) \omega \quad (7.12)$$

Using $\omega = u_1 / r_1 = u_2 / r_2$, we obtain the **Euler's Equation for Turbomachinery**:

$$\boxed{P = \dot{m} (c_{u2} u_2 - c_{u1} u_1)} \quad (7.13)$$

Euler's formula is used to evaluate the power needed by a compressor stage. This can be applied to all types of turbomachinery.

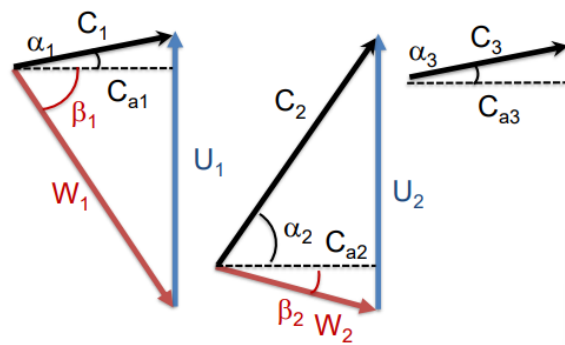
For an axial compressor a stage can be depicted as follows:



Note the direction of rotation w.r.t. the camber of the airfoils, since work is being done on the flow. To study turbomachinery blades velocity triangles are used, which represent the following velocity relation:

$$W + U = C$$

Where U is the rotational speed of the blade section we're considering, W is the flow relative velocity to the blade and C is the flow's absolute velocity. For an axial compressor:



We assume that the angle of the relative velocity of the flow leaving a blade is the same as the angle of the trailing edge of the blade.

To repeat stages the absolute velocity coming out of the stator needs to be equal to the incoming absolute velocity to the rotor.

Looking at velocity triangles of the rotor and assuming that the axial velocity has the same value, we obtain:

$$u = c_{u1} + c_{w1} = c_a (\tan \alpha_1 + \tan \beta_1) = c_{u2} + c_{w2} = c_a (\tan \alpha_2 + \tan \beta_2) \Rightarrow$$

$$\Rightarrow (\tan \alpha_1 + \tan \beta_1) = (\tan \alpha_2 + \tan \beta_2) \quad (7.14)$$

Using the Euler's formula divided by the mass flow and with $u_1 = u_2 = u$ we obtain the specific power:

$$P_s = \dot{W}_s = u (c_{u2} - c_{u1}) \quad (7.15)$$

Because of

$$c_{u1,2} = c_a \tan \alpha_{1,2} \Rightarrow \dot{W}_s = u \cdot c_a (\tan \alpha_2 - \tan \alpha_1) \quad (7.16)$$

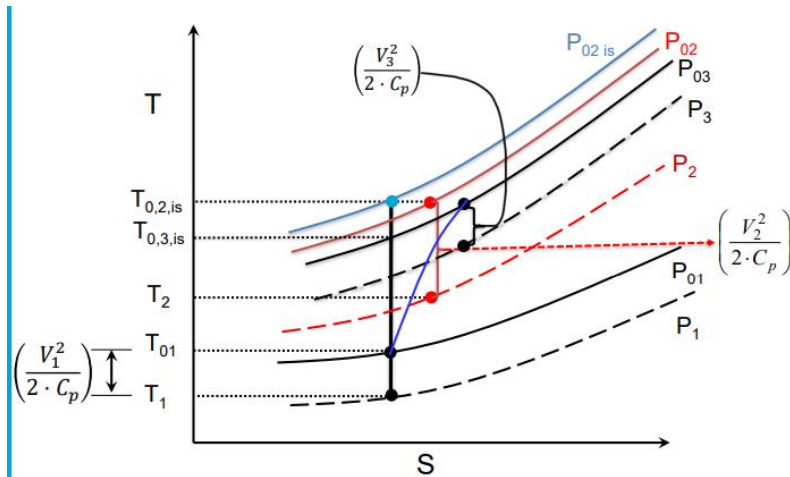
and

$$(\tan \alpha_1 + \tan \beta_1) = (\tan \alpha_2 + \tan \beta_2) \quad (7.17)$$

It follows that

$$P_s = \dot{W}_s = u \cdot c_a (\tan \beta_1 - \tan \beta_2) \quad (7.18)$$

From the T-S diagram of a compressor stage we have:



The air is coming in with a velocity V_1 and is being compressed by the rotor to a total pressure $p_{0,2}$, which is slightly lower than the isentropic $p_{0,2,is}$. We see from the graph that the dynamic pressure at point 2 is also higher than in point 1 since the rotor has also increased the flow velocity. Then from 2-3 the stator increases the static pressure even more by decreasing the dynamic one. The total pressure slightly decreases due to friction losses. Overall, V_1 should be equal to V_3 .

From the evaluation of the power needed by each stage and the isentropic efficiency of the stage, the pressure and temperature ratios can be evaluated.

■ Increase in total temperature per stage: $T_{0,3} - T_{0,1} = T_{0,2} - T_{0,1} = \frac{\dot{W}_s}{c_p}$

■ Pressure ratio per stage: $\frac{p_{0,3}}{p_{0,2}} = \left(1 + \eta_{is} \frac{\Delta T_{0,1-3}}{T_{0,1}} \right)^{\frac{k}{k-1}} =$

$$= \left(1 + \frac{\eta_{is} \cdot \lambda \cdot u \cdot c_a (\tan \alpha_2 - \tan \alpha_1)}{c_p T_{0,1}} \right)^{\frac{k}{k-1}}$$

We would like to have high u to achieve high pressure ratios, but the centrifugal stresses do not allow too much high rotational velocities; we would like to have high axial flow velocity, but flow separation and losses problems limit the axial velocity; we would like to have $\tan \alpha_2 - \tan \alpha_1$, but there is a limit to the blade curvature caused by flow separation problems.

As diffusion takes place in both rotor and stator and there will be an increase in static pressure, the degree of reaction Λ provides a measure of the extent to which the rotor contributes to the overall static pressure rise in the stage. It is normally defined as:

■ Definition of degree of reaction:

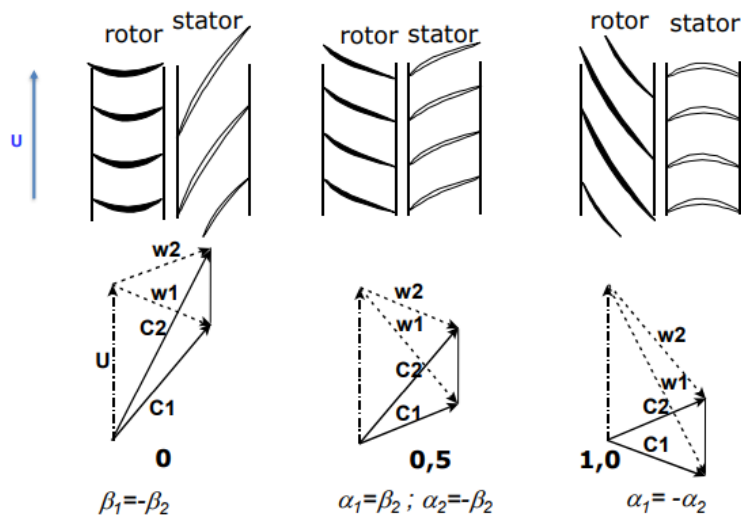
$$\Lambda = \frac{\text{static enthalpy rise in the rotor}}{\text{static enthalpy rise in the stage}}$$

$$\Lambda = \frac{\Delta T_{\text{static rotor}}}{\Delta T_{\text{static rotor}} + \Delta T_{\text{static stator}}}$$

■ So for an axial compressor:

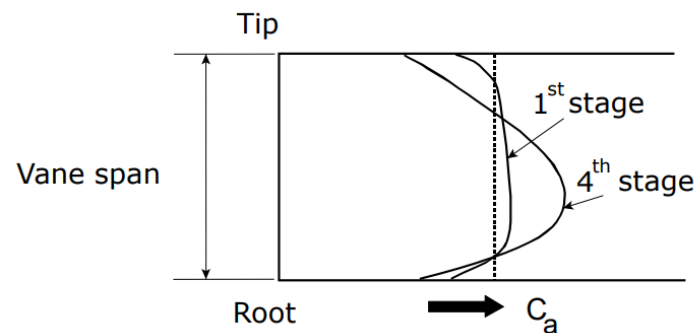
$$\Lambda = \frac{c_a}{2u} (\tan \beta_2 + \tan \beta_1) = 1 - \frac{c_a}{2u} (\tan \alpha_1 + \tan \alpha_2)$$

For degree of reaction close to zero, the rotor is only increasing the velocity of the flow, while all of the diffusion occurs on the stator, while for degree of reaction close to 1 most of the diffusion occurs in the rotor. Typically we want a degree of reaction of around 0.5. The higher is the degree of reaction, the more is the amount of energy increased in the rotor and so the higher is the temperature rise in that part of the compressor. Different values of the degree of reaction are responsible for a different shape of the vanes.



All the considerations that have been made refer to an imaginary plane of the blade. The velocity triangles have been drawn in a tangential plane at the mean radius. This simplified analysis is reasonable for the later stages where the blade speeds at root and tip are similar. At the front end of the compressor, however, the blades are much longer, there are marked variations in blade speed from root to tip, and it becomes essential to consider three-dimensional effects in analyzing the flow. Moreover, in the real condition the axial velocity changes between root and tip of the blades because of the presence of the boundary layer along the annulus walls. This effect becomes more evident in the later stages, because the boundary layer thickens as the flow progresses and the area available for the flow is reduced below the geometric area.

Also the clearance between the blade tips and the blades themselves is constant in absolute value. However, the chord of the later stages is smaller so relatively the space in between is bigger and tip effects losses are greater.

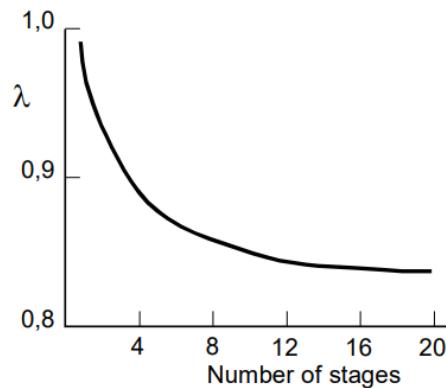


A so called “Work-Done” factor is introduced to account for the reduction in work capacity caused by the change in axial velocity. Integrating the specific power along the blade height and using the “Work-Done” factor λ , we obtain:

- Integrate specific power along the blade height → “work done”-factor, λ

- Power:

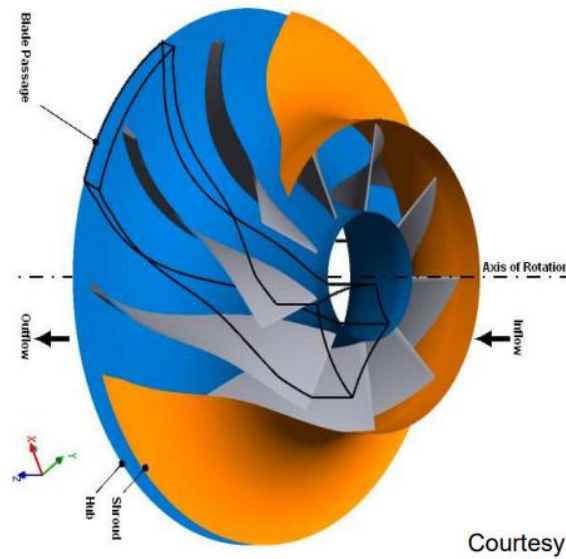
$$\dot{W} = \lambda \cdot \dot{m} \cdot W_s$$



As the number of stages increases the work being done on the fluid is being wasted more and more on losses and increasing the total temperature of the fluid instead of the total pressure.

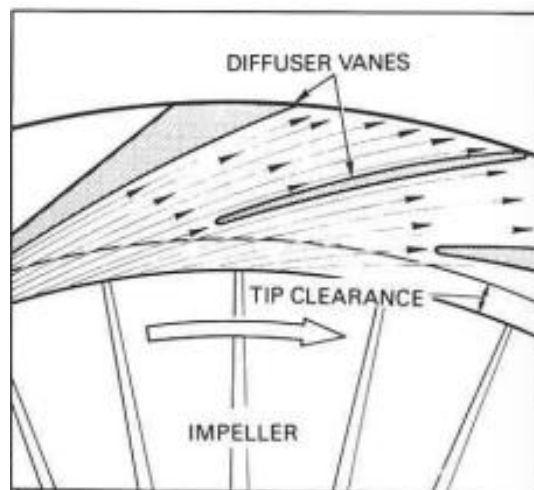
Centrifugal compressors

Centrifugal compressors, unlike axial compressors, change the direction of the incoming axial flow. The flow which is parallel to the axis of rotation leaves the compressor in a radial manner. The first part of the compression occurs due to work being done on the flow by the compressor blades, however most of the compression occurs due to the centrifugal forces that act on the fluid particles when the flow becomes radial.

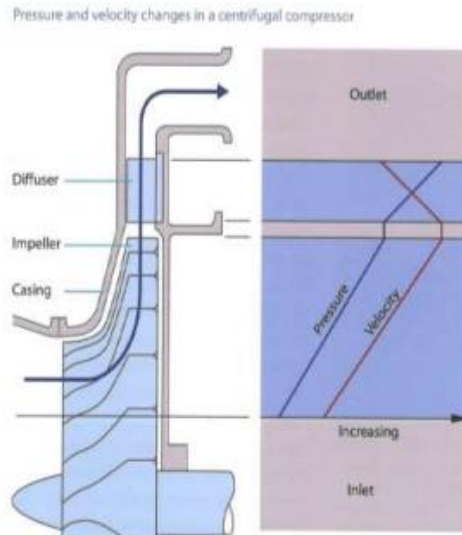


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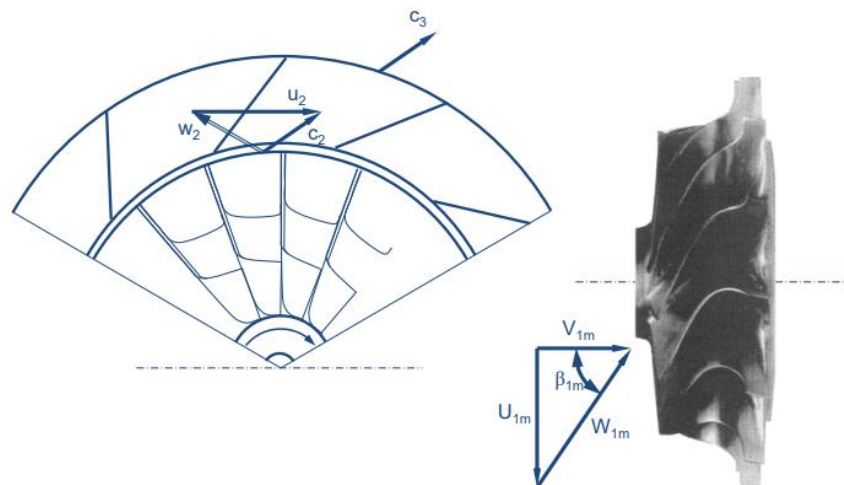
When the flow comes out of the impeller it goes into a radial diffuser which slows down the fluid and further increases pressure.



Similarly to an axial compressor, in the rotor the flow's velocity and pressure are increased thanks to the work being done on the fluid, while in the stator the velocity decreases as the pressure increases.

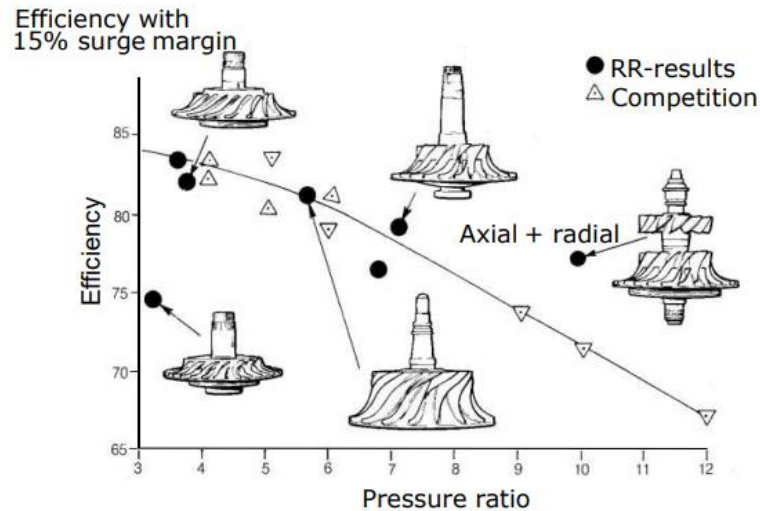


The velocity triangles are as follows:

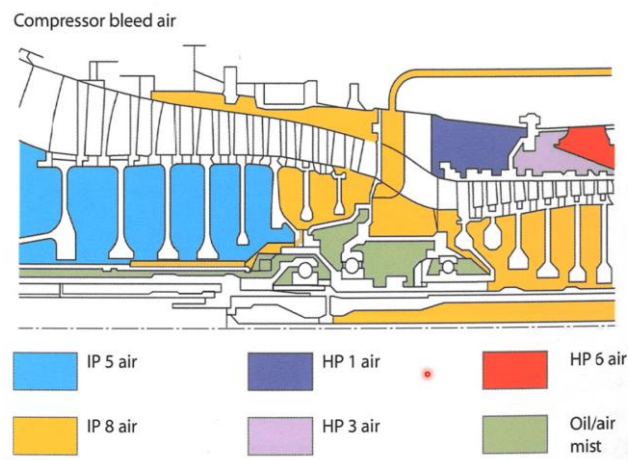


Centrifugal compressors are never used in civil aircraft engines but are sometimes used for turboshafts for helicopters.

The advantages of the centrifugal compressors is that they are compact with respect to axial compressors, however they have a smaller obtainable pressure ratio. Typical pressure ratios for modern centrifugal compressors are around 7-8 for one stage. They can be put in succession but after a while axial compressors become much more efficient from a space and weight point of view.



Note: compressor bleed air can be up to 20% of the total mass flow rate through the engine. This spilled air can be used for many function like: cabin environmental control, tip clearance control, thrust balancing, ... One of the main functions of the compressor bleed air is to pressurize the cavities in that are present in the engine. This is done so that the oil present in the shaft, which is at extremely high temperatures, does not get inside the engine and stays inside the shaft. Since it's always at a high temperature it can evaporate and the mist produced is highly inflammable. This mist is typically expelled at the back of the engine.



The bleed air taken from later stages is much more expensive from an energy POV, therefore typically we try to use lower pressure bleed.

Compressor characteristics

For any gas compressor we could express the dependence of the stagnation pressure at the compressor outlet total pressure p_{02} and adiabatic efficiency on the other important physical variables, in the following form:

$$\Delta h_s, P = f(\dot{m}, \Omega, k, R, \nu, \rho, \text{design}, D, a)$$

The gas properties significant to the compression process are specified by stating the kinematic viscosity ν , the density ρ , the specific heat ratio k , and the gas constant R that appears in the perfect

gas law. Design is the complete specification of the geometric shape of the machine and D is the characteristic size. a is the speed of sound. Δh is the variation of enthalpy and P is the output of the compressor.

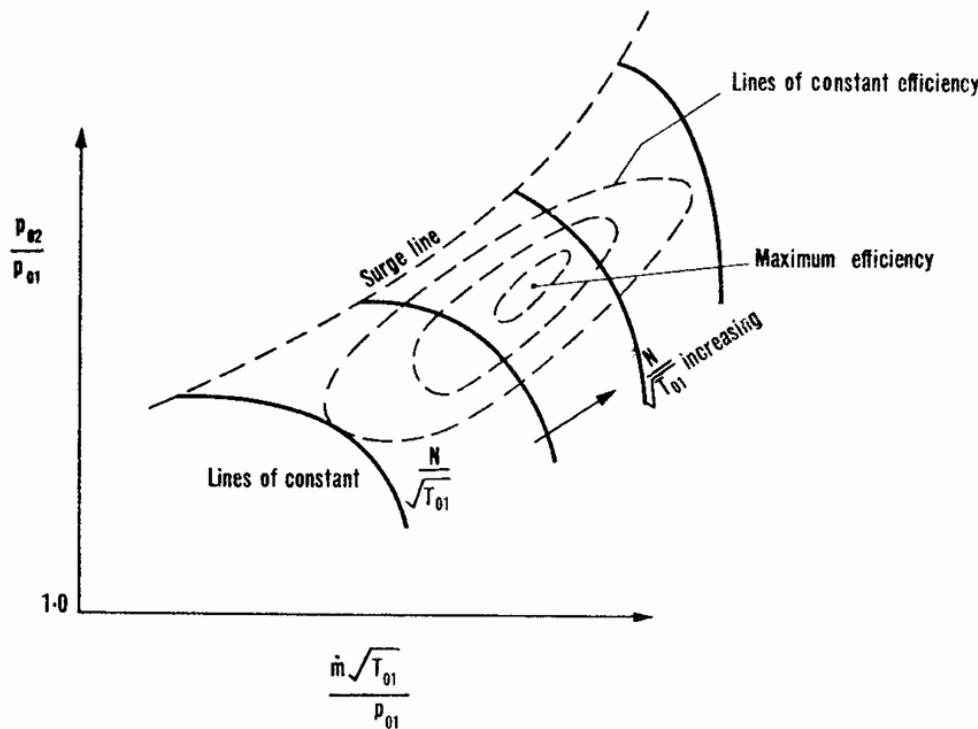
Any attempt to allow for full variations of all these quantities over the working range would involve an excessive number of experiments and make a concise presentation of the results impossible. Much of this complication may be eliminated by using the technique of dimensional analysis, by which the variables involved may be combined to form a smaller and more manageable number of dimensionless groups.

At the first step, we check that all the variables on the right-hand side of the relation are independent. At the second step, we have to reduce the statement of the physical dependence to one in which each (dimensionless) dependent variable is seen to be a function of a lower number of independent dimensionless variables.

By following these steps we get:

$$\frac{p_2}{p_1}, \frac{\Delta T}{T} = f\left(\frac{\dot{m}\sqrt{RT_{0,1}}}{D^2 p_{0,1}}, \frac{\Omega D}{\sqrt{RT_{0,1}}}\right)$$

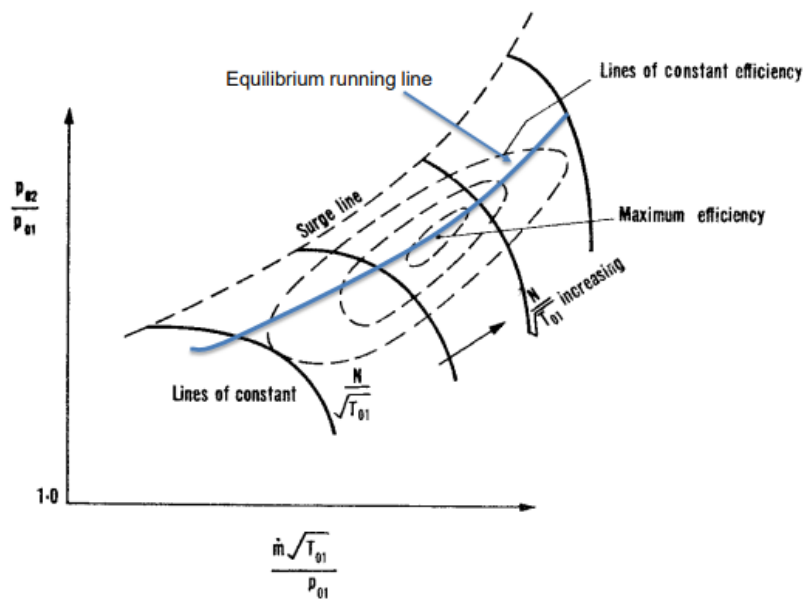
Radius and diameter can all be taken out this relations since they are constant. From this relation we can see that the pressure ratio and temperature ratio (related to the efficiency of the compressor) in the compressor can be expressed as a function of the non-dimensional mass flow rate, the non-dimensional rotation and the flow properties. Assuming constant flow properties we get that we can represent this relation in a diagram called the compressor map. This “map” represents the performance of the compressor.



The compressor characteristic shows the most important quantities that influence the performance of the compressor. These quantities are grouped in dimensionless parameter groups:

- a) $\frac{\dot{m}\sqrt{RT_{0,2}}}{P_{0,2}D^2}$; mass flow parameter group
- b) $\frac{P_{0,3}}{P_{0,2}}$; the pressure ratio
- c) $\frac{ND}{\sqrt{RT_{0,2}}}$; shaft speed parameter group
- d) η ; efficiency (isentropic or polytropic)

The design point is typically close to the maximum efficiency region. By changing the rpms of the compressor, the operating point shifts along the blue line beneath. This design operating line needs to be distant enough from the surge line. This distance is called the surge margin.



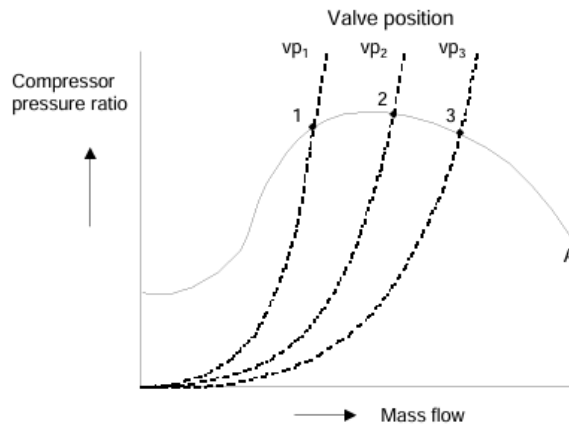
Operational limits

Surge

To explain “surge”, assume a compressor running at a fixed spool speed having an adjustable valve at the exit channel. By adjusting the position of the valve, the mass flow through and the pressure ratio over the compressor can be controlled. Plotting the pressure over the compressor as function of the mass flow through the compressor by slowly adjusting (opening) the valve positions a curve similar to the “curve A”. As the mass flow through the compressor increases, the pressure ratio reaches a maximum when the compressor is working under optimal conditions. After this the velocity of the fluid becomes too much and the pressure ratio starts decreasing again.

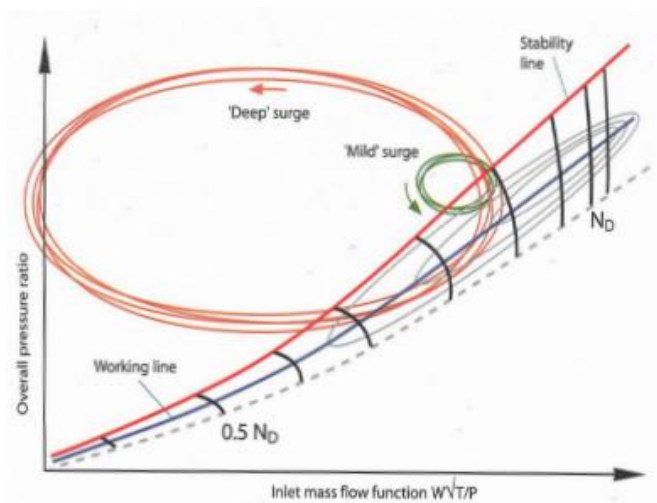
When the valve is opened slowly, the pressure ratio will increase at first since the stator vanes contribute to the pressure ratio and inflow to the blades will be better.

From a certain valve position the pressure ratio will decrease because the efficiency decreases as a result of an increasing (more negative) inflow angle of incidence.



The curve enclosed by points 2 and 3 is considered to represent stable operation of the compressor in contrast to the “unstable” curve up to point 1. The curve before point 1 is unstable because a disturbance causing the mass flow to decrease results in a pressure drop. If the downstream air pressure does not fall quickly enough, the air will tend to flow in the direction of the resulting pressure gradient (to the lower pressure part of the compressor), causing reverse flow in the compressor. If this occurs, the pressure will drop suddenly. In the mean time, the pressure downstream of the compressor also drops, facilitating the compressor to pick up again to repeat the whole cycle again. The whole cycle of events taking place at high frequency is called “surge”. The difference with the stall phenomenon is that the stall is a local disturbance of airflow with local flow reversal, while surge is characterised by a flow reversal of the entire compressor.

The position of the surge line is highly dependent on the high dynamic character of the surge phenomenon and is not only compressor dependent but also depends on the whole engine system. The surge line defines a hard lower limit for compressor operations that cannot be crossed.



We can also see that at lower rotational speeds the operating margin of the compressor reduces quite drastically and surge becomes much more likely.

Surge in compressors has to be avoided because surge causes periodically oscillating forces of high magnitude, which can set individual blades into vibration. The vibration of the blades will ultimately result in fracture (fatigue). Since the flow is reverted continuously during surge, compressor air of the end of the compressor gets to the front (which is then compressed again) causing the compressor to warm up. The warming up also contributes to the fracture in the form of thermal fatigue.

Some ways to avoid surge are:

- Bleed air from the compressor: blow-off reduces the flow resistance downstream and increases the airflow in front of the blow-off.

- Variable stator vanes: The front stages are fitted with variable stator vanes to provide optimal angles of airflow incidence for low spool speeds.
- Splitting up compressor to have different rotating speeds

Stall

Airflow separation occurs when the angle of incidence of the inflow in blades becomes too large for the airflow to follow the blade profile. Separation of flow at the convex part of the blade is referred to as “stall”.

When a disturbance in the flow or geometry is observed stall can occur. Let's observe for example channel B, in which flow breakdown results in deflection of the inflow incidence at channel A and C. Channel A receives the fluid at a reduced angle of incidence (positive direction), while channel C receives the fluid at an enlarged angle of incidence (negative direction). The reduced angle of inflow at channel A will result in a breakdown of flow in that channel. The flow to channel B will now be more favourable due to the deflection around the blockage in channel A. It appears that the blockage (“stall”) has moved opposite to the direction of rotation. This phenomenon is called “rotating stall”.

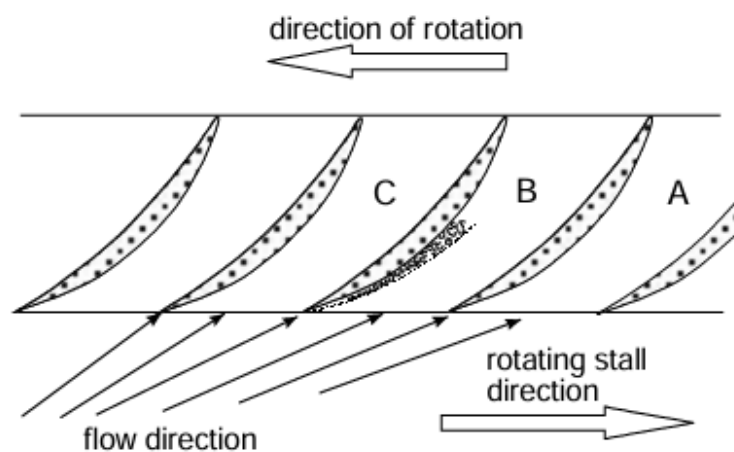
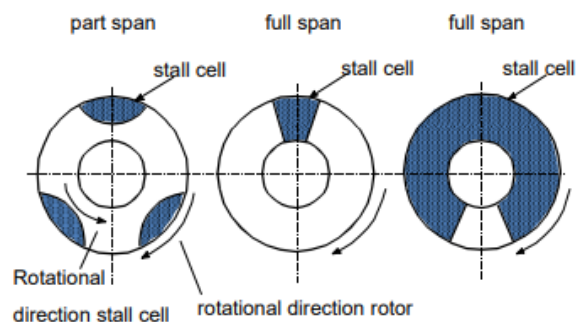
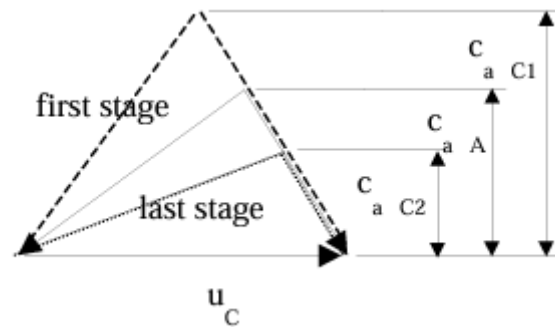


Figure 8-4 - Rotating stall

“Rotating stall” appearing at the blade tips (since the speeds are at maximum), often is referred to as “part span stall”. In case the flow worsens the stall areas will expand to grow to the length of the blade, which is called “full span stall”.



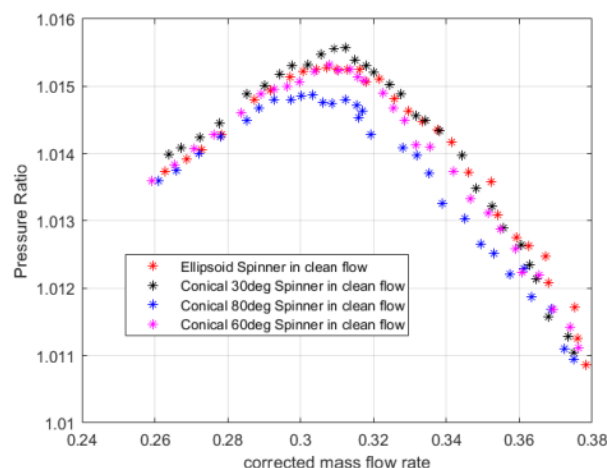
Stall can occur most likely at low rotor speeds because in these conditions we have the following velocity triangles:



When the rotor velocity is low, the pressure ratio is lower than what it typically is, which means that the density of the fluid at the final stages is much lower than usual. For the same mass flow rate, this means that the last stages see an axial velocity which is much higher than normal and this can decrease the AoA of the blades that negative stall can occur. The opposite occurs in the first few stages, leading to positive stall.

The solution for these issues are the same as the ones for the surge.

Note: one important element that determines the inflow conditions for the compressors is the shape of the spinner of the fan and the amount of distortion present at the fan face. We see that typically the spinner has a conical or elliptical shape, since these shapes help reduce the amount of distortion. Also we see that they have very similar performances in clean flow.



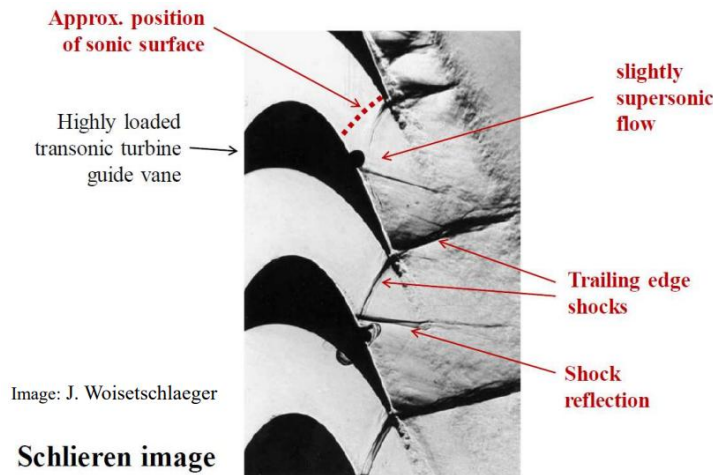
Lecture 11

Axial turbine

The axial turbine has a very similar architecture to the axial compressor. In this case however, it is the fluid that does work to move the turbine blades. This work is then used to run the compressor.

A typical turbine stage is made up:

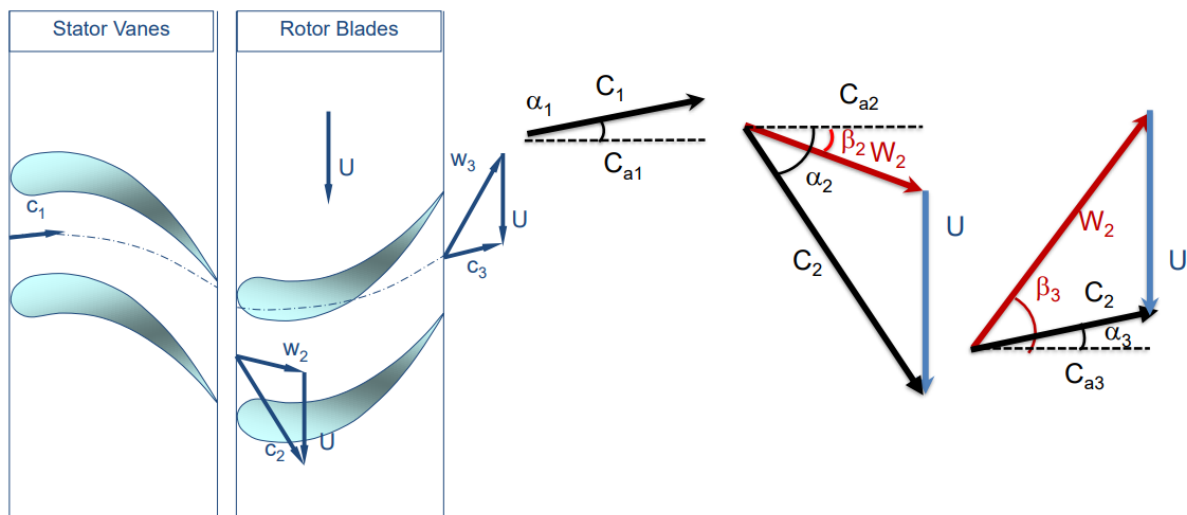
- Stator: fluid expands and accelerates. Pressure decreases while velocity increases. Instead of the compressor stators, which act as diffusers, turbine stators act as nozzles. This is why they are sometimes called nozzle guide vanes. The channels within the stator are converging, accelerating the flow, usually to sonic and in some cases, supersonic velocities. This occurs because the flow in the turbine is already in the transonic regime and after the expansion it can reach sonic conditions, leading to the possible formation of shocks.



- Rotors: the fluid moves the turbine blades, so its velocity is decreased.

The flow inside the turbine is generally less complex than the flow in a compressor, since in this case the flow is expanding, so a favourable pressure gradient is present. This pushes the air through the turbine, unlike the adverse pressure gradient that is present in the compressor stages.

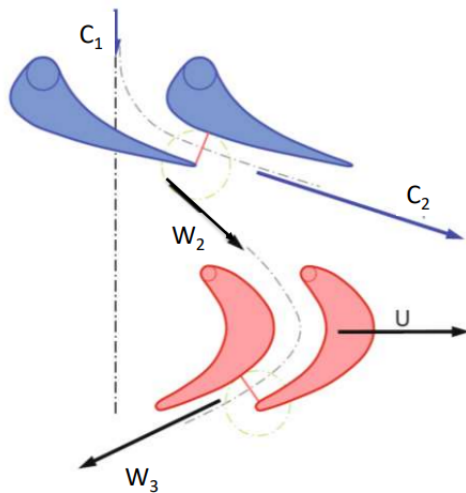
The velocity triangles of a turbine stage are as follows:



Since in this case the rotor is being “pushed” by the air and not the other way around, the relative velocity it sees is much lower. This is the main reason why the turbine blades are much blunter than the compressor blades, since stall is less of an issue. Also the airfoils can be thicker, allowing for the cooling channels that bring the compressor bleed air to form a thermal protective film around the turbine blades.

Just like for the compressor, since we typically want repeating stages, the exiting velocity of one stage has to be the same as the entering velocity.

Finally we have to remember that the axial component of the absolute velocity is constant throughout and this is ensured by having an increasing cross section.



- Airfoil shapes are determined initially from flow inlet and exit angles, which essentially fix the camber line.
- For a vane, the inlet metal angle is obtained from the absolute inlet flow angle; the exit metal angle follows from the turning needed.
- For a rotor blade, the inlet metal angle matches the relative inlet flow angle closely; the exit metal angle aligns with the relative exit flow angle, which is determined by the turning angle needed to provide the power

The variation of total pressure, static pressure and absolute velocity we can see that:

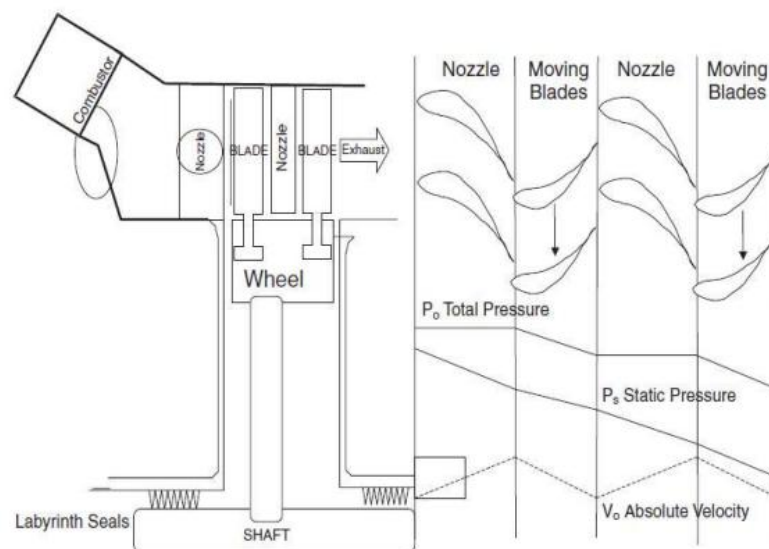
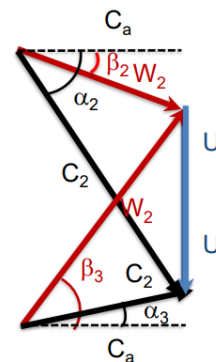


Figure 9-10. Schematic of a reaction-type turbine showing the distribution of the thermodynamic and fluid mechanic properties.

The total pressure decreases slightly in the stator due to losses and then decrease in the rotor due to the extraction of work, the static pressure decreases overall (the amount by which it decreases in each section is determined by the degree of reaction) and the velocity increases in the stator and decreases in the rotor.

Let's now observe the T-S diagram of a turbine stage:



Other important turbine parameters:

■ Total temperature drop per stage: $T_{0,1} - T_{0,3} = T_{0,2} - T_{0,3} = \frac{W_s}{c_p}$

■ Pressure ratio per turbine stage:
$$\Pi_{urb} = \frac{p_{0,1}}{p_{0,3}} = \left(1 - \frac{1}{\eta_{is}} \frac{\Delta T_{0,1-3}}{T_{0,1}} \right)^{\frac{k}{k-1}} =$$

$$= \left(1 - \frac{UC_a (\tan \alpha_2 + \tan \alpha_3)}{\eta_{is} c_p T_{0,1}} \right)^{\frac{k}{k-1}}$$

The first relation can be simplified only because the total temperature does not change in the stator. The total temperature drop can then be used, together with the isentropic efficiency term to determine the pressure ratio per turbine stage.

To study turbine performance two important parameters need to be defined:

- Flow coefficient: ratio between the axial velocity of the flow and its tangential velocity. In a way it's a measure of the amount of flow that can pass through the turbine stage.

$$\phi = \frac{C_a}{U}$$

- Loading factor: ratio between the enthalpy drop for a stage and the square of the tangential velocity. It defines the “skew” of the velocity triangles.

$$\psi = \frac{\Delta h}{U^2}$$

$$\psi = \frac{(C_{U2} \cdot U_2 - C_{U3} \cdot U_3)}{U^2}$$

$$\psi = \frac{(C_{U2} - C_{U3})}{U}$$

$$\psi = \frac{C_a (\tan \alpha_2 - \tan \alpha_3)}{U}$$

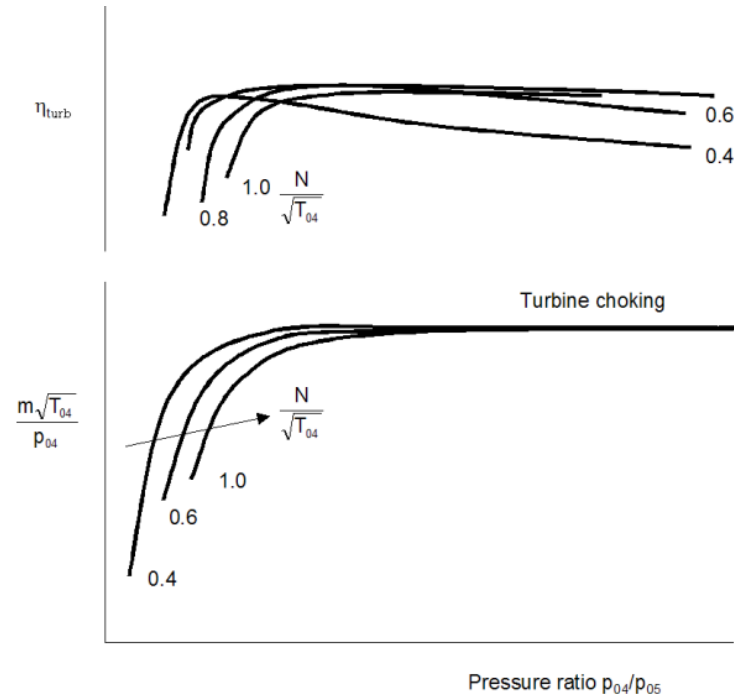
- Degree of reaction: same definition as for the compressor. For an axial turbine a relation can be derived to evaluate it from the velocity triangles:

$$A = \frac{C_a}{2u} (\tan \beta_3 - \tan \beta_2) = 1 - \frac{C_a}{2u} (\tan \alpha_2 - \tan \alpha_3)$$

Typical aero-engine turbines have a degree of reaction of around 0.5.

Just like for the compressor, the turbine performance can be evaluated from dimensionless parameters. In particular, the corrected mass flow rate and the pressure ratio are used and are represented as a function of the corrected rotational speed.

The turbine map is:



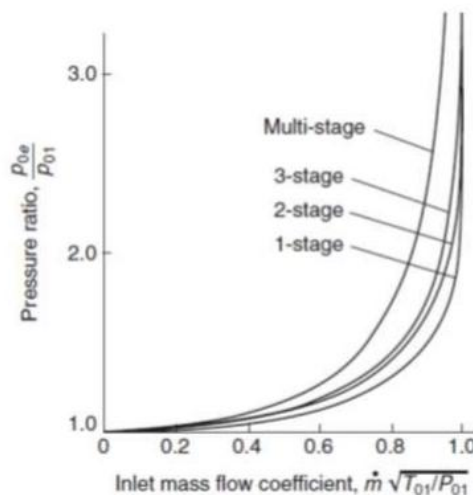
Differently from the compressor however, the performance characteristics of a turbine are determined by the choking of the stator vanes. As the pressure ratio increases, the corrected mass flow rate increases until the turbine becomes choked and the mass flow rate plateaus. As the rotation of the turbine increases, more work is being extracted by the rotors and a higher pressure ratio is needed for the mass flow to become choked.

Since the turbine is operating in a favourable pressure gradient we also see that its efficiency is not as dependent on the operating conditions as the efficiency of the compressor.

Since the turbine and the compressor are connected to each other and have drastically different performance characteristics, a compromise has to take place, while keeping in mind that the rotational speed and the corrected mass flow rate are the same for both. Also the power balance has to be taken into account.

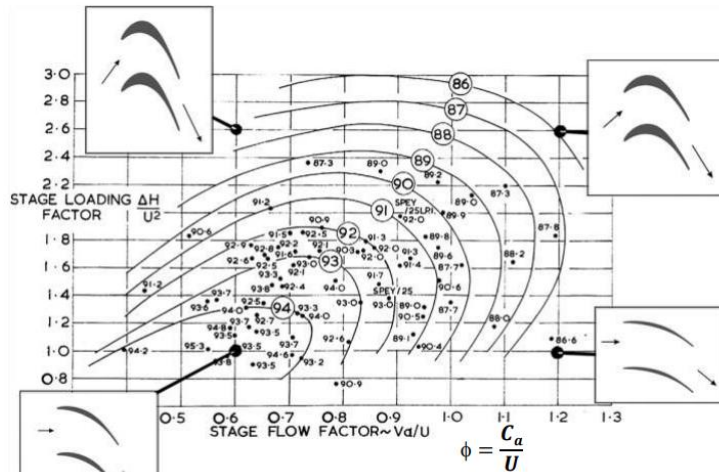
These three constraints determine the operating point of the turbomachinery.

As we increase the number of stages in a turbine increases we see that the corrected mass flow rate tends to decrease, since it is determined by the stage that can handle the lowest mass flow rate. This also applies in the choked condition.



This is not really relevant for the high pressure turbine, since it typically has 1-2 stages, however it is relevant for the low pressure turbine since it usually is a multistage turbine.

One way to determine the performance of turbines is to look at the smith chart of the turbine, which represents the stage efficiency as a function of loading factor and flow factor:



- Reduction in U increases stage loading and flow coefficient, thereby reducing the stage efficiency
- Or for same stage loading, increases the number of stages in the turbine

A lower RPM increases the stage loading, while a higher RPM reduces it. For a high pressure turbine the RPM is very high so it has no issues with having high efficiency. However, the low pressure turbine typically has to rotate at a lower speed since it is also connected to the fan, which cannot have high rotational speed due to tip shocks.

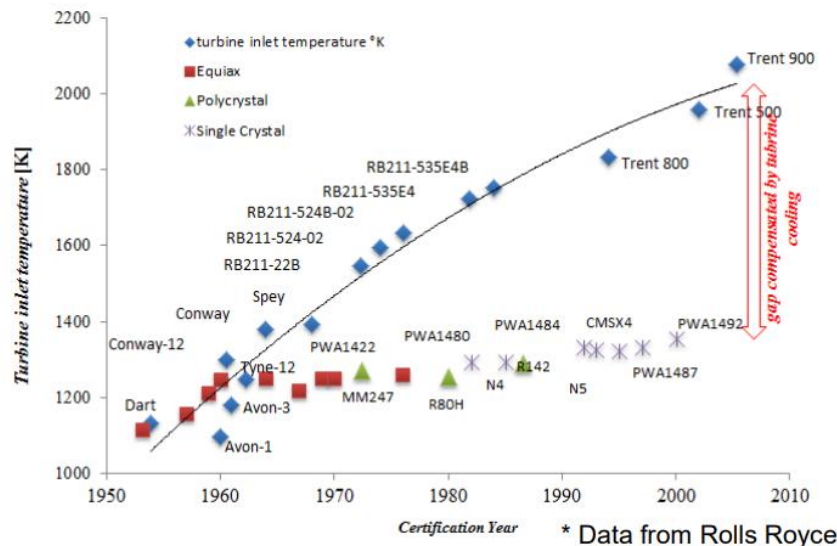
From the chart we can observe that a high loading factor corresponds to the flow being “turned” a lot, therefore extracting a lot of work per each stage while also introducing losses. If the flow is turned too much losses occur due to the presence of shocks. On the other hand a high flow factor corresponds to high passage area.

High stage loading means that the number of stages needed can be reduced, thus reducing the total weight of the engine. However this does come at the disadvantage of lower turbine efficiency. In aero-engines, since the weight is one of the most critical design parameters, blades that turn the flow a lot are typically used to have higher loading factors.

The reason why the GTF is such an efficient engine is that the tangential velocity of the low pressure turbine can be much higher, since it is basically decoupled from the fan rotational speed. This allows to have much higher turbine efficiencies even with high loading factors.

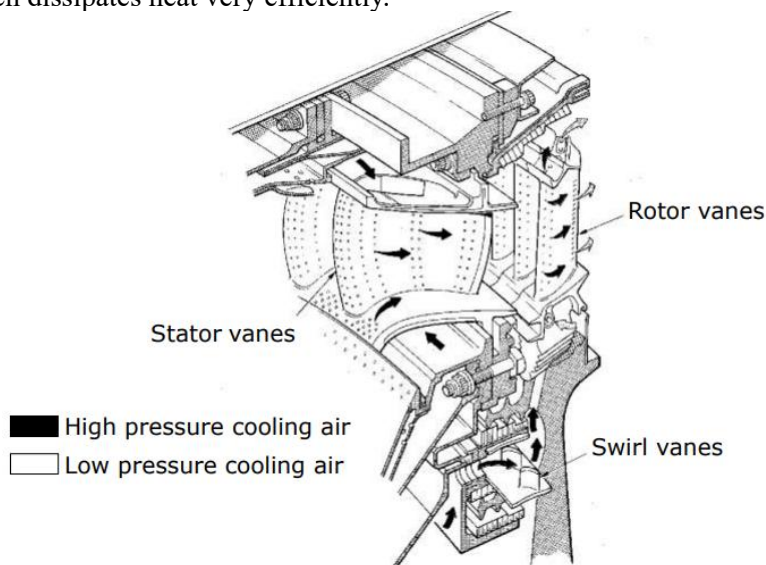
Turbine cooling

The maximum temperature that is allowed inside the engine is determined by the high pressure turbine blades, that receive the hot gasses from the combustion chamber at the TIT. Since we want to have the highest temperature possible to increase the thermodynamic efficiency as much as we can, the turbine blades are built to withstand incredibly high temperatures. From the chart below we actually see that the TIT is typically much higher than the melting point of the material out of which the turbine blades are made out of.



Advancements in material technology have made possible the manufacturing of turbine blades that have a single crystal structure without any grain boundaries. This allows to eliminate any imperfections in the molecular structure of the material and greatly decreases the effects of high temperature creep which can lead to failure at high temperatures and high stresses (centrifugal forces).

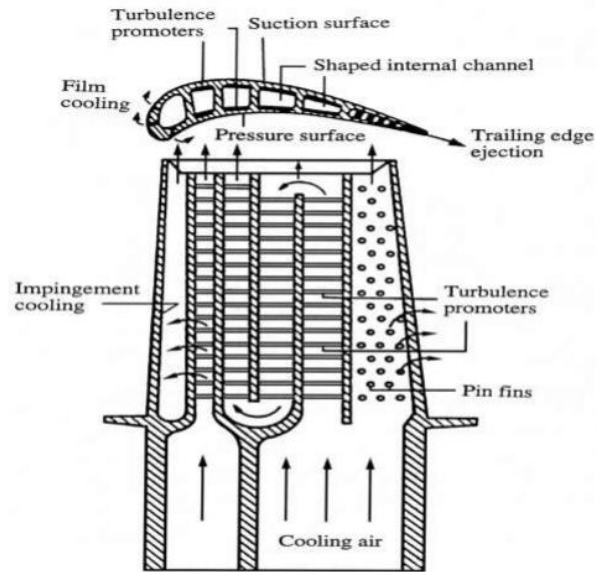
The main reason why the discrepancy between the melting point of the material and the operating temperature is possible is cooling, which is achieved by supplying compressor bleed air to the turbine blades. This air is released from tiny holes on the surface of the turbine blades and forms a protective thermal film which dissipates heat very efficiently.



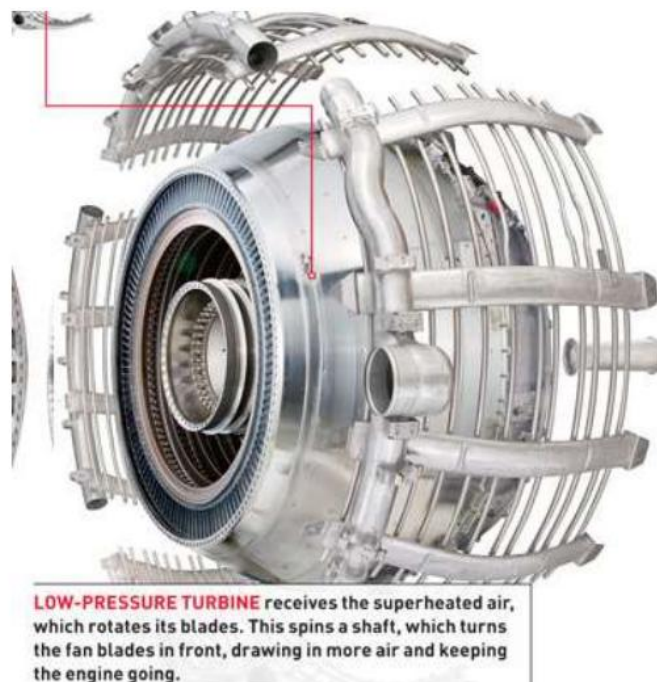
One element present in some turbine blades is the shroud, which interlocks with the shroud of different turbine blades forming a platform which reduced tip losses. It is not always used since it does increase the centrifugal force that acts of the blades, effectively reducing their lifespan.

The most critical region of the blade that needs to be cooled is the leading edge, since that is where the stagnation point of the incoming flow is and where the temperature reaches the maximum. To cool this part of the blade impingement cooling is used, where a channel dedicated to the leading edge only blows air on the back of the leading edge to promote heat dissipation.

Note: the channels inside the blades that transport bleed air have all turbulence promoters to create turbulence and enhance heat transfer.



Jet impingement is also used in the active clearance control system, which is made up of small ducts in which compressor bleed air is run through. These ducts blow air on the casing of the turbine blades and can either expand the casing, allowing for more tip clearance when the engine is running at cruise (high temperatures and high centrifugal forces) or contract the casing, allowing for less tip clearance when the engine is running idle for example. The heating/cooling of the casing occurs through jet impingement.



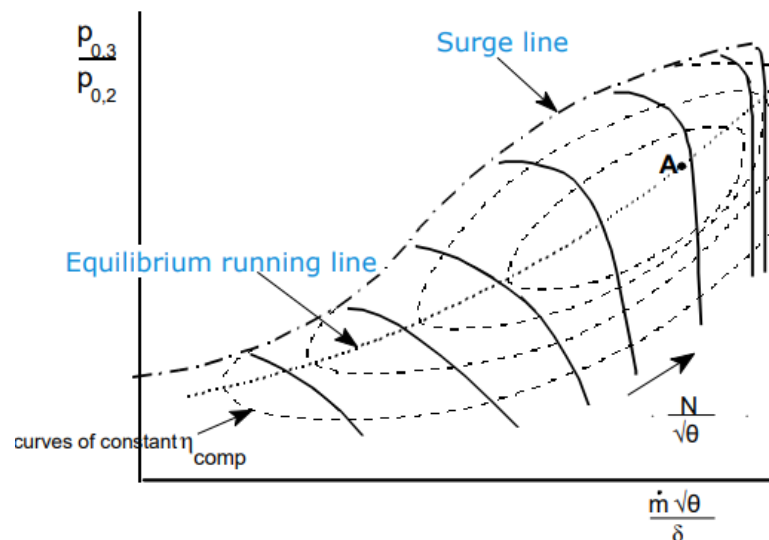
Lecture 12

The performance of a gas turbine aero engine is determined by:

- Components characteristics: performance is determined by all the components we have seen so far as well as their limits (e.g. stall, temperature, ...)
- Engine characteristics: the operating conditions obviously determine the engine's performance.
- Ambient conditions

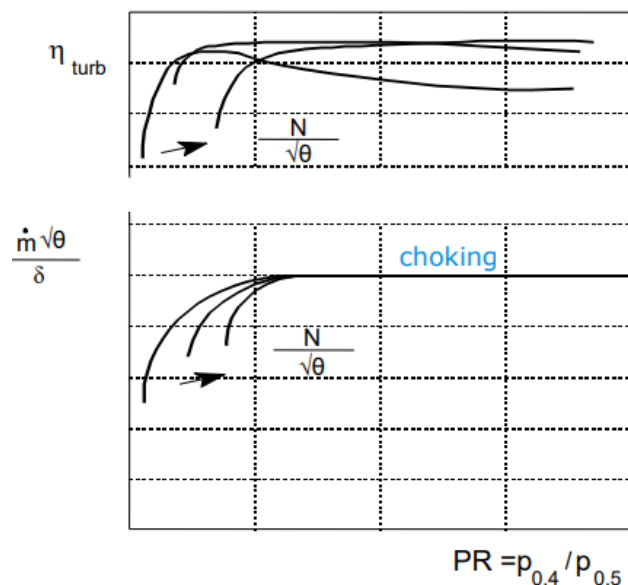
The engine operating condition is determined by the superimposing of the operational envelopes of all the components that make up the engine. The FADEC (Full Authority Digital Engine Control) keeps the engine within its operational limits. It can do this by controlling different variables and systems like: the fuel flow, variable stator vanes, variable bleed valves, active clearance control, nozzle geometry, after burner fuel flow, thrust vectoring, ...

Let's start with the compressor. As previously seen, the compressor's characteristics are represented by the compressor map.



The operating conditions of the compressor are typically represented along an equilibrium line. However, this is an ideal representation, which is valid only if the compressor is changing speed very slowly without strong accelerations.

The operating conditions of a turbine is determined from the turbine map.



The turbine characteristics are less complex than that of the compressor since the turbine becomes choked after a certain pressure ratio, independently of the speed. During normal operations, the turbine is almost always choked. Moreover, the efficiency of the turbine does not change much once it becomes choked.

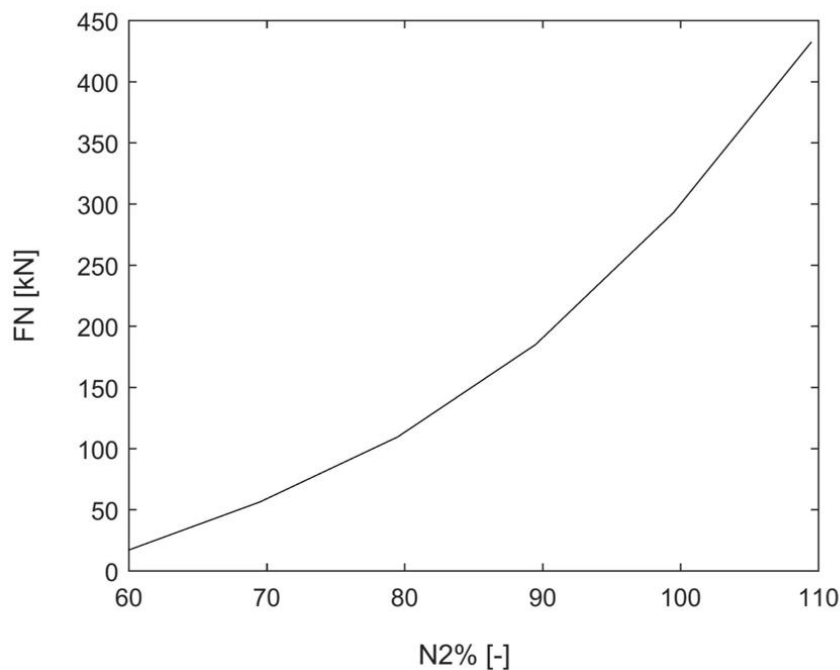
Other components such as the ducts and inlets have much simpler characteristics. In this type of component, the only thing that matters is the pressure loss, which is a function of the corrected flow or the Mach number.

For all components heat loss effects are also taken into account through sophisticated models to evaluate their characteristic performance.

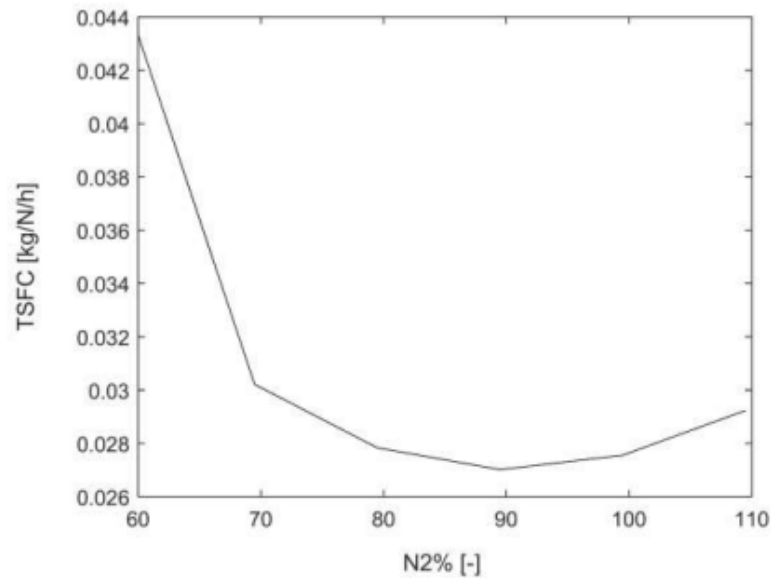
The characteristic performance of a component can be adapted to the operating condition mainly by adapting the geometry of the component. For example in a compressor it is possible to avoid the surge line when changing the operational point of the compressor by implementing variable stator vanes (adapt the angle of attack of both the stators and the rotors avoiding stall and surge) and variable bleed valves (opening these valves reduces the back pressure and allows for an increase in mass flow rate, moving away from the surge line)

Although stall is not very common in turbines (since there is a favourable pressure gradient), the same thing can be applied to turbine through variable stator vanes. This has been tried in the past, leading to improved SFC but it is not implemented widely.

Let's now see how the performance of the whole engine (thrust and fuel consumption) change with different component operating conditions. For the compressor, an increase in the speed directly correlates to an increase in thrust.

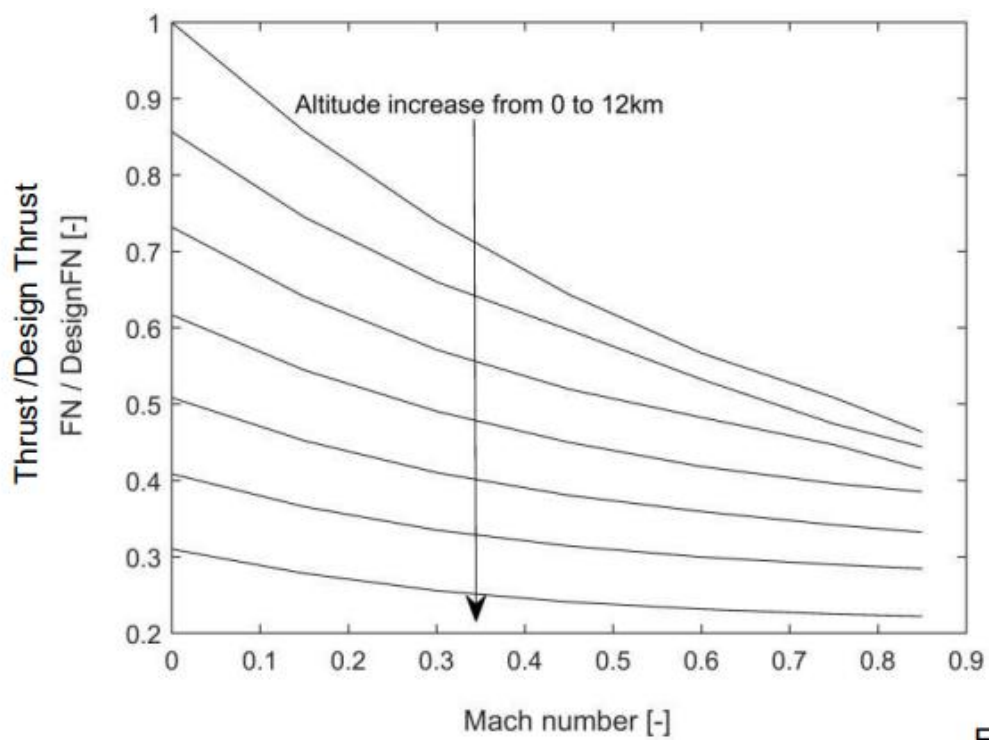


This is due to the fact that an increase in spool speed is directly correlated to an increase in pressure ratio and therefore energy available (i.e. thrust). For the TSFC we also see improvements as the spool speed increases at first, however after a certain threshold the trend changes.



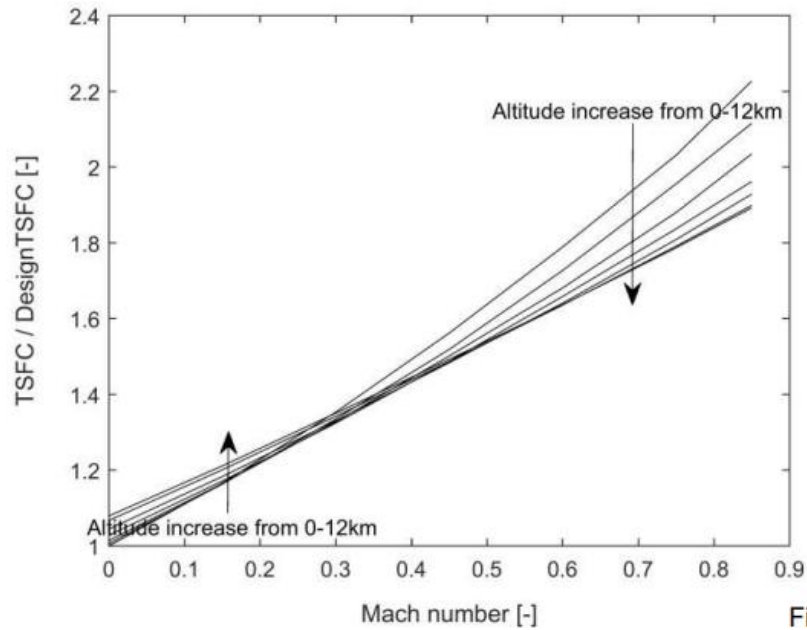
The cruise spool speed is determined by this graph and is set at around 80-90%.

With respect to Mach number and altitude:



As we can see, when the altitude increases the thrust produced also reduces w.r.t. the design thrust. This is mainly due to the reduction in air density and therefore mass flow rate. The trend with increasing Mach number is the same. This is instead due to the increase in ram drag. From the thrust equation we know that the thrust is directly proportional to the difference in jet velocity and flight velocity. As the flight velocity increases it becomes harder and harder to provide the same velocity increase to the exhaust gasses.

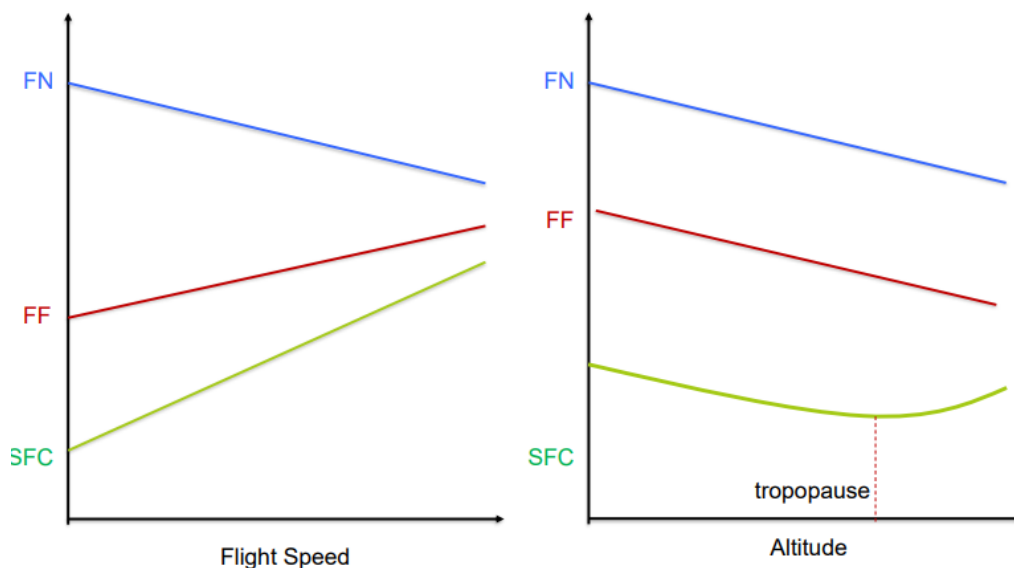
For the TSFC:



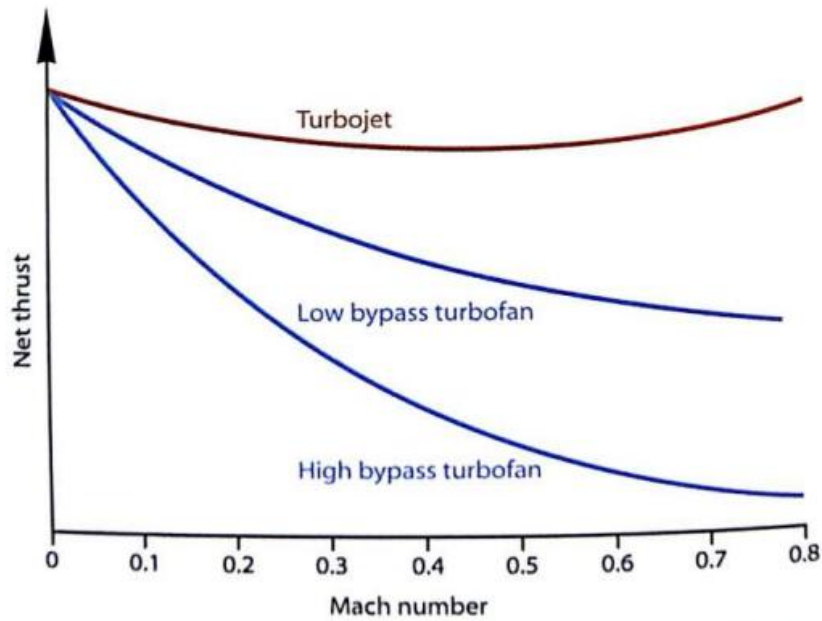
The behaviour of the thrust specific fuel consumption is very different from the thrust. The TSFC increases with Mach number always since it becomes harder to provide the same exhaust velocity increase w.r.t. the flight speed at higher Mach.

However, it decreases with altitude until the tropopause is reached (i.e. the external temperature stays roughly constant). This decrease in external temperature corresponds to an increase in the temperature ratio between the turbine inlet temperature and the ambient temperature. From a thermodynamic point of view this is very beneficial for the efficiency, which leads to a decrease of TSFC.

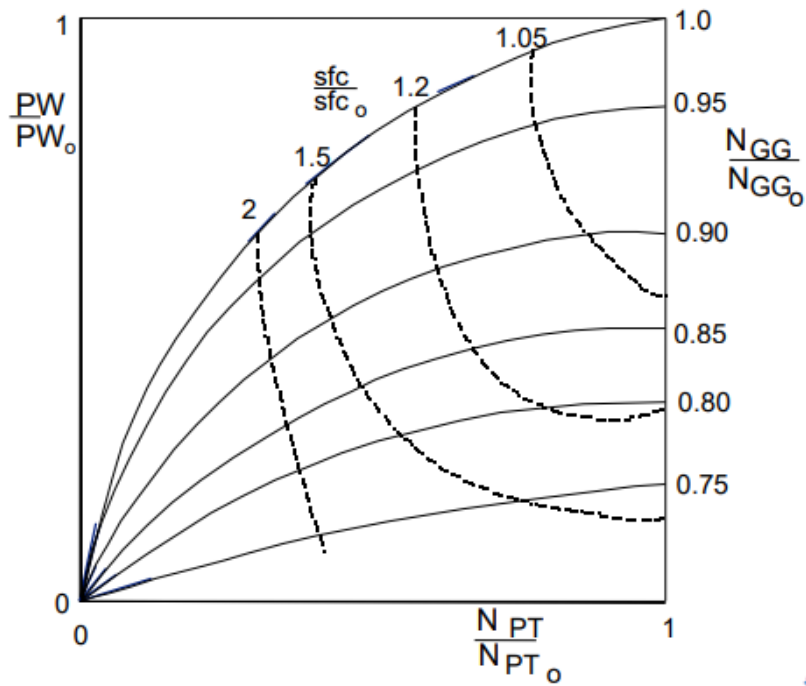
In summary:



To evaluate engine performance w.r.t. ambient conditions we can define the thrust lapse rate. This parameter quantifies the reduction in thrust of an engine w.r.t. to its design value at sea level and at $M = 0$. It can be defined w.r.t. Mach number or altitude:

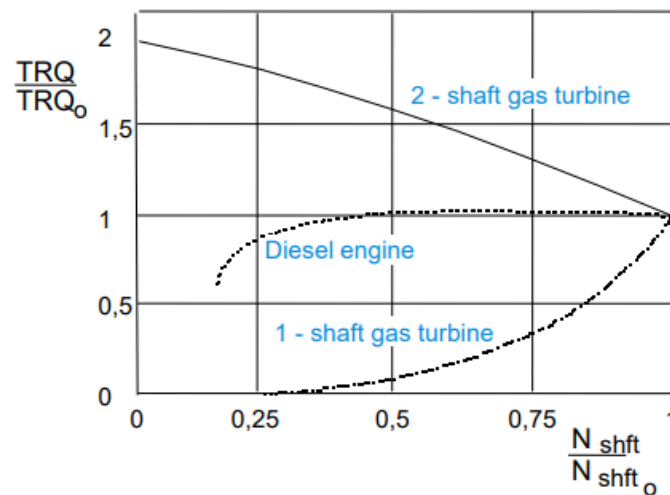


For a turbofan the thrust reduces with Mach number, while for a turbojet it first reduces and then increases again (after a certain point all the compression needed comes from ram compression). Let's now consider the characteristics of the whole engine for a 2 shaft power turbine:

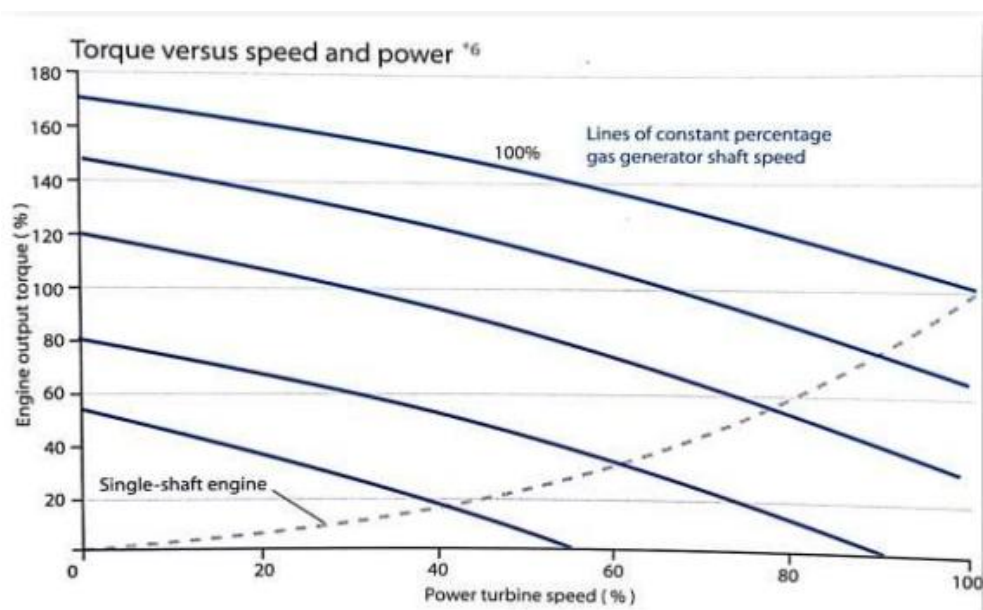


Where N_{PT} is the speed of the power turbine that drives the load (the propeller) and N_{GG} is the speed of the gas generator (other turbine and compressor). We can see that at low speeds for the power turbine the specific fuel consumption is quite high and it gets much better at higher N_{PT} values. Same goes for the N_{GG} .

Another advantage of having a two shaft turboshaft is that it's torque characteristics (the amount of torque needed to spin the load) is much better than a one shaft turboshaft, for which the propeller is mounted on the same shaft as the gas generator.



The gas turbine shaft can generate power (or torque) when it's running at high speeds to drive the loaded shaft even if the power turbine is not running. This is the great advantage of two spool turboshafts. For one spool turboshafts starting up the power turbine takes a great amount of torque, which the turboshaft struggles to generate at low speeds.



Transient operations

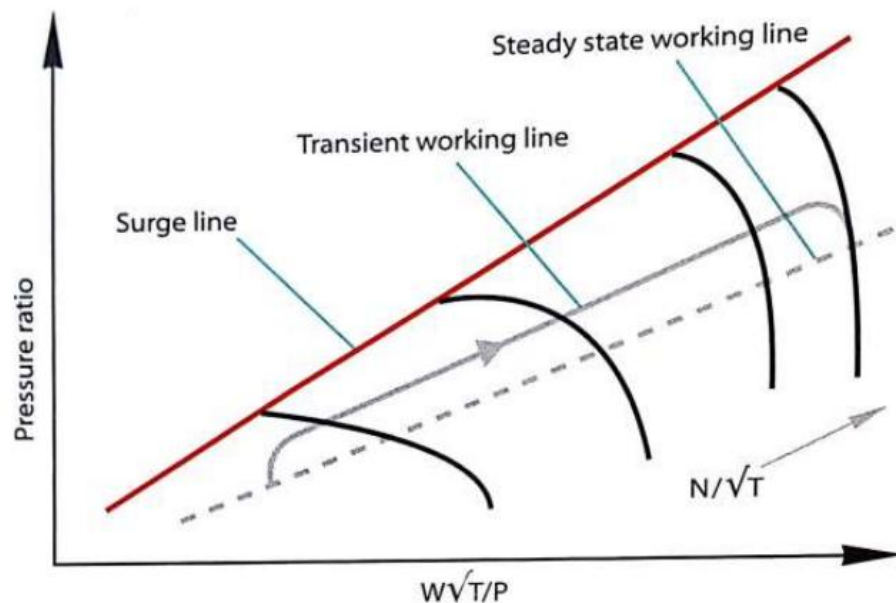
Although all the ideal operational conditions lie on the engine equilibrium line (steady-state) in reality during normal operations the transition between one operating state and another does not occur slowly enough so that all states in between are steady states. In reality the operating line moves closer/ away from the surge line depending on the change of speed.

Each spool obeys a rotational balance, where the change in rotational speed is proportional to the difference in power extracted by the turbine and the power consumed by the compressor. However, spool speed cannot change instantaneously, while the turbine power changes quite rapidly.

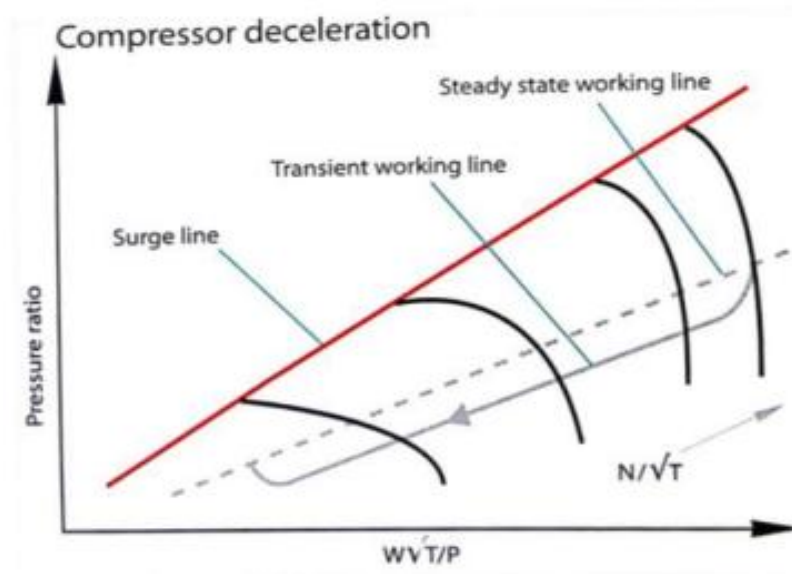
To increase the thrust generated by the engine, the fuel flow rate is increased, the TIT increases and therefore the turbine speed increases. This means that the compressor starts working at a higher speed, while the mass flow rate does not immediately increase as well and lags behind the speed increase.

This means that during the transient the operating point of the compressor moves up (higher pressure ratio due to higher rotational speed) but at the same mass flow rate.

This is the acceleration line, which occurs when there is overfueling (more fuel flow than needed to maintain the current steady state):

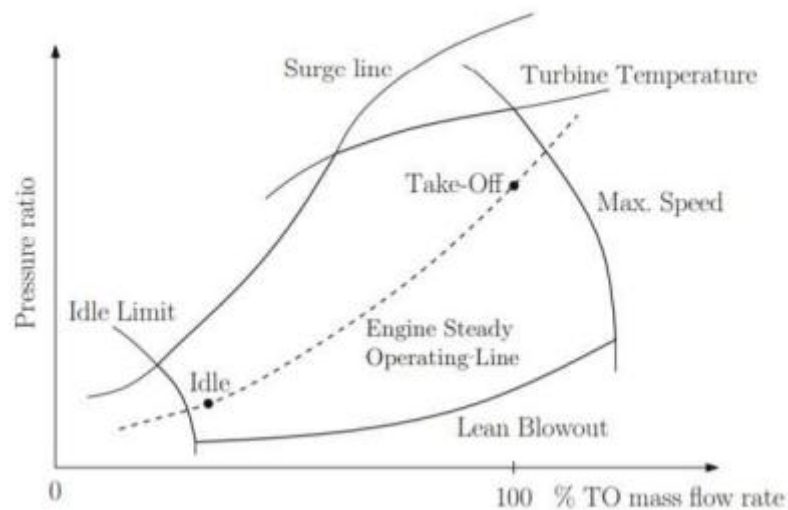


The exact opposite occurs when the thrust needs to be reduced. In this case the mass flow rate in the compressor lags behind the reduction in pressure ratio due to the reduction of spool speed. This is the deceleration line, which occurs when there is underfueling (less fuel flow is provided than needed to maintain the current steady state).



Aero engine limitations

In the chart below the main aero engine operational limitations are represented. The ideal operational conditions all lie on the engine equilibrium line between the take off condition (maximum mass flow rate for max thrust) and idle (minimum mass flow rate).



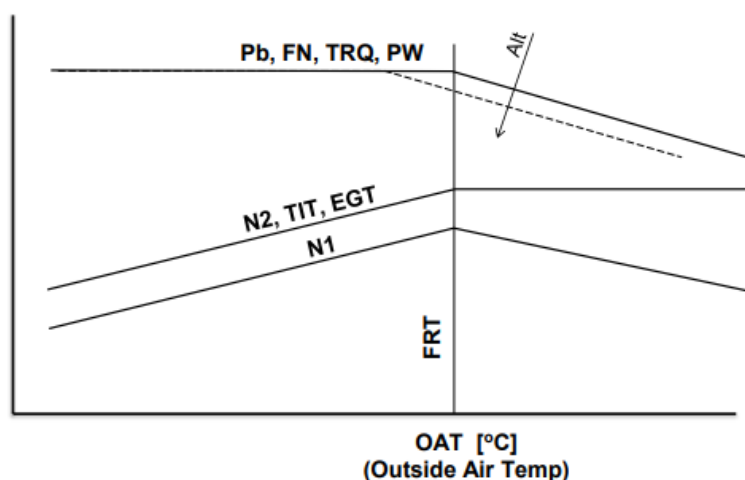
These limits are taken into account by the FADEC, especially during transient operations. The increase of fuel flow needs to be limited to avoid surge or surpassing the maximum turbine inlet temperature (for lower mass flow rates surge is the limiting factor, since stall is more likely to occur due to higher incidence angles. For higher mass flow rates the turbine inlet temperature becomes the limiting factor since an increase in fuel flow corresponds to an increase in the final combustion temperature).

The decrease of fuel flow needs to be limited to avoid lean blowout in the combustor.

The idle limit is defined as the minimum speed required for the core gas generator to sustain itself, while the max speed limited is imposed by the centrifugal forces acting on the disks of the turbines.

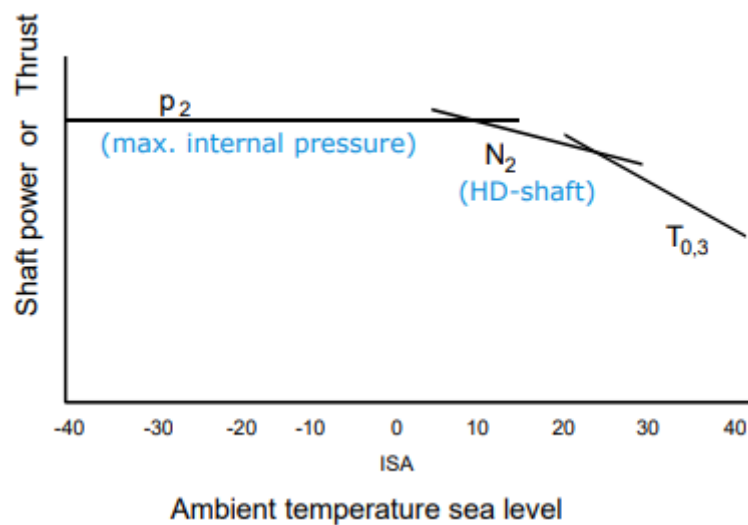
Another way to determine the performance of an engine is to evaluate the flat rating, or the outside air temperature after which the thrust generated by the engine reduces.

We say that an engine is flat rated if it delivers constant rated thrust (or shaft power) over a range of ambient temperatures and pressures, up to a specified “flat-rating temperature”, and becomes thermally limited only above that point.



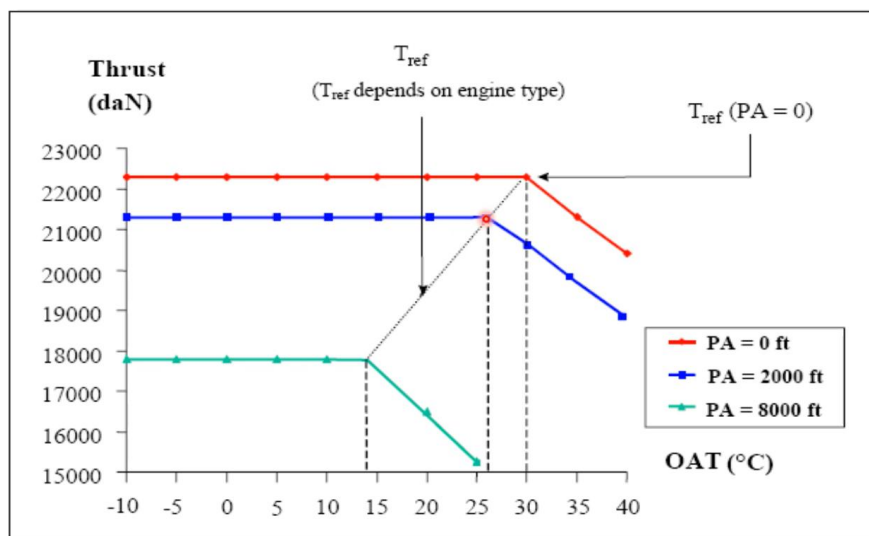
For lower temperatures the engine performance improves: the same thrust can be achieved with lower angular speeds and lower TIT. In this region it is the maximum pressure ratio that limits the maximum thrust that can be achieved. Nevertheless the maximum thrust can be obtained by increasing the TIT and the spool speeds.

For higher temperatures the engine performance decays: to obtain a lower thrust the TIT and power turbine speed still need to increase. In this region the engine becomes temperature limited, meaning that the maximum thrust is now limited by the maximum TIT (or the maximum spool speed).



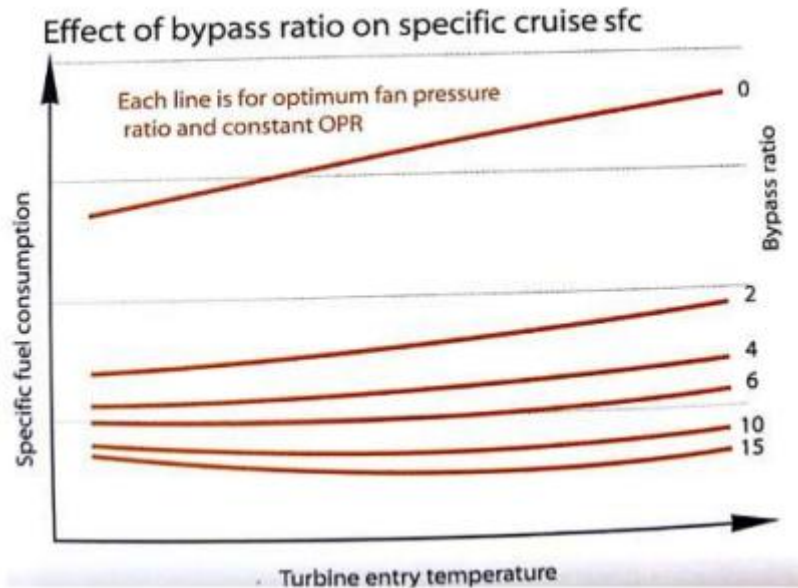
Under the flat rating of the engine a lower TIT than the maximum TIT that the turbine can withstand is needed to obtain the temperature ratio required for max thrust. However, after that rating, the temperature ratio becomes limited by the TIT,max.

The flat rating temperature reduces with altitude since the density of the incoming air is also reducing.

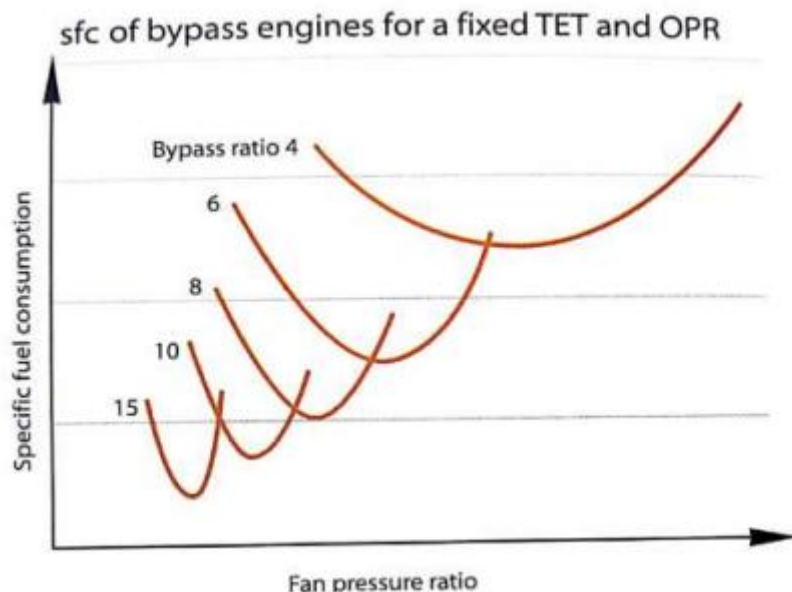


Effect of design parameters on performance

For a turbofan the 4 main design parameters are the ByPass Ratio BPR, the Fan Pressure Ratio FPR, the Overall Pressure Ratio OPR and the Turbine Inlet Temperature TIT. Let's now look at their impact on aero engine performance.



As we can see from the figure above, the fuel consumption decreases with increasing BPR. Additionally the dependence of the fuel consumption on the TET decreases with increasing BPR.

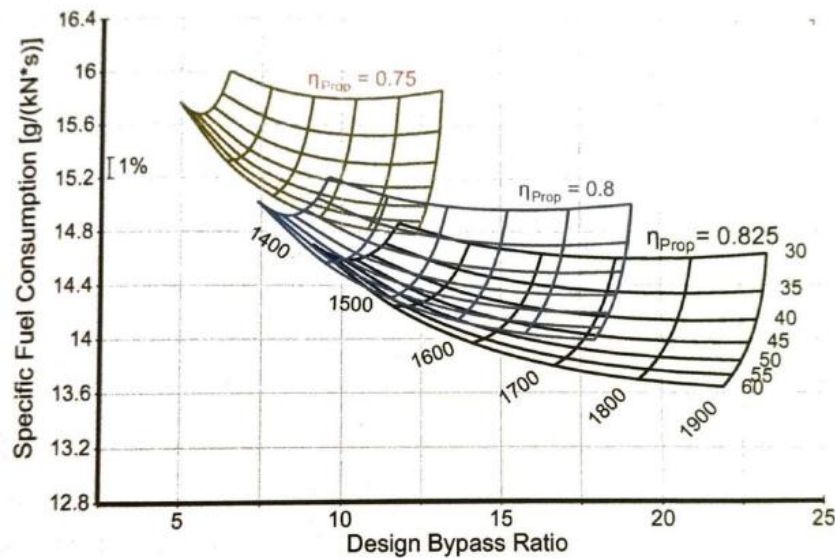


For any given BPR there is an optimum FPR which minimizes the fuel consumption, and this optimum decreases with increasing BPR. This is due to the fact that with increasing BPR more and more thrust is generated by the bypass flow, whose velocity increase is mainly dictated by the FPR. If the BPR increases there will be more mass flow rate and therefore to produce the same amount of thrust, a lower pressure increase is required.

There is a caveat however; when the FPR reduces a lot (typically to a value lower than 1.4) the bypass nozzle might not be choked in the cruise condition.

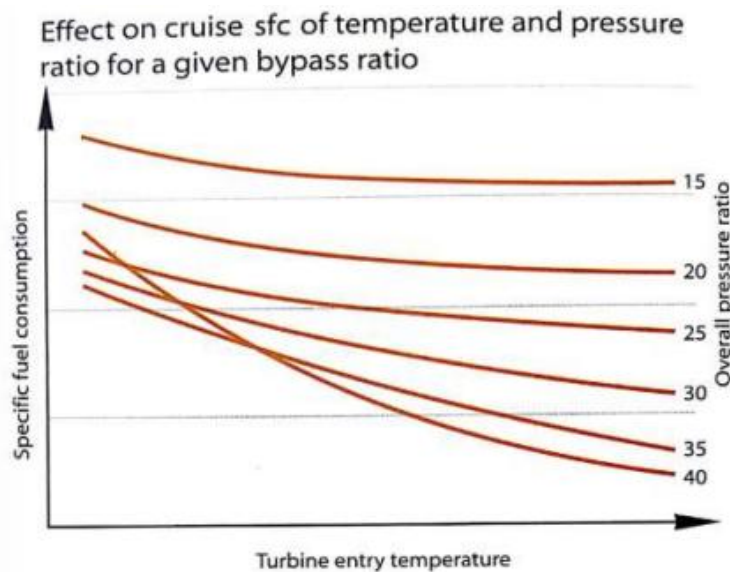
To avoid surge and stall, especially during take off variable blades for the fan or variable geometry for the nozzle are required.

If we also take into account the propulsive efficiency:



The propulsive efficiency is determined by the BPR and the FPR. From the graph above we can also see that the TSFC can also be reduced by increasing the OPR (at a given propulsive efficiency).

Now looking at the effect of the other design parameters:



The fuel consumption decrease with both the TIT and the OPR. This has to do with the thermodynamic efficiency of the cycle.

Random lecture notes

T-S diagram (from lecture 8 IRL)

Temperature (which can be both static and total) - entropy diagrams are a useful tool to represent the thermodynamic processes that occur inside an aero-engine. The temperature used (static or total) depends on if we can measure the velocity of the flow: total temperature is used when flow velocity is not known (since it's related to total quantities and energy coming into/out from the system), while static is used when the flow velocity is known.